

The constant, the curve and a paradox

Arithmetic information in digits, probability and graphs

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Symbols at a glance

Each object is introduced where it first appears; this page is a place to return to. Here p_j is the j th prime. The lane of p consists of integers whose least prime factor is p .

Symbol	Meaning
$\mathcal{G}$	The original Goldbach constant.
C_g	A constant carrying a binary message g : $g_j = 0$ closes the lane of p_j ; $g_j = 1$ opens it.
$x_g(n)$	The n th binary digit of C_g .
$\ell(n)$	For $n \geq 2$, the index of its least prime factor: $\text{lpf}(n) = p_{\ell(n)}$.
$\lambda(n)$	The Liouville sign, $(-1)^{\Omega(n)}$; Ω counts prime factors with multiplicity.
δ_j	The density of positions in the lane of p_j .
S_g	The total density of disabled lanes.
$M_k(g)$	The correlation of k signs $1 - 2x_g(n)$, conditioned on pairwise coprime digit positions.
$K(k)$	The limiting probability that k sampled integers are pairwise coprime.
Φ_g, Ψ_g	The generating functions for conditioned and unconditioned correlations, respectively.
W, W_c	Independent prime and composite random factors; WW_c has the exponential law of mean one.
f, F_{sym}	The density of W on $(0, \infty)$, extended to the complex plane, and the symmetric cumulative distribution function derived from it.
G_g	The cumulative distribution function from sampling equally spaced digit positions with coprime start and step.
G_N, Y_N	The clean and observed labelled coprimality tables in the finite recovery experiment.
T_N	Effective signal $N(1 - 2\varepsilon_N)^2$, where ε_N is the probability of flipping each pair answer.
$D(n)$	The effective count of prime-power parts $p^{v_p(n)}$, measured by the entropy of their logarithmic shares.
B_k	The set of primes still present after deletion round k .

All logarithms are natural unless a base is specified. Local cutoff variables are defined where used.

1 The constant: repeating the Goldbach questions

Goldbach's conjecture asks whether every even integer greater than two is a sum of two primes. We begin by making a real number that remembers all of these questions. List the primes as $p_1 = 2, p_2 = 3, p_3 = 5, \dots$, and attach the question about $2(j+1)$ to p_j . Thus the prime 2 carries the question about 4, the prime 3 the question about 6, and so on. Write

$$g_j = \mathbf{1}_{\{2(j+1) \text{ is a sum of two primes}\}}.$$

Putting g_j only at digit position p_j would hide the answers in a set of density zero. Instead, we repeat each answer along a whole *prime lane*: the positive integers whose least prime factor is p_j .

One further ingredient gives the number its internal arithmetic. Let $\Omega(n)$ count prime factors with multiplicity, let $\lambda(n) = (-1)^{\Omega(n)}$, and set $\lambda(1) = 1$. For $n \geq 2$ let $\ell(n)$ be the index of its least prime factor. Our constant is

$$x_1 = 0, \quad x_n = \frac{1 - \lambda(n)}{2} g_{\ell(n)} \quad (n \geq 2), \quad \mathcal{G} = \sum_{n \geq 1} x_n 2^{-n}. \quad (1.1)$$

Every digit is computable by a finite calculation. In an enabled lane, the digit is one when the number has an odd number of prime factors, counting multiplicity. In a disabled lane every digit is zero. At prime positions the Liouville factor equals -1 , so

$$x_{p_j} = g_j. \quad (1.2)$$

The prime digits therefore list the Goldbach answers exactly, while the other digits repeat those answers in a structured way.

It is useful to permit an arbitrary binary message $g = (g_1, g_2, \dots)$ in the same definition. We write the resulting digits as $x_g(n)$ and the resulting real as C_g . Unless explicitly stated otherwise, the family has $g_1 = 1$; the original $\mathcal{G}$ is its Goldbach member. We shall use

$$h_g(1) = 0, \quad h_g(n) = 1 - g_{\ell(n)} \quad (n \geq 2), \quad a_g(n) = 1 - 2x_g(n) = h_g(n) + (1 - h_g(n))\lambda(n). \quad (1.3)$$

For the Goldbach member, h_g marks the disabled lanes corresponding to hypothetical exceptions. A general mask lets us study the construction without assuming any particular pattern of answers.

1.1 A fixed arithmetic signal inside every message

The lane of 2 is enabled because $4 = 2 + 2$. Every even position belongs to that lane, and complete multiplicativity gives

$$x_g(2m) = \frac{1 + \lambda(m)}{2}, \quad x_g(2m) + x_g(4m) = 1 \quad (m \geq 1). \quad (1.4)$$

All members of the family have exactly the same even-position digits. The second identity already rules out rationality.

Proposition 1.1. Every C_g with $g_1 = 1$ is irrational. Moreover, for $n \geq 2$ and any prime $q > n$,

$$x_g(n) + x_g(nq) = g_{\ell(n)}. \quad (1.5)$$

Proof. If the binary expansion had an eventual period T , choose a sufficiently large multiple m of T . The digits at $2m$ and $4m$ would agree, contradicting (1.4). A rational binary expansion is eventually periodic, so C_g is irrational. For the second assertion, nq has the same least prime factor as n , while $\lambda(nq) = -\lambda(n)$. Adding the two defining digits gives (1.5). $\square$

The paired reading shows how the parity rule repeats information: one of the two positions exposes an enabled lane even when the other does not. It also avoids any ambiguity between the two binary expansions of a dyadic rational, since none of the numbers under discussion is rational.

1.2 Goldbach is a balance law

Put

$$\eta_j = \prod_{r < j} (1 - p_r^{-1}), \quad \delta_j = \frac{\eta_j}{p_j}, \quad S_g = \sum_{j: g_j=0} \delta_j. \quad (1.6)$$

The number δ_j is the natural density of the j th lane. For example, $\delta_1 = 1/2$, $\delta_2 = 1/6$, and $\delta_3 = 1/15$. The identity $\delta_j = \eta_j - \eta_{j+1}$ and the divergence of $\sum_p 1/p$ give $\sum_j \delta_j = 1$.

Theorem 1.2 (Digit balance). For every binary mask,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n \leq N} x_g(n) = \frac{1}{2} \sum_j g_j \delta_j = \frac{1 - S_g}{2}. \quad (1.7)$$

Consequently C_g has limiting one-frequency $1/2$ if and only if all its message bits equal one. In particular,

$$\text{Goldbach} \iff \mathcal{G} \text{ is simply normal in base 2.} \quad (1.8)$$

Proof. Write $L(u) = \sum_{n \leq u} \lambda(n)$. We use the classical consequence $L(u) = o(u)$ of the prime number theorem; see, for example, [19, Section 2]. For fixed j , set $Q_j = \prod_{r < j} p_r$. An integer belongs to lane j exactly when it is $p_j m$ with $(m, Q_j) = 1$. Inclusion–exclusion gives

$$\begin{aligned} \sum_{\substack{m \leq u \\ (m, Q_j)=1}} \lambda(m) &= \sum_{d|Q_j} \mu(d) \lambda(d) L(u/d) \\ &= \sum_{d|Q_j} L(u/d) = o(u). \end{aligned}$$

Here Q_j is squarefree, so $\mu(d)\lambda(d) = 1$ for its divisors. The same inclusion–exclusion without λ gives the lane density δ_j . Since $\lambda(p_j m) = -\lambda(m)$, exactly half this lane contributes ones when it is enabled.

Sum first over the first J lanes. All omitted indices are coprime to $p_1 \cdots p_J$, a periodic set with density η_{J+1} . The upper limiting contribution of the omitted digits is at most η_{J+1} , which tends to zero. Taking $N \rightarrow \infty$ and then $J \rightarrow \infty$ proves (1.7). Every δ_j is strictly positive, so the limiting frequency is $1/2$ precisely when no lane is disabled. $\square$

A single missing answer therefore changes a global limiting statistic. For an illustration of the size of this effect, artificially turning off the lane of 5 lowers the one-frequency by $\delta_3/2 = 1/30$. This is an illustrative mask: the actual question attached to 5 is settled by $8 = 3 + 5$.

The amplification also obstructs long words. If $g_j = 0$, every position congruent to p_j modulo $P_j = \prod_{r \leq j} p_r$ has digit zero. Every interval of P_j consecutive positions contains one of these zeros. Thus arbitrarily long runs of ones, and in particular disjunctivity, would force every mask bit to be one. Their occurrence in the unmasked Liouville sequence remains a separate question.

1.3 RH measures the rate of balance

Simple normality specifies a limiting proportion. The Riemann hypothesis enters when we ask how quickly the even digits balance. The classical Liouville criterion [19, Theorem 1.1] is

$$\text{RH} \iff L(N) = O_\varepsilon(N^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (1.9)$$

Combining it with (1.4) proves

$$\text{RH} \iff \sum_{m \leq N} x_g(2m) = \frac{N}{2} + O_\varepsilon(N^{1/2+\varepsilon}) \quad (\varepsilon > 0). \quad (1.10)$$

For the whole Goldbach number we obtain the equally exact statement

$$\text{Goldbach and RH} \iff \sum_{n \leq N} x_n = \frac{N}{2} + O_\varepsilon(N^{1/2+\varepsilon}) \quad (\varepsilon > 0). \quad (1.11)$$

Indeed, the bound first forces limiting frequency $1/2$, hence Goldbach. Then every mask bit is one and the discrepancy equals $-L(N)/2$. The reverse implication follows from the same identity.

There is an analytic version of the common channel. Absolute convergence of its Euler product for $\Re s > 1$ gives

$$\sum_{m \geq 1} \frac{2x_g(2m) - 1}{m^s} = \sum_{m \geq 1} \frac{\lambda(m)}{m^s} = \frac{\zeta(2s)}{\zeta(s)}. \quad (1.12)$$

Thus RH is also equivalent to convergence of the left-hand series for every real $s > 1/2$. Under RH this follows by partial summation from (1.9). Conversely, those convergence statements make the Dirichlet series holomorphic in $\Re s > 1/2$. In that half-plane $\zeta(2s)$ is nonzero, so a zero of $\zeta(s)$ would give a pole in (1.12). The functional symmetry of zeta's nontrivial zeros then gives RH.

The distinction between proportion and rate matters. A binary law of the iterated logarithm gives discrepancy $O(\sqrt{N \log \log N})$, and therefore would imply the applicable RH bound above. Ordinary normality supplies no such rate. We use these equivalences to describe the information in the digits; none assumes the required cancellation has been proved.

1.4 Every fourth digit, and a rational percentage

The arithmetic repetition has a particularly short self-similarity test. Write $D_m C_g = \sum_{n \geq 1} x_g(mn) 2^{-n}$ for the number obtained by keeping every m th digit. Since $x_g(4n) = (1 - \lambda(n))/2$ for all n ,

$$\text{Goldbach} \iff D_4 \mathcal{G} = \mathcal{G}. \quad (1.13)$$

Equality is equality of binary streams, by irrationality; at a prime position it forces the corresponding Goldbach bit to be one.

More generally, let $p = \text{lpf}(m)$ and define $G_p(1) = g_{\ell(p)}$, with

$$G_p(n) = \begin{cases} g_{\ell(n)}, & \text{lpf}(n) < p, \\ g_{\ell(p)}, & \text{lpf}(n) \geq p \end{cases} \quad (n \geq 2).$$

Then

$$x_g(mn) = \frac{1 - \lambda(m)\lambda(n)}{2} G_p(n). \quad (1.14)$$

The function G_p is periodic, depending only on divisibility by primes smaller than p . In particular, the two irrational numbers $D_p C_g$ and $D_{p^2} C_g$ have a rational sum. For $p = 2$ their sum is one.

For the original Goldbach mask, there are infinitely many certainly successful lanes: whenever r is prime, the lane indexed by $r - 1$ carries the true equation $2r = r + r$. Testing (1.14) at its second digit first forces $\lambda(m) = 1$ if $D_m \mathcal{G} = \mathcal{G}$. Testing at prime digit positions then forces the Goldbach bits from $\text{lpf}(m)$ onward to be constant, hence equal to one. Conversely, that successful tail and $\lambda(m) = 1$ make (1.14) equal the original stream. We have proved

$$\exists m \geq 2 : D_m \mathcal{G} = \mathcal{G} \iff \text{there are only finitely many Goldbach exceptions.} \quad (1.15)$$

If the exceptions are nonempty and their last lane has prime p , the smallest reproducing stride is q^2 , where q is the next prime after p . With no exceptions the smallest stride is 4.

There is also a numerical reading of the message that will be useful when we later lose the digit addresses. For a prime p , write $\eta_p = \prod_{r < p} (1 - 1/r)$ and $\delta_p = \eta_p/p$; write $f_p = 1 - g_{\ell(p)}$ for its failure bit. Let z_m be the limiting fraction of zeros in the dilated stream $x_g(m), x_g(2m), \dots$

Proposition 1.3 (A percentage decoder). If $m \geq 2$ and $p = \text{lpf}(m)$, then

$$2z_m - 1 = \sum_{r < p} f_r \delta_r + f_p \eta_p. \quad (1.16)$$

This is rational and determines all message bits at primes at most p , provided $g_1 = 1$.

Proof. The failed-lane indicator at mn is

$$h_g(mn) = \sum_{r < p} f_r \mathbf{1}_{\{\text{lpf}(n)=r\}} + f_p \mathbf{1}_{\{(n, \prod_{r < p} r)=1\}}.$$

It is periodic. Its product with $\lambda(n)$ has mean zero by the same finite inclusion–exclusion argument used in Theorem 1.2. Averaging the signed digit $h_g(mn) + \lambda(m)(1 - h_g(mn))\lambda(n)$ proves (1.16).

Set $P = \prod_{r < p} r$, $A_r = \prod_{q < r} (q - 1)$, and $M = P(2z_m - 1)$. For odd primes $r \leq p$, define

$$C_r = \begin{cases} A_r P / \prod_{q \leq r} q, & r < p, \\ A_p, & r = p. \end{cases}$$

Then $M = \sum_{3 \leq r \leq p} f_r C_r$. The power of two dividing C_r is exactly the power dividing A_r , and these powers strictly increase as r runs through the odd primes. Divide by the smallest remaining power of two and read parity: this recovers the first remaining f_r . Subtract $f_r C_r$ and repeat. The bit at two was fixed in advance, so every required bit is recovered. $\square$

This is an exact-information statement. Recovering M by rounding an approximation to z_m requires error less than $1/(4P)$. The demanding precision is part of the theorem’s meaning. Later, coprimality frequencies will give a different decoder with statistical recovery guarantees.

Remark 1.4 (Many messages in a small set). The set $\{C_g : g_1 = 1\}$ is a compact perfect set of Hausdorff and upper box dimension zero. Indeed, the map from masks to reals is continuous and injective, with inverse on its image read at prime positions. Changing a sufficiently late bit gives a distinct arbitrarily close member. For $N \geq 2$, there are exactly $2^{\pi(N)-1}$ possible prefixes of length N , so the set is covered by that many intervals of length 2^{-N} . Since $\pi(N) = o(N)$, the dimension assertions follow. There are uncountably many irrational members, but exactly one simply normal member, the all-one-mask Liouville constant.

2 A divisor reading: Euler’s sum or infinity

Digits can be read through multiplication as well as in their original order. For the Goldbach member write $a_n = 1 - 2x_n$ and $b_n = h_g(n)(1 - \lambda(n))/2$. Then $a_n = \lambda(n) + 2b_n$. The elementary identity

$$\sum_{d|m} \lambda(d) = \mathbf{1}_{\{m \text{ is a square}\}}$$

follows by factoring the divisor sum prime by prime. Thus the signed divisor sums

$$c_m = \sum_{d|m} a_d = \mathbf{1}_{\{m \text{ is a square}\}} + 2 \sum_{d|m} b_d \quad (2.1)$$

are nonnegative integers. A cancellation rule in the digits has become a positive arithmetic spectrum.

Theorem 2.1 (The Basel alternative). The series below has nonnegative summands and satisfies

$$\sum_{m \geq 1} \frac{1}{m} \sum_{d|m} (1 - 2x_d) = \begin{cases} \pi^2/6, & \text{if Goldbach holds,} \\ +\infty, & \text{if Goldbach fails.} \end{cases} \quad (2.2)$$

Proof. Under Goldbach, every b_d is zero, so only squares contribute: $\sum_m c_m/m = \sum_{r \geq 1} r^{-2} = \pi^2/6$. If a lane with prime p fails, then $b_p = 1$. Equation (2.1) gives $c_{pk} \geq 2$ for every $k \geq 1$, and hence $\sum_m c_m/m \geq (2/p) \sum_{k \geq 1} 1/k = +\infty$. $\square$

One failed Goldbach case changes Euler's value into infinity. The proof's strength comes from positivity: the harmonic contribution of one disabled lane cannot be cancelled by any other lane.

All-enabled digits	0	1	1	0	1	0	1	1	0	0	1	1	1	0	0	0	1	1	1	1	0	0	1	0
Every fourth digit	0	1	1	0	1	0	1	1	0	0	1	1	1	0	0	0	1	1	1	1	0	0	1	0
Toy: lane 7 blocked	0	1	1	0	1	0	0	1	0	0	1	1	1	0	0	0	1	1	1	1	0	0	1	0

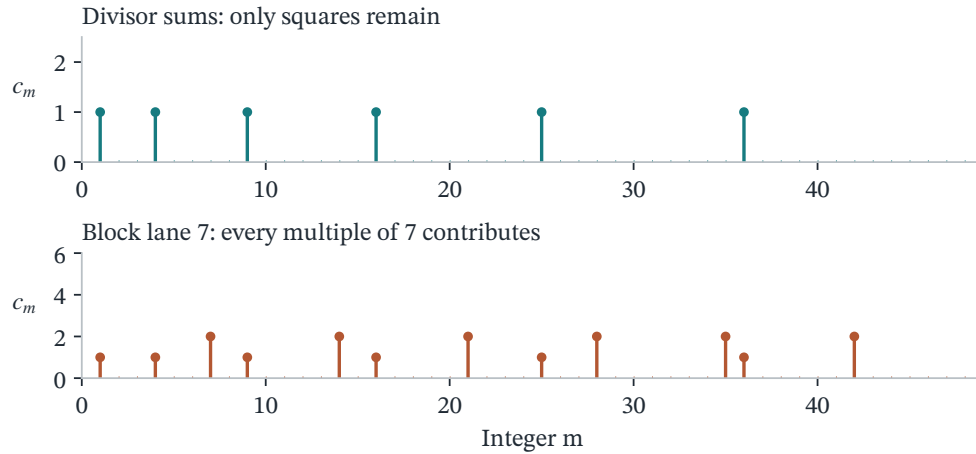


Figure 1: Exact finite digit and divisor readings. The all-enabled mask is compared with an artificial mask that disables the lane of 7. The actual Goldbach question assigned to that lane is already settled by $10 = 5 + 5$. The lower plot illustrates how a failed lane would add a harmonic subseries.

2.1 A dimension change and a broken symmetry

Exponentially damping the same divisor sums gives, for $t > 0$,

$$\Theta_{\mathcal{G}}(t) = 1 + 2 \sum_{m \geq 1} c_m e^{-\pi m t} = \theta(t) + 4 \sum_{n \geq 1} \frac{b_n}{e^{\pi n t} - 1}, \quad \theta(t) = \sum_{k \in \mathbb{Z}} e^{-\pi k^2 t}. \quad (2.3)$$

Absolute convergence follows, for example, from $0 \leq c_m \leq \sum_{d|m} 1$; the second equality follows by summing over multiples. It is legitimate to view $\Theta_{\mathcal{G}}$ as the heat trace of a diagonal operator: there is one zero state and energy πm has multiplicity $2c_m$ whenever $c_m > 0$.

Proposition 2.2 (Heat-growth dimension). The limit

$$d_{\mathcal{G}} = 2 \lim_{t \downarrow 0} \frac{\log \Theta_{\mathcal{G}}(t)}{\log(1/t)}$$

exists, and equals one if Goldbach holds and two otherwise. More precisely, in the latter case,

$$\Theta_{\mathcal{G}}(t) \sim \frac{2S_g}{\pi t} \log \frac{1}{t}. \quad (2.4)$$

Proof. Under Goldbach, $\Theta_{\mathcal{G}} = \theta$, and Jacobi's transformation $\theta(t) = t^{-1/2} \theta(1/t)$ gives $\theta(t) \sim t^{-1/2}$; see [30, p. 20.7.32]. If a lane p fails, its single term in (2.3) gives a lower bound of order t^{-1} . Since $b_n \leq 1$, splitting the remaining sum at $n = 1/t$ gives an upper bound $O(t^{-1} \log(1/t))$. These estimates prove the dimension dichotomy without a density theorem.

For the sharper equivalent, Theorem 1.2 applied lane by lane gives $\sum_{n \leq y} b_n = (S_g/2)y + o(y)$. Partial summation yields $\sum_{n \leq y} b_n/n = (S_g/2) \log y + o(\log y)$. Consequently

$$\sum_{m \leq y} c_m = \lfloor \sqrt{y} \rfloor + 2 \sum_{n \leq y} b_n \lfloor y/n \rfloor \sim S_g y \log y,$$

the floor error being $O(y)$. Integrating this counting function against $\pi t e^{-\pi t y}$ proves (2.4). Domination follows from $\sum_{m \leq y} c_m = O(y \log(y+2))$. $\square$

The word dimension here means the heat-growth exponent just defined. Under Goldbach the operator agrees with the Laplacian spectrum of a circle of circumference $2\sqrt{\pi}$. The extra logarithm in (2.4) is retained explicitly; no ordinary smooth two-dimensional surface is being asserted in the other case.

Proposition 2.3 (A single inversion identity). For every fixed $t > 1$,

$$t^{-1/2} \Theta_{\mathcal{G}}(1/t) - \Theta_{\mathcal{G}}(t) \geq 0,$$

and equality holds if and only if Goldbach holds.

Proof. Jacobi's identity cancels the two theta terms in (2.3). What remains is

$$4 \sum_{n \geq 1} b_n \left[\frac{t^{-1/2}}{e^{\pi n/t} - 1} - \frac{1}{e^{\pi n t} - 1} \right].$$

Every bracket is positive. In fact, for $u = \pi n/t$, convexity gives $e^{t^2 u} - 1 \geq t^2(e^u - 1)$, while $t^{-1/2} > t^{-2}$. Thus equality forces all b_n to vanish; a failed prime lane would have $b_p = 1$. $\square$

The same transform can locate the first failure, if one exists. Put $E_g(t) = \Theta_g(t) - \theta(t)$. If p_* is the smallest failed-lane prime, its first nonzero exponential coefficient is four:

$$E_g(t) = 4e^{-\pi p_* t} (1 + O(e^{-\pi t})) \quad (t \rightarrow \infty), \quad p_* = -\frac{1}{\pi} \lim_{t \rightarrow \infty} \frac{\log E_g(t)}{t}. \quad (2.5)$$

To justify the error, divide the series in (2.1) by its first exponential term and bound its remaining coefficients by the divisor function. The resulting tail is $O(e^{-\pi t})$ for $t \geq 1$. Moreover $-E'_g(t)/E_g(t)$ decreases to πp_* : its derivative is minus the variance of the energies under their exponentially weighted positive coefficients. There are at least two energies, πp_* and $2\pi p_*$, so the decrease is strict.

2.2 A finite exception list in a heat coefficient

If there are finitely many disabled lanes, a further exact reading is available. Let $F = \{j : g_j = 0\}$ be finite, and set

$$A(s) = \sum_{j \in F} p_j^{-s} \prod_{r < j} (1 - p_r^{-s}), \quad C(s) = \sum_{j \in F} p_j^{-s} \prod_{r < j} (1 + p_r^{-s}), \quad B = \sum_{j \in F} 2^{j-1}.$$

Here B is the integer whose binary digits list the disabled lanes. These are finite Dirichlet polynomials, and $A(1) = S_g$.

Proposition 2.4. For a finite failed-lane set in the Goldbach construction,

$$\begin{aligned} \Theta_g(t) = \frac{2}{\pi t} & \left[S_g \log(1/t) + (\gamma - \log \pi) S_g + A'(1) \right] \\ & + \frac{1 + C(1/2)}{\sqrt{t}} - B + O_F(\sqrt{t}) \quad (t \downarrow 0). \end{aligned} \quad (2.6)$$

Thus the constant term after the three divergent terms have been removed records the entire finite exception list in binary.

The mechanism is a residue calculation: the growing terms come from the poles at 1 and 1/2, while the constant term comes from the pole of the gamma factor at zero. The cancellation at that last pole leaves a sum of distinct powers of two. The full calculation is in Appendix A.1.

For example, an artificial message that disables lanes with indices 2 and 4 has remaining constant term $-(2 + 8) = -10$. Read the magnitude as binary 1010: the two ones recover

those exact indices. The actual Goldbach questions attached to these lanes, 6 and 10, have known representations; the example illustrates the encoding for a general mask. The limiting growth detects that something was removed. Its finite part can remember exactly what was removed.

The finiteness hypothesis belongs to this last proposition. The Basel alternative, the dimension dichotomy, and the first-failure reading required no such assumption. Each was obtained from the same nonnegative divisor sums, so a single construction supplies several different levels of information.

3 A staircase made by sampling the digits

Reading the digits at equally spaced positions gives another real number:

$$Y_g(a, d) = \sum_{j \geq 0} x_g(a + jd) 2^{-j-1}.$$

Now choose the starting point a and the step d uniformly from $1, \dots, N$, subject to $\gcd(a, d) = 1$. This last restriction prevents the entire progression from being trapped in a common prime lane. Let $\nu_{g,N}$ be the distribution of $Y_g(a, d)$. Its limiting distribution gives a curve that rises across every interval, even in examples where its derivative is zero almost everywhere. The apparent contradiction disappears once we distinguish total rise from the slope at a typical point.

Write $\mathcal{F} = \{p_j : g_j = 0\}$ for the disabled primes. The hypothesis needed below says that the successful primes retain a divergent reciprocal sum:

$$\sum_{p \notin \mathcal{F}} \frac{1}{p} = \infty. \quad (3.1)$$

It holds unconditionally for the actual Goldbach mask. Indeed, let $E(Y)$ count even Goldbach exceptions in $[4, Y]$ and let $A(R) = \#(\mathcal{F} \cap [2, R])$. Pintz's Theorem 1 [32] states that $E(Y) < Y^{18/25}$ beyond an ineffective threshold. The prime number theorem therefore gives

$$A(R) = E(2(\pi(R) + 1)) \ll (R/\log R)^{18/25} \quad (R \geq 3). \quad (3.2)$$

Partial summation shows $\sum_{p \in \mathcal{F}} 1/p < \infty$, whereas the sum over all primes diverges. We use this established estimate as an input. Its precise exponent is useful later; here any power strictly smaller than one would suffice.

Theorem 3.1 (Primitive sampling). Assume (3.1). Then $\nu_{g,N}$ converges weakly to a nonatomic probability measure ν_g of full support $[0, 1]$ and Hausdorff dimension one. Its Lebesgue decomposition is

$$\nu_g = L_g \text{Leb} + \nu_{g,\text{sc}}, \quad L_g = \prod_{p \in \mathcal{F}} \frac{1}{p+1}, \quad (3.3)$$

where $\nu_{g,\text{sc}}$ is singular continuous. The empty product is one; an infinite product here is zero. Consequently

$$G_g(y) := \nu_g([0, y]) \quad (0 \leq y \leq 1)$$

is a homeomorphism of $[0, 1]$ onto itself and $G'_g(y) = L_g$ for Lebesgue-almost every y .

Full support means that every nonempty open interval receives positive probability. Singular means that a set of ordinary length zero carries the measure. Dimension one means that every set carrying its full mass has Hausdorff dimension one. These properties can coexist: small in length need not mean small in dimension. We do not claim that the measure has the same local dimension at almost every point.

The mechanism is easier to see than the full convergence proof. Choose, independently at every prime p , a residue pair $(U, V) \bmod p$ uniformly among the $p^2 - 1$ pairs other than $(0, 0)$. These simultaneous residues are a primitive profinite pair. Independently choose fair bits $B_0, B_1, \dots$. The limiting sampled real has the law of

$$Z_g = \sum_{j \geq 0} (1 - h_g(U + jV)) B_j 2^{-j-1}.$$

Thus an open lane contributes a fair bit and a disabled lane contributes zero. This is a theorem about the average over starting points and steps; it does not assume that the original consecutive digits are independent.

A lane p is visited precisely when p does not divide V . The probability that it is avoided is $1/(p + 1)$. Consequently every bit is free precisely on an event of probability L_g ; that event supplies the uniform part of the measure. Once a disabled lane is visited, zeros repeat on an infinite arithmetic progression. Such digit strings form a single Lebesgue-null set, which carries the remaining part.

This explains the decomposition and its product formula. Full support comes from making any finite prefix open by finitely many compatible congruences. Full dimension requires an additional limiting argument: there are positive portions of the law with a proportion of free bits arbitrarily close to one. The complete proof, including the Frantzikinakis–Host variable-step cancellation that produces the fair bits, is in Appendix B.

Corollary 3.2. For the original Goldbach constant, G_g is the identity if and only if Goldbach holds. If there are finitely many exceptions and at least one, then

$$G'_g(y) = \frac{1}{\prod_{p \in \mathcal{F}} (p + 1)} \in (0, 1) \quad \text{almost everywhere.}$$

There are infinitely many Goldbach exceptions if and only if $G'_g = 0$ almost everywhere. In this last case G_g is a singular homeomorphism whose associated measure has dimension one.

This phenomenon also has an unconditional explicit example. Disable exactly the lanes

$$\mathcal{F}_* = \{p_{2^j} : j \geq 2\}. \quad (3.4)$$

This is a concrete mask, defined without asking any unresolved arithmetic question. Its disabled-prime reciprocal sum converges because $p_{2^j} > 2^j$. The associated G_* is therefore a strictly increasing singular homeomorphism, with full-support measure of dimension one. Thus the geometry exists independently of which Goldbach alternatives occur.

4 Reading several digits at coprime positions

The divisor reading led to a positive spectrum. We now try a different experiment: select several digit positions, insisting that no two have a common prime factor. This prevents two positions from drawing their least prime from the same lane. Consequently a correlation of k digits can ask whether there are at least k disabled lanes.

For independent uniform integers $n_1, \dots, n_k$ in $[1, N]$, let $\mathcal{C}_{k,N}$ be the event that they are pairwise coprime. Define $M_0(g) = 1$ and, for $k \geq 1$,

$$M_k(g) = \lim_{N \rightarrow \infty} \mathbb{E} \left[\prod_{i=1}^k a_g(n_i) \middle| \mathcal{C}_{k,N} \right]. \quad (4.1)$$

We will prove that this limit exists. Throughout this section an arbitrary binary mask is allowed, including $g_1 = 0$.

At one prime p , pairwise coprimality permits either no divisible coordinate or exactly one. Their total probability is $(1 - 1/p)^k + kp^{-1}(1 - 1/p)^{k-1}$. It follows that

$$K(k) = \lim_{N \rightarrow \infty} \mathbb{P}(\mathcal{C}_{k,N}) = \prod_p (1 - p^{-1})^k \left(1 + \frac{k}{p-1} \right) > 0. \quad (4.2)$$

To justify the infinite product, first impose the restrictions only at primes at most R and use the Chinese remainder theorem. The probability of a collision at a larger prime is at most $\binom{k}{2} \sum_{p > R} p^{-2}$, uniformly in N . This tends to zero, and the local factors differ from one by $O_k(p^{-2})$. In particular $K(0) = K(1) = 1$ and $K(2) = 1/\zeta(2) = 6/\pi^2$.

The Euler product in (4.2) is classical. Written as $\prod_p (1 - 1/p)^{s-1} (1 + (s-1)/p)$, it is the arithmetic factor appearing in mod-Poisson descriptions of the number of distinct prime divisors; see [21, Section 4]. Here its role is dictated by the coprimality experiment itself.

Proposition 4.1 (Correlations count available lanes). The limits in (4.1) exist and satisfy

$$0 \leq M_k(g) \leq 1, \quad M_1(g) = S_g, \quad M_k(g) > 0 \iff \#\{j : g_j = 0\} \geq k. \quad (4.3)$$

They have the following probability description. Independently at each prime p , choose one of k coordinates with probability $1/(p+k-1)$ each, or choose none with probability $(p-1)/(p+k-1)$. A coordinate is labelled by the first prime that chooses it. Then $M_k(g)$ is the probability that all k labels belong to disabled lanes.

Each factor $a_g(n_i)$ is a sign, so its product can be negative. Yet the limit is a probability. Averaging cancels the Liouville oscillation on every enabled lane; the disabled lanes leave a persistent contribution. Coprimality forces the first prime labels to be distinct. Those two facts explain both the nonnegativity and the exact counting test. The truncation argument that justifies them is in Appendix A.4.

We collect the correlations into two entire functions,

$$\Phi_g(z) = \sum_{k \geq 0} M_k(g) \frac{z^k}{k!}, \quad \Psi_g(z) = \sum_{k \geq 0} K(k) M_k(g) \frac{z^k}{k!}. \quad (4.4)$$

Both series converge absolutely everywhere, with modulus at most $e^{|z|}$. The first averages after conditioning on coprimality; the second gives noncoprime tuples score zero and averages over all tuples. If there are exactly r disabled lanes, each function is a polynomial of degree r . Under Goldbach both are identically one. When there are infinitely many disabled lanes, every coefficient is positive.

4.1 A positive operator behind the correlations

The change from conditional to unconditional averages has a strong spectral effect. Its proof also gives an explicit construction. A positive operator is a matrix with a nonnegative quadratic form, possibly on an infinite-dimensional space. Its eigenvalues are nonnegative. In the theorem below they also have a finite sum. Such an operator has a determinant $\prod_j (1 + \alpha_j z)$, so its zeros can lie only on the negative real axis. The assertion is that the unconditional correlation function has precisely this structure. Write $\mathcal{F} = \{p_j : g_j = 0\}$ for its disabled primes.

Theorem 4.2 (The positive mask operator). For every prime mask there is a positive trace-class operator A_g with

$$\Psi_g(z) = \det(I + zA_g), \quad \text{tr } A_g = S_g, \quad \text{rank } A_g = \#\mathcal{F}. \quad (4.5)$$

The rank may be infinite. Every zero of Ψ_g is negative and real.

The construction processes primes one at a time. Each disabled lane adds one positive spectral direction, while the trace records the total disabled-lane density. Appendix A.2 gives the explicit differential recurrence and the limiting operator.

This explains the rank statement in ordinary terms: counting the operator's nonzero directions counts the missing answers. If exactly three lanes are disabled, the infinite construction produces only three such directions.

The proof establishes a summable spectrum for every mask, without requiring any estimate for Goldbach's exceptional set. It also gives the useful bound

$$K(k)M_k(g) \leq S_g^k. \quad (4.6)$$

Indeed, $K(k)M_k(g)/k!$ is the k th elementary symmetric function of the α_j , and the distinct-index terms in $(\sum_j \alpha_j)^k$ contribute exactly $K(k)M_k(g)$.

4.2 Two missing lanes, two geometries of zeros

The distinction between the two correlation functions becomes visible in a small example. Artificially disable just the lanes of 3 and 5. Their densities add to $1/6 + 1/15 = 7/30$. Two coprime positions can use those two lanes in either order; the prime-assignment model gives $M_2 = 2(1/3)(1/4)(1/6) = 1/36$. Consequently

$$\Phi_g(z) = 1 + \frac{7}{30}z + \frac{1}{72}z^2, \quad \Psi_g(z) = 1 + \frac{7}{30}z + \frac{1}{12\pi^2}z^2.$$

The discriminant of the first polynomial is $-1/900$, so its two zeros are nonreal. The second has two negative real zeros, as the operator theorem requires. The same two disabled lanes have produced both geometries. Conditioning on coprimality changes the zeros, even though both functions retain the exact count of missing lanes.

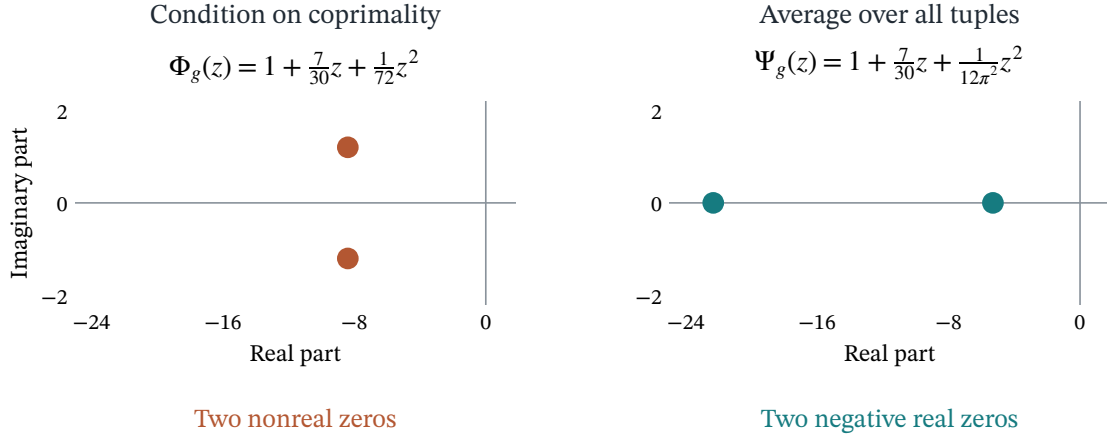


Figure 2: The same artificial message disables the lanes of 3 and 5. Conditioned and unconditioned correlations give two quadratic polynomials with different zero locations. The coefficients are exact; coordinates in the plot are their numerical root locations.

This is the question that will lead to the curve: can we describe that change of average by introducing an independent random scale?

4.3 What the growth and zeros count

The operator behind Ψ_g controls the location of its zeros. The companion Φ_g records how quickly disabled lanes accumulate. If those primes grow sparse, the function grows slowly; if they grow more numerous, it grows faster. For the Goldbach member this comparison also turns the conjecture into a statement about complex zeros.

Put $a_k = M_k(g)/k!$ and $B_g(R) = \#(\mathcal{F} \cap [2, R])$. If all k labels in the prime-assignment model of Proposition 4.1 are masked, their first assigned primes are k distinct members of $\mathcal{F}$. For a chosen set of k primes, sum over its $k!$ assignments to the labels and discard the first-assignment restrictions. Independence between primes gives the useful bound

$$a_k \leq e_k \left(\frac{1}{p+k-1} : p \in \mathcal{F} \right) \leq e_k \left(\frac{1}{p} : p \in \mathcal{F} \right), \quad (4.7)$$

Here e_k means the sum of products over all k -element subsets; for example $e_2(w) = \sum_{p < q} w_p w_q$. Thus, when $\sum_{p \in \mathcal{F}} 1/p < \infty$,

$$|\Phi_g(z)| \leq \Phi_g(|z|) \leq P_{\mathcal{F}}(|z|) := \prod_{p \in \mathcal{F}} (1 + |z|/p). \quad (4.8)$$

The product bounds the function from above. Its factors are not being identified with the zeros of Φ_g .

The *order* of an entire function records the power in its exponential growth: roughly, growth like $\exp(r^\alpha)$ on disks of radius r gives order α . The following theorem matches that analytic exponent with a counting exponent for the disabled primes.

Theorem 4.3 (Counting exponent and entire-function order). Suppose the disabled-prime counting exponent

$$\eta_g = \limsup_{R \rightarrow \infty} \frac{\log \max(1, B_g(R))}{\log R}$$

is strictly less than one. Then Φ_g has order exactly η_g . Formally the order is the limsup of $\log \log \max_{|z| \leq r} |\Phi_g(z)|$ divided by $\log r$; constants and polynomials have order zero by convention. For the actual Goldbach mask,

$$\text{ord } \Phi_g = \limsup_{Y \rightarrow \infty} \frac{\log \max(1, E(Y))}{\log Y} \leq \frac{18}{25}, \quad (4.9)$$

and, as $r \rightarrow \infty$,

$$\log \max_{|z| \leq r} |\Phi_g(z)| = \log \Phi_g(r) \ll (r/\log r)^{18/25}. \quad (4.10)$$

In particular, for an absolute constant C and $k \geq 2$,

$$M_k(\mathcal{G}) \leq \left(\frac{C}{k^{7/18} \log(k+1)} \right)^k. \quad (4.11)$$

The proof is in Appendix C. Its two estimates have opposite roles. Summing over every possible set of disabled primes bounds the function from above. Requiring the first k disabled primes to hit distinct labels supplies a matching lower bound on its order. The analytic exponent therefore cannot lose the arithmetic counting exponent.

Corollary 4.4 (Zeros count the exceptions). Under the hypothesis of Theorem 4.3, list the zeros of Φ_g with multiplicity as (ζ_j) . Then

$$\Phi_g(z) = \prod_j (1 - z/\zeta_j), \quad \sum_j |\zeta_j|^{-1} < \infty, \quad \sum_j \zeta_j^{-1} = -S_g. \quad (4.12)$$

Counting multiplicity, the number of zeros equals the number of disabled lanes, allowing infinity. If $n_g(R)$ counts the zeros in $|z| \leq R$, with multiplicity, then

$$\limsup_{R \rightarrow \infty} \frac{\log \max(1, n_g(R))}{\log R} = \eta_g. \quad (4.13)$$

For the actual Goldbach function,

$$\text{Goldbach} \iff \Phi_g \text{ has no complex zeros,} \quad n_g(R) \ll (R/\log R)^{18/25}.$$

If there are infinitely many exceptions, every complex value is attained by Φ_g infinitely often.

Proof. Hadamard's factorization theorem [15] says that an entire function of order less than one has a genus-zero product and a constant exponential prefactor. Since $\Phi_g(0) = 1$, this gives the first two assertions of (4.12); differentiation at zero gives its last assertion because $\Phi'_g(0) = M_1(g) = S_g$. For a finite mask the polynomial degree is $\#\mathcal{F}$. For an infinite mask all coefficients are positive, so the function is not a polynomial; the factorization then forces infinitely many zeros. The same argument applied to $\Phi_g - w$ shows infinite attainment of each $w \in \mathbb{C}$ when the mask is infinite.

Jensen's formula and $\Phi_g(0) = 1$ give $n_g(R) \log 2 \leq \log \Phi_g(2R)$, which bounds the zero-counting exponent by η_g and gives the displayed quantitative estimate. Conversely, if that exponent were $\lambda < \eta_g$, choose $\lambda < \beta < \eta_g < 1$. The product (4.12), together with $n_g(t) \ll_\beta t^\beta$, implies

$$\log \max_{|z| \leq r} |\Phi_g(z)| \leq \sum_j \log(1 + r/|\zeta_j|) = O_\beta(r^\beta)$$

by the same integration used in (C.1), starting below the smallest zero modulus. This contradicts its order. The zero-order case follows from the upper bound and nonnegativity. $\square$

These are zeros of the mask function. The worked example above shows that they can be nonreal. The sparsity hypothesis is essential to the factorization argument. Also, order zero alone does not mean finitely many exceptions: an infinite mask with subpower counting function has the same order. The improved moment estimate measures a statistic of our construction; it does not improve the imported bound for $E(Y)$.

4.4 One bias can hide infinitely many changes

There is a sharp reason to keep the higher correlations rather than only the proportion of zero digits. Put $\delta_p = p^{-1} \prod_{q < p} (1 - 1/q)$, so $S_g = \sum_{p \in \mathcal{F}} \delta_p$.

Proposition 4.5. Among masks known to have finitely many disabled lanes, the exact bias S_g determines the mask. Nevertheless, for every sufficiently large prime p , there is an infinite set $\mathcal{H} = \{q_1 < q_2 < \dots\}$ of primes greater than p such that

$$\sum_{q \in \mathcal{H}} \delta_q = \delta_p, \quad q_{j+1}/q_j \longrightarrow \infty, \quad \#(\mathcal{H} \cap [2, R]) = o(\log R). \quad (4.14)$$

Thus two masks can have exactly the same digit bias while one has one disabled lane and the other has infinitely many.

Proof. If P is the largest prime of a finite nonempty mask, then δ_P has P -adic valuation -1 : its numerator contains only factors $q - 1 < P$, and its denominator is the primorial through P . Every earlier δ_q has nonnegative P -adic valuation. Consequently P is the largest prime dividing the reduced denominator of S_g . Recover P , subtract δ_P , and repeat.

List all primes as p_i and set $d_i = \delta_{p_i}$. These weights decrease strictly to zero and

$$d_i/d_{i+1} = p_{i+1}/(p_i - 1) \longrightarrow 1$$

by the prime number theorem. Choose p beyond a point where every consecutive ratio is less than two. Starting with residual $r_0 = \delta_p$, repeatedly subtract the largest weight at a prime greater than p which does not exceed the residual. The first choice is the successor of p . If q is a later choice and q^- its immediate prime predecessor, the residual after subtraction satisfies

$$0 \leq r_{\text{new}} \leq \delta_{q^-} - \delta_q = o(\delta_q).$$

The ratio bound makes $r_{\text{new}} < \delta_q$, so no weight is used twice and selected primes strictly increase. The process cannot terminate: a finite sum at primes greater than p has a reduced denominator containing its largest prime, unlike δ_p . It therefore makes infinitely many selections, and $r_j < \delta_{q_j} \rightarrow 0$ proves the sum identity.

The next selected weight is at most the preceding residual, hence $\delta_{q_{j+1}}/\delta_{q_j} \rightarrow 0$. Mertens' product estimate $\delta_q \sim e^{-\gamma}/(q \log q)$ forces $q_{j+1}/q_j \rightarrow \infty$. For any fixed $C > 1$, these ratios eventually exceed C , so the counting function is at most $\log R/\log C + O_C(1)$. Letting C tend to infinity proves (4.14). $\square$

The equality is exact, so it survives arbitrarily accurate measurement of the digit frequency. For these two masks $\Phi'_g(0)$ agrees, but their functions are respectively linear and transcendental entire. The second has order zero and infinitely many zeros by the preceding results. This is an example within the mask family; it asserts no actual Goldbach exception, and no collision between the whole functions.

5 The curve selected by coprimality

The positive operator belongs to a particular message. The factor $K(k)$ that connected its determinant with the original correlations does not. We now isolate that universal factor. Define

$$K(s) = \prod_p (1 - p^{-1})^s \left(1 + \frac{s}{p-1}\right), \quad s \in \mathbb{C}, \quad (5.1)$$

using the real logarithm of $1 - 1/p$. The local factor is $1 + O_B(p^{-2})$ on $|s| \leq B$, so the product is entire. Its zeros are simple and occur precisely at $s = 1 - p$: away from these points the logarithm of the tail converges absolutely, and at each such point exactly one factor vanishes.

The reciprocals of its nonnegative integer values are moments of a positive random variable. This kind of reciprocal-Mellin construction, including its role in cancelling arithmetic factors by an auxiliary random scale, occurs in Barhoumi–Andréani's dissertation [2, Section 3.6.1]. Its general mechanism is therefore prior work. Our aim here is the concrete coprimality law and its actual density.

Proposition 5.1 (The universal scale). Let $(U_p)_p$ be independent uniform variables on $(0, 1)$. The product

$$W = \prod_p \frac{p}{p-1} U_p^{1/(p-1)} \quad (5.2)$$

converges almost surely to a finite positive random variable, and

$$\mathbb{E}W^s = \frac{1}{K(s)} \quad (\Re s > -1). \quad (5.3)$$

This is the exact half-plane of finite absolute moments. In particular, $\mathbb{E}W = 1$ and $\text{Var } W = \pi^2/6 - 1$.

Proof. Write $E_p = -\log U_p$, so the E_p are independent exponentials of mean one. The logarithm of the p th factor is $\log(p/(p-1)) - E_p/(p-1)$. Its mean is $O(p^{-2})$ and its variance is $(p-1)^{-2}$. The centered independent series converges almost surely and in L^2 , while its means sum absolutely. Exponentiation proves the product assertion.

For a finite prime cutoff R , direct integration gives $\mathbb{E}W_R^s = 1/K_R(s)$ for $\Re s > -1$. If $\sigma = \Re s$, choose $r > 1$ with $r\sigma > -1$; this is always possible. The product of moments $\mathbb{E}W_R^{r\sigma}$ is uniformly bounded, since its local factors are $1 + O_{s,r}(p^{-2})$. Thus $(W_R^s)_R$ is uniformly integrable, and almost-sure convergence proves (5.3). For $\sigma \leq -1$, write $W = 2U_2V$ with $V > 0$ independent of U_2 . Conditional integration over U_2 makes $\mathbb{E}W^\sigma$ infinite. Finally $K(1) = 1$ and $K(2) = 6/\pi^2$ give the stated mean and variance. $\square$

In terms of the cited general construction, the logarithm of W is a deterministic constant plus $\sum_p (1 - E_p)/(p-1)$. This identifies the specialization directly. The same scale now reverses the coprimality weighting of every message.

Corollary 5.2 (The exact bridge). For every mask and every $z \in \mathbb{C}$,

$$\Phi_g(z) = \mathbb{E}\Psi_g(zW) = \mathbb{E}\det(I + zWA_g). \quad (5.4)$$

Proof. All coefficients of Ψ_g are nonnegative. Tonelli and (5.3) give

$$\mathbb{E}\Psi_g(|z|W) = \sum_{k \geq 0} K(k)M_k(g) \frac{|z|^k}{k!} \mathbb{E}W^k = \Phi_g(|z|) \leq e^{|z|}.$$

Absolute integrability therefore permits the same termwise exchange for complex z , proving the identity. $\square$

We have reached a division of roles. The operator carries the message; W carries the universal effect of changing the average. It is this universal law that produces the curve in the title.

5.1 The entire density theorem

A probability density is initially a real function, defined on the possible values of its random variable. The density here has an additional analytic structure: it continues to an entire function, and the positions of its nonzero Taylor coefficients recover the primes exactly. Reciprocal transforms of real-zero entire products have a classical theory, notably Schoenberg's [36]; the following proof establishes this specific density and the required convergence directly.

Theorem 5.3 (The prime density). The random variable W has a density on $(0, \infty)$ which extends to the entire function

$$f(z) = \sum_{p \text{ prime}} \frac{z^{p-2}}{K'(1-p)}. \quad (5.5)$$

Consequently

$$f^{(n)}(0) \neq 0 \iff n+2 \text{ is prime.} \quad (5.6)$$

With $c_p = 1/K'(1-p)$, the nonzero coefficients have signs $\text{sgn}(c_p) = (-1)^{\pi(p)-1}$ and satisfy

$$\log |c_p| = -p \log \log p + O(p). \quad (5.7)$$

Why a probability density contains this pattern. With only finitely many primes, take logarithms of the defining random product. We obtain a sum of independent exponential variables, whose rates are $p-1$. Returning from logarithms turns these rates into powers z^{p-2} in a finite density formula. As the prime cutoff grows, the right endpoint of that formula moves to infinity and its coefficients converge. This is how the primes become derivative positions.

There are two things to prove: convergence of the entire series, and identification of its restriction to the positive axis with the actual probability density. Appendix A.3 proves both, including the coefficient estimate in the theorem.

The support of the Taylor series begins

$$f(z) = c_2 + c_3 z + c_5 z^3 + c_7 z^5 + c_{11} z^9 + c_{13} z^{11} + \dots$$

The construction did not begin by writing down a prime-supported series. It began with coprimality probabilities, then produced a positive random scale and finally its density. The theorem says that positivity, normalization, entire continuation and this exact prime pattern coexist in the same curve.

Consider what the derivative test says at its first few positions. The second derivative vanishes at zero because $2+2=4$ is composite. The third derivative does not vanish because $3+2=5$ is prime. The fourth vanishes, and the fifth does not. The test continues forever. These are exact zeros amid exact nonzeros, rather than a tendency for prime positions to have larger coefficients.

The source of the curve is equally important. An observer can describe the sampling experiment and ask for the resulting probability density before asking any question about its derivatives. The arithmetic pattern then appears inside that density. This is what distinguishes the result from merely choosing a list of coefficients indexed by primes.

5.2 Reflection, a symmetric distribution, and the twin constant

Let

$$\mathfrak{C}_2 = \prod_{p>2} \frac{p(p-2)}{(p-1)^2} \quad (5.8)$$

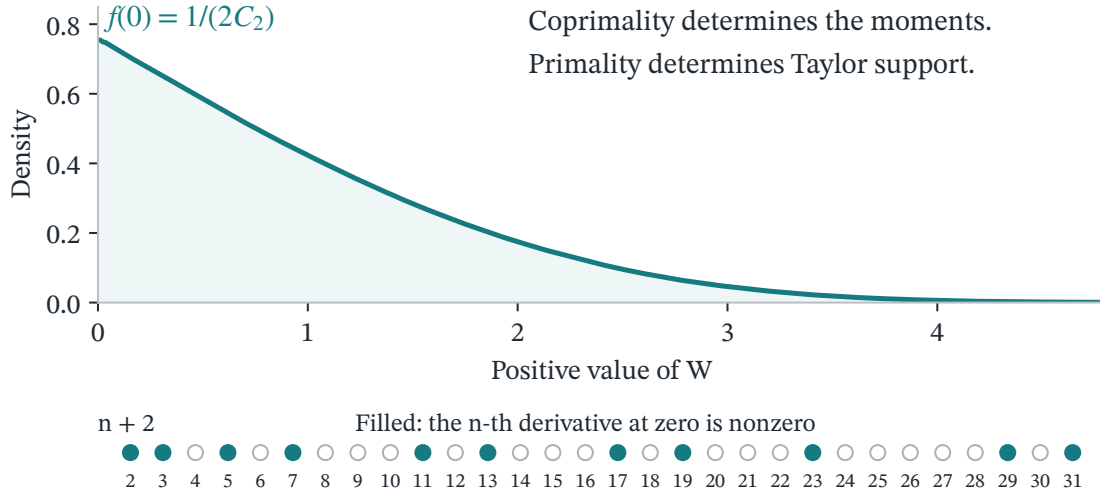


Figure 3: The density on the positive axis and its exact Taylor-support pattern. The curve is an explicitly approximate conditional Monte Carlo preview with 180,000 samples, primes through 2,000 and seed 271828; no numerical error certificate is claimed. The marked derivative positions represent Theorem 5.3, not a numerical derivative test.

be the usual twin-prime constant. Its appearance below uses only this convergent product, with no assertion about twin-prime infinitude.

Corollary 5.4. The entire density satisfies

$$f(0) = \frac{1}{2\mathfrak{C}_2}, \quad f(z) + f(-z) = \frac{1}{\mathfrak{C}_2} \quad (z \in \mathbb{C}). \quad (5.9)$$

It is nonincreasing on the real line, with limits $1/\mathfrak{C}_2$ and zero at negative and positive infinity, respectively. Consequently

$$F_{\text{sym}}(x) = \mathfrak{C}_2 f(-x)$$

is a symmetric cumulative distribution function on $\mathbb{R}$. The entire function f is bounded on the real axis and has infinite order in the complex plane.

Proof. Writing $W = 2U_2V$ and conditioning on V gives

$$f(w) = \mathbb{E} \left[\frac{1}{2V} \mathbf{1}_{\{2V > w\}} \right], \quad w > 0. \quad (5.10)$$

The independent prime-3 factor makes the distribution of V nonatomic. Once its inverse moment below is finite, dominated convergence makes the right-hand side continuous, so this identity holds pointwise for the entire density, not merely almost everywhere. The inverse moment of V is $\mathbb{E}V^{-1} = \prod_{p>2} (p-1)^2/[p(p-2)] = 1/\mathfrak{C}_2$; the moment passage follows as in Proposition 5.1, now with domain $\Re s > -2$. Thus $f(0) = 1/(2\mathfrak{C}_2)$ and f is nonincreasing on the positive half-line. Integrability makes its limit there zero. Every nonconstant exponent $p-2$ in (5.5) is odd, proving the reflection identity. This extends the monotonicity and gives the two endpoint limits, from which the CDF assertion follows.

For completeness, the coefficient formula for the order of an entire series is $\rho = \limsup_{n \rightarrow \infty} n \log n / \log(1/[z^n]f|)$, with zero coefficients omitted. Along $n = p - 2$, (5.7) makes this ratio asymptotic to $\log p / \log \log p$, which tends to infinity. $\square$

Thus a smooth bounded curve on the real line can have extremely large growth in other complex directions. For every fixed $a > 0$, there are arbitrarily large R for which $\max_{|z|=R} |f(z)| > \exp(R^a)$, even though f stays bounded on the whole real axis. The prime pattern is local analytic information at zero; ordinary plotting resolution does not reveal its vanishing derivatives. The reflection identity provides a simple real counterpart of that analytic structure.

5.3 Put the composites back

The product defining W uses only prime indices. The missing composite factors complete it to a familiar distribution. On an independent family of uniform variables define

$$W_c = \prod_{\substack{m \geq 4 \\ m \text{ composite}}} \frac{m}{m-1} U_m^{1/(m-1)}. \quad (5.11)$$

Its logarithm converges by the same summable-mean and summable-variance argument as before. We can now ask a simple question: what happens if the prime contribution and the composite contribution are put back together? The answer is the standard exponential distribution, whose density is e^{-w} for $w > 0$.

Proposition 5.5 (Prime-composite completion). For independent W and W_c defined above,

$$WW_c \stackrel{d}{=} \text{Exp}(1), \quad \mathbb{E}W_c^s = \Gamma(1+s)K(s) \quad (\Re s > -1). \quad (5.12)$$

Proof. Combine all integer factors up to M :

$$V_M = \prod_{m=2}^M \frac{m}{m-1} U_m^{1/(m-1)}.$$

Its Mellin transform is

$$\mathbb{E}V_M^s = M^s \prod_{j=1}^{M-1} \frac{j}{j+s} = M^s \frac{\Gamma(M)\Gamma(1+s)}{\Gamma(M+s)}.$$

This is also the Mellin transform of $M \min(U_1, \dots, U_{M-1})$, as follows directly by integrating the density of the minimum. Characteristic functions of the logarithms identify the two laws. For fixed $x \geq 0$ and $M > x$, the survival probability of the latter variable is $(1 - x/M)^{M-1} \rightarrow e^{-x}$. Meanwhile V_M converges almost surely to the product WW_c . Thus WW_c is exponential. Uniform integrability of fixed moments with real part greater than -1 follows from a nearby moment exactly as in Proposition 5.1. Independence and (5.3) now give the second identity. $\square$

This completion is a short specialization of classical beta and gamma product identities. It will also supply the composite random scale in the later matrix construction. The path from the constant to the curve has therefore produced more than an encoding: the same coprimality factor can change conditional averages, generate a positive spectrum, and connect two complementary random products.

The second moments give a small numerical glimpse of this cancellation:

$$\mathbb{E}W^2 = \frac{\pi^2}{6}, \quad \mathbb{E}W_c^2 = \frac{12}{\pi^2}, \quad \mathbb{E}(WW_c)^2 = 2.$$

The complete law, and not only this moment, becomes exponential. The prime and composite factors separately retain their arithmetic structure; their independent product reconstructs the distribution obtained by allowing every integer factor.

6 Two questions to ask the curve

The density has already given us a way to recognize individual primes. It is natural to ask whether familiar conjectures can be read from the probability law without extracting its Taylor coefficients one at a time. Two observations do this. One changes the way we sample the law and watches its variance. The other compares two moments and watches the signs of a matrix's eigenvalues.

These are criteria: the relevant sign inequality and the unbounded matrix index remain to be proved. Their interest is that the observables are ordinary probabilistic and linear-algebraic objects belonging to the same law.

6.1 The Riemann hypothesis as a variance comparison

Write $Y = \log W$. For $t > -1$, change its distribution by giving an outcome of size $Y = y$ a weight proportional to e^{ty} . More formally,

$$\mathbb{E}_t h(Y) = \frac{\mathbb{E}[W^t h(\log W)]}{\mathbb{E}W^t}.$$

This is exponential tilting. Increasing t favors the larger values of W . If $\kappa(t) = \log \mathbb{E}W^t = -\log K(t)$, differentiation gives

$$\text{Var}_t(Y) = \kappa''(t) = \sum_p \frac{1}{(p-1+t)^2}. \quad (6.1)$$

The variance is a softened count of primes. The prime number theorem suggests replacing the prime atoms by the continuous density $du/\log u$, giving

$$V_0(t) = \int_2^\infty \frac{du}{(u-1+t)^2 \log u}.$$

Does the actual variance eventually lie above or below this smooth comparison?

Theorem 6.1 (An eventual variance deficit). The Riemann hypothesis holds if and only if

$$\text{Var}_t(\log W) < V_0(t) \quad \text{for all sufficiently large } t.$$

The word *eventually* matters. This is an inequality along the positive real axis, with no assumed simplicity or spacing of zeta zeros. Eventual-sign criteria for RH have a long history; Robin's work [33] is an important example. The statistic here is the tilted variance of the law already constructed.

What would an experimenter see? The parameter t changes which outcomes matter most: increasing it favors larger values of W . We then measure the spread of their logarithms. The comparison curve $V_0(t)$ is obtained by replacing the discrete primes with their smooth predicted density. The criterion says that RH is exactly an eventual ordering of these two spreads. No individual zero of zeta appears in the statement.

The proof exposes why a one-sided comparison is possible. Prime squares give a negative correction to the smooth approximation. On RH, that correction is larger than all the possible oscillations from the nontrivial zeros combined. If RH fails, oscillations force crossings in both directions arbitrarily far out. The complete argument and its quantifiers are in Appendix D.1.

This is an exact reformulation attached to a probability law we already met. Its usefulness as a route to RH would require a new way to compare those variances. The theorem supplies the comparison to aim for.

6.2 Twin primes hidden in matrices with positive entries

For this observation we separate the prime 2, writing

$$W = 2U_2V, \quad K_o(s) = \prod_{p>2} (1 - p^{-1})^s \left(1 + \frac{s}{p-1}\right).$$

Then $\mathbb{E}V^s = 1/K_o(s)$ for $\Re s > -2$. Compare moments three units apart:

$$F(t) = \frac{\mathbb{E}V^t}{\mathbb{E}V^{t+3}} = \frac{K_o(t+3)}{K_o(t)}, \quad t > 0.$$

The number three has a purpose. Differences between odd primes are even; a shift of three crosses precisely the gap of two before it reaches the next possible gap.

For a symmetric matrix, its *negative index* is the number of negative eigenvalues, counted with multiplicity. Equivalently, it is the greatest dimension of a subspace on which its quadratic form is strictly negative.

Theorem 6.2 (A matrix count of twin pairs). Fix $a > 0$, and form

$$H_N(a) = \left[\frac{(-1)^{i+j} F^{(i+j+2)}(a)}{(i+j+2)!} \right]_{0 \leq i, j < N}.$$

Every entry is strictly positive. Nevertheless,

$$\sup_{N \geq 1} n_-(H_N(a)) = \#\{p > 2 : p \text{ and } p-2 \text{ are prime}\}.$$

Both sides are allowed to be infinite. The negative index is nondecreasing in N .

The surprise is the exact arithmetic count. A quotient of moments of the probability law gives a sequence of positive-entry matrices whose negative directions count twin pairs.

The general relation between signed moments and negative squares belongs to indefinite moment theory [3]. We prove the compact signed-measure step directly in the appendix, which also shows exactly where the primes enter.

Why a negative direction means a twin pair. The moment quotient has one pole associated with each odd prime p . Its residue is negative precisely when $p-2$ is also prime. This sign test is elementary: the shift by three crosses the preceding odd prime only when the two primes are twins.

The derivatives assemble those signed residues into a moment matrix. Polynomials can separate any finite collection of its negative atoms, turning each chosen twin pair into a negative direction in a sufficiently large matrix. Conversely, the matrix cannot have more negative directions than there are negative atoms. Appendix D.2 makes this argument precise and controls the infinite tail.

Thus the matrices provide a counter whose reading can rise as their size increases. The twin-prime conjecture says that this counter never reaches a final value. All its entries are positive throughout; the signs being counted belong to quadratic forms, which combine entries using both positive and negative weights.

For comparison, the positive-entry matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has quadratic form -2 at the vector $(1, -1)$, and therefore one negative direction. Our matrices turn this familiar linear-algebraic possibility into an exact count of prime pairs.

6.3 What a finite calculation can certify

The theorem turns any rigorously negative k -dimensional subspace into a certificate of k twin pairs. One can build the matrices from ordinary reciprocal-prime sums. At $a = 1$, write

$$\frac{F(1-z)}{F(1)} = \exp\left(\sum_{k \geq 1} \frac{d_k z^k}{k}\right) = \sum_{n \geq 0} c_n z^n, \quad d_k = \sum_{p > 2} \{p^{-k} - (p+3)^{-k}\}.$$

Then

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{k=1}^n d_k c_{n-k}, \quad H_N(1) = F(1)[c_{i+j+2}]_{0 \leq i, j < N}.$$

For an integer cutoff Y ,

$$0 \leq d_k - d_k^{(Y)} \leq 3Y^{-k}.$$

These are explicit bounds for every uncomputed prime, not a decision to ignore the tail.

An exact rational certificate supplied with the manuscript verifies that $[c_{i+j+2}]_{0 \leq i, j < 15}$ has at least four negative directions using only primes through 100 and the displayed tail bounds. We give its specification in Appendix J. It recovers already known pairs; it demonstrates a verification mechanism rather than a new twin-prime lower bound.

There is a reason to resist turning this calculation into a promise of an easy proof of infinitude. The tail bounds allow all omitted d_k -increments to be zero. Any conclusion valid uniformly for every such tail must also hold for the finite prime product, whose negative index counts only its existing twins. A proof forcing further negative directions needs additional arithmetic information. The matrix formulation tells us precisely what that information would have to accomplish.

The next question is different. Instead of asking a probability law to answer a conjecture, we can ask whether its observable relationships reveal the hidden integers that produced them.

7 What the relationships remember

The curve led us to $K(k)$, the limiting probability that k sampled integers are pairwise coprime. Let us reverse that experiment. Hide the integers and keep only the coprimality answers. A single answer contains very little: “not coprime” does not say whether the common divisor is 2, 7, or 1009. A whole collection of answers contains much more. It can identify the *actual numerical primes* shared by a pair.

Here are the rules of the finite game we shall eventually play.

1. Put independent uniform integers from 1 to N^2 in N numbered boxes. The box numbers remain visible; the contents are hidden.
2. For each pair, write one bit: 1 if the contents are coprime, and 0 otherwise. Independently reverse each bit with the same unknown probability.
3. Give the resulting table to a decoder. Its task is to name every prime shared by each pair of boxes, with a fraction of mistaken pairs tending to zero as the experiment grows.

There is just one observed bit per pair. The decoder cannot ask for a second, cleaner answer. We will allow the error probability to approach one half, making individual answers approach fair coin tosses.

Before introducing noise, we can see the arithmetic mechanism in three ordinary fractions.

7.1 Three percentages name the shared primes

Temporarily open two boxes and put 210 and 462 inside them. Test each against a third integer chosen uniformly from 1 to M , and let $M \rightarrow \infty$. The exact limiting frequencies are

the third integer is coprime to	frequency
210 = $2 \cdot 3 \cdot 5 \cdot 7$	8/35
462 = $2 \cdot 3 \cdot 7 \cdot 11$	20/77
both	16/77

For the last row it must avoid 2, 3, 5, 7, 11. Multiply the first two frequencies and divide by the third:

$$\frac{(8/35)(20/77)}{16/77} = \frac{2}{7} = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{7}\right).$$

The factors belonging to 5 and 11 cancel. Each belongs to one box, so it occurs once above and once below. A shared prime occurs twice above and once below, leaving one factor behind. Closing the boxes again does not change the calculation: the three frequencies alone identify the common-prime product.

There is a second small surprise. The reduced fraction $2/7$ has hidden the primes 2 and 3 through ordinary cancellation, but has not lost them. Its denominator gives away 7; removing the factor $6/7$ exposes the next prime:

$$\frac{2}{7} \xrightarrow{\times 7/6} \frac{1}{3} \xrightarrow{\times 3/2} \frac{1}{2} \xrightarrow{\times 2} 1.$$

Thus the three percentages recover the entire set $\{2, 3, 7\}$.

Lemma 7.1 (Reading a finite prime set from a rational number). For a finite set A of primes, the rational number $q(A) = \prod_{p \in A} (p-1)/p$ determines A uniquely. If A is nonempty, its largest prime is the largest prime factor of the denominator of $q(A)$ in lowest terms. Remove this prime p by multiplying the ratio by $p/(p-1)$, and repeat until the ratio is one.

Proof. Let $p = \max A$. No numerator factor $r-1$, with $r \in A$, is divisible by p , so p remains in the reduced denominator. No larger prime can occur there. Removing the factor $(p-1)/p$ reduces the question to a smaller set. $\square$

This is the classical injectivity of the totient ratio on squarefree integers; Tao [38, Lemma 2.1 and Remark 2.3] records the same descending-prime mechanism. We will use it as a decoder.

7.2 A rational relation among transcendental frequencies

To turn the three percentages into observable quantities, give each box infinitely many independent reference boxes. The natural limiting model keeps only prime divisibility. At prime p , a box contains that feature with probability $1/p$; all these choices are independent.

For each vertex $i \geq 1$ and prime p , independently let $B_{i,p}$ have Bernoulli distribution with parameter $1/p$. Write $S_i = \{p : B_{i,p} = 1\}$ and join i to j when $S_i \cap S_j$ is empty. We denote this answer by E_{ij} , and put $E_{ii} = 0$. These are the prime-divisibility profiles of independent Haar-random profinite integers. No profinite terminology is needed for the arguments: the independent bits just specified define the entire model. Each box carries infinitely many primes almost surely, by independence and the divergence of $\sum_p 1/p$. Two boxes share only finitely many almost surely, since $\sum_p 1/p^2$ converges. That difference between a single profile and a shared profile is what makes a finite rational answer possible.

For a fixed number of sampled ordinary integers, their coprimality graph converges to this model as the sampling range tends to infinity. Indeed, the Chinese remainder theorem gives convergence of every fixed finite prime profile. For k vertices, ignoring primes above R changes some edge with probability at most $\binom{k}{2} \sum_{p>R} p^{-2}$, in either model. First take the range limit and then $R \rightarrow \infty$. This prime-feature description belongs to the established probabilistic theory of coprimality; see, for example, Martineau [24].

The familiar kernel has a direct interpretation here:

$$\mathbb{P}(E_{ij} = 1 \text{ for all } 1 \leq i < j \leq k) = \prod_p (1 - 1/p)^{k-1} (1 + (k-1)/p) = K(k). \quad (7.1)$$

At each prime, at most one of the k profiles may contain it. In particular the edge probability is

$$\kappa = K(2) = \prod_p (1 - p^{-2}) = \frac{6}{\pi^2}. \quad (7.2)$$

Thus the same K that determined the moments of the prime probability law now describes a sampling experiment.

For a fixed box, let d_i be the fraction of reference boxes coprime to it. For two boxes, let c_{ij} be the fraction coprime to both. Formally,

$$d_i = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k \leq M} E_{ik}, \quad c_{ij} = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k \leq M} E_{ik} E_{jk}.$$

The omitted diagonal terms have no effect on these limits. The same cancellation as in the example survives even though each individual profile now contains infinitely many primes almost surely.

Theorem 7.2 (Frequency reconstruction). Almost surely all d_i, c_{ij} exist and are positive, every pair shares only finitely many primes, and

$$\frac{d_i d_j}{c_{ij}} = \prod_{p \in S_i \cap S_j} \left(1 - \frac{1}{p}\right). \quad (7.3)$$

Consequently the ordered infinite graph determines every set S_i , including the numerical labels of its primes. The random variables $d_1, d_2, \dots$ are almost surely algebraically independent over $\mathbb{Q}$, and each c_{ij} is almost surely transcendental.

Proof. Tonelli's theorem gives $\mathbb{E} \sum_{p \in S_i} p^{-1} = \sum_p p^{-2} < \infty$. Thus, almost surely, the following products converge to positive values. Conditional strong laws, with the displayed profiles held fixed, give

$$d_i = \prod_{p \in S_i} (1 - 1/p), \quad c_{ij} = \prod_{p \in S_i \cup S_j} (1 - 1/p). \quad (7.4)$$

Also $\mathbb{E}|S_i \cap S_j| = \sum_p p^{-2} < \infty$, so the intersection is finite almost surely. Cancelling the convergent logarithms in (7.4) proves (7.3), and Lemma 7.1 recovers each intersection. Every prime present in S_i occurs in infinitely many other profiles almost surely, so their intersections recover all of S_i . Countability makes these assertions simultaneous for all vertices and primes.

For the last assertions consider $Z = \prod_p (1 - 1/p)^{C_p}$, where independent C_p have parameters $q_p = 1/p$ or $q_p = 2/p - 1/p^2$. Both products converge positively by the same argument. By Lemma 7.1, distinct assignments of the bits through R give distinct prefix products. The largest prefix atom is therefore

$$M_R = \prod_{p \leq R} \max(q_p, 1 - q_p) \longrightarrow 0,$$

since $\sum_p \min(q_p, 1 - q_p) = \infty$. Conditional on the independent positive tail, the probability that the whole product equals any specified real number is at most M_R . Hence Z has no atoms. This applies to d_i and c_{ij} , and the countability of algebraic numbers proves their almost-sure transcendence.

The d_i are independent. A nonzero polynomial in finitely many of them vanishes with probability zero: induct on the number of variables, condition on all but the last, and use the finitely many roots of the remaining nonzero one-variable polynomial. Taking the countable union over rational polynomials proves algebraic independence. $\square$

The single-box frequencies d_i are almost surely algebraically independent: no nonzero polynomial with rational coefficients vanishes at any finite list of them. Each common-neighbor frequency is also transcendental. Yet multiplying the two individual frequencies and dividing by the joint frequency gives an exact rational code for the shared primes. The arithmetic is in the dependence between the observations. None of the three percentages needs to be rational for their combination to name 2, 3, and 7.

For a fixed box, every prime it contains eventually appears in another box. Taking the union of the recovered intersections therefore recovers its entire prime profile. This is a statement about the random infinite experiment; it supplies almost-sure transcendence, rather than an explicit new named transcendental number. The visible box labels keep the sampling order needed to form the frequencies.

7.3 Calibrating a corrupted infinite table

Now corrupt the answers. There is initially an extra unknown: a low observed coprimality frequency might reflect arithmetic or a tendency of the channel to reverse answers. The

average over the whole table separates them. Its clean value is already known, $6/\pi^2$, so the same constant that counted coprime pairs calibrates the noise.

Suppose each unordered answer is independently flipped with a fixed probability ε , independently of the profiles. Put $a = 1 - 2\varepsilon$. The corrupted neighbor and common-neighbor frequencies u_i, v_{ij} satisfy

$$u_i = \varepsilon + ad_i, \quad v_{ij} = \varepsilon^2 + \varepsilon a(d_i + d_j) + a^2 c_{ij}. \quad (7.5)$$

The limiting edge density q is $q = \varepsilon + a\kappa$, so the unknown error rate is recovered by

$$\varepsilon = \frac{\kappa - q}{2\kappa - 1}. \quad (7.6)$$

The denominator is nonzero. The conditional strong law proves (7.5); the edge-density strong law follows from bounded differences in the independent vertex profiles and then in the independent edge flips. For example the resulting exponentially small tail probabilities are summable, so Borel–Cantelli applies.

If $a \neq 0$, equations (7.5) recover d_i, c_{ij} , then all prime profiles by Theorem 7.2. An error rate above one half is still usable: for example, an answer that is always reversed can simply be reversed again. The calibration discovers which side of one half we are on. Exactly at one half, every observed answer is an independent fair coin and all arithmetic information disappears. An arbitrarily small fixed distance from one half can be overcome by infinitely many boxes. The finite game asks how small that distance may become as the number of boxes grows.

8 The recovery paradox

Return now to the finite boxes. The exact ratio is no longer available: we have estimated frequencies, corrupted observations, and only finitely many reference boxes. We must explain how an approximation to a real number can still reveal an exact prime set. The proof uses the repeated patterns across the whole table to strengthen its frequency estimates.

For each N , independently sample $X_1, \dots, X_N$ uniformly from $\{1, \dots, N^2\}$. Let G_N be the labelled coprimality graph with adjacency matrix E , and observe

$$Y_{ij} = E_{ij} \oplus A_{ij}, \quad A_{ij} \sim \text{Bernoulli}(\varepsilon_N), \quad i < j. \quad (8.1)$$

All flips are mutually independent and independent of the integers. The common error probability is unknown to the recovery algorithm and may vary with N ; $\oplus$ denotes addition modulo two. Write

$$\begin{aligned} a_N &= 1 - 2\varepsilon_N, & T_N &= Na_N^2, \\ R_{ij} &= \text{rad gcd}(X_i, X_j). \end{aligned} \quad (8.2)$$

The quantity a_N is the signed strength of one answer; T_N combines that strength with the number of available boxes. The integer R_{ij} records every shared prime once. For the pair 210, 462, it equals $42 = 2 \cdot 3 \cdot 7$. Coprimality does not record prime-power exponents.

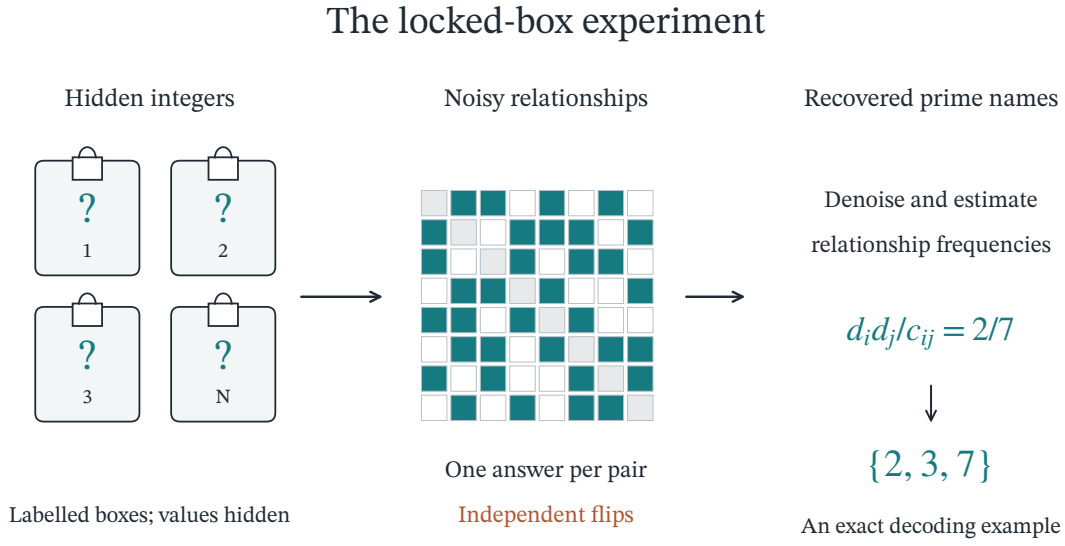


Figure 4: A schematic of the recovery experiment. The box identities remain aligned while their integer values are hidden. One independently corrupted coprimality bit is observed per unordered pair. The theorem recovers exact shared-prime sets on a fraction of pairs tending to one; this diagram makes no finite-sample accuracy claim. The exact ratio $2/7$ illustrates the decoder for the prime set $\{2, 3, 7\}$.

Theorem 8.1 (The threshold for recovery on almost every pair). There is a single procedure using Y alone such that, whenever $T_N \rightarrow \infty$, for every fixed pair $i \neq j$,

$$\mathbb{P}(\hat{R}_{ij} = R_{ij}) \rightarrow 1, \quad \frac{1}{\binom{N}{2}} \sum_{i < j} \mathbf{1}_{\{\hat{R}_{ij} \neq R_{ij}\}} \rightarrow 0 \quad \text{in probability.}$$

If T_N does not tend to infinity, neither a vanishing error probability for a fixed pair nor a vanishing fraction of pair errors is possible, even if ε_N is supplied. The same criterion holds in the independent-prime model.

This is a sharp distinction. If the distance of the error rate from one half is a constant times $N^{-1/2}$, some uncertainty about a fixed pair persists. Multiply that distance by *any* factor tending to infinity, however slowly, and consistency becomes possible. The theorem names the primes correctly on a proportion of pairs tending to one. It gives no guarantee that every pair in a given table is correct.

How an approximate table gives exact names. There are three steps. First calibrate the noise using the average coprimality frequency. Then use the repeated prime features to denoise the whole matrix. Finally estimate the three frequencies for each pair and search for the nearest admissible rational code. With a bound on their denominators, distinct rationals have a positive separation; the allowed bound grows slowly enough for the estimation error to fall below that separation. A typical pair has a bounded common divisor with high probability, so eventually its correct code enters the search.

The matrix step uses established singular-value thresholding [6, 40]. The full construction and all error estimates are in Appendix E. The rational search there deliberately grows very slowly; the theorem is an asymptotic result, with no practical sample-size guarantee.

There is also an elementary reason a weaker signal cannot suffice. Imagine being told everything except whether one hidden odd number u has been replaced by $2u$. If its partner is even, that one choice changes whether they share the prime 2. Only the answers involving the uncertain box can distinguish the choices. Their combined testing strength is bounded when T_N is bounded. Even this almost-completely opened version of the game retains a positive error probability.

8.1 Parity and every fixed finite prime profile

The decoder identifies numerical prime names, so it can distinguish even from odd. This is more concrete than recovering an unnamed group of boxes that happen to resemble one another. A box containing a multiple of 2 shares 2 with roughly half the other boxes. An odd box shares 2 with none. The decoded pair sets turn that difference into a vote.

Corollary 8.2. Under $T_N \rightarrow \infty$, for each fixed prime cutoff P the divisibility bits $\mathbf{1}_{\{p|X_i\}}$, $p \leq P$, can be recovered correctly for a fraction of vertices tending to one. For each fixed vertex their joint recovery probability tends to one. In particular the table reveals the parity of almost every hidden integer.

Proof. For a fixed p , declare that $p \mid X_i$ when more than $N/(2p)$ of the estimated R_{ij} are divisible by p . The true number is zero if $p \nmid X_i$, and otherwise it is the number of other sampled multiples of p , which is $N/p + o_{\mathbb{P}}(N)$ simultaneously for all such rows. On this event, a mistaken vertex bit requires at least $N/(3p)$ pair errors in its row. Summing over rows proves a vanishing vertex-error fraction; a finite union handles $p \leq P$. Equivariance gives the fixed-vertex assertion. $\square$

For parity, the rule in the proof uses a threshold of $N/4$ shared-2 reports. The same idea recognizes multiples of 3, of 5, or of any other fixed prime: the two limiting row frequencies are 0 and $1/p$. Putting finitely many of these tests together reveals patterns such as “divisible by 3 and 7, but not by 2 or 5.” These will shortly recover the prime lanes of the original constant.

The fixed cutoff in the corollary matters. A prime occurring in only one sampled integer has no shared-prime witness in the finite table. And replacing a prime factor by a higher power changes none of its coprimality relationships. The recovered arithmetic is the presence of the specified primes.

Question 15.2 asks for a practical version of this recovery: how the required number of boxes depends on the prime, the desired accuracy and the noise.

8.2 Two ways to count what survives

To compare recovery with information, use natural logarithms and let $H(U) = -\sum_u \mathbb{P}(U = u) \log \mathbb{P}(U = u)$ for a discrete random object. Mutual information is $I(U; V) = H(U) - H(U | V)$: the average reduction in uncertainty about U after observing V . With natural logarithms, one fair binary choice has entropy $\log 2$. All graph entropies below include the vertex labels.

There are now two scores for the game. Pair accuracy gives one point for each completely correct shared-prime set. The information fraction

$$\frac{I(G_N; Y_N)}{H(G_N)} = 1 - \frac{H(G_N | Y_N)}{H(G_N)}$$

asks what proportion of the clean table's uncertainty the noisy table has removed. A small value means that most of that original uncertainty remains. The two scores need not move together.

Theorem 8.3 (Information in noisy coprimality). In the ordinary-integer model (8.1), uniformly in $\varepsilon_N \in [0, 1]$,

$$I(G_N; Y_N) = N \log(1 + T_N) + O(N \log \log N). \quad (8.3)$$

In particular

$$H(G_N) = N \log N + O(N \log \log N). \quad (8.4)$$

The clean table has roughly $N^2/2$ entries but only about $N \log N$ units of uncertainty. Its entries are tied together by the hidden integers. For comparison, the ordered list of those integers has entropy exactly $2N \log N$. Since the clean table is determined by the list, (8.4) says that it retains asymptotically half the list's Shannon information.

At any fixed error rate different from one half, the noisy table retains a fraction tending to one of the *clean table's* information. Thus even 49% independent reversals leave nearly all of that fraction in the limit. Letting the error rate approach one half gives a continuous range of outcomes. For every fixed $0 < \theta \leq 1$,

$$T_N = N^\theta \implies \frac{I(G_N; Y_N)}{H(G_N)} \longrightarrow \theta.$$

Meanwhile, prime-set recovery on almost every pair succeeds for every one of these choices. Taking $\theta = 1/100$ leaves asymptotically one percent of the information, with the fraction of correctly decoded pairs still tending to one. We can push the information fraction all the way to zero by making T_N grow more slowly.

The error in (8.3) may exceed its displayed main term for very small T_N . The proof also gives $I(G_N; Y_N) = O(N)$ when T_N stays bounded. We shall only use the uniform formula below to obtain an upper bound, for which its error term suffices.

Corollary 8.4 (Almost every answer, a vanishing information fraction). Take $\varepsilon_N = (1 - \sqrt{\log N/N})/2$, so $T_N = \log N$. Then a procedure recovers the exact shared-prime set for a fraction of pairs tending to one, while

$$\frac{I(G_N; Y_N)}{H(G_N)} \longrightarrow 0. \quad (8.5)$$

Proof. The recovery conclusion follows from Theorem 8.1. The numerator in (8.5) is $O(N \log \log N)$ by Theorem 8.3, whereas its denominator is $N \log N + O(N \log \log N)$. $\square$

This is the paradox named in the title. On almost every pair we recover an exact list of numerical primes, although a fraction tending to one of the clean table’s original uncertainty remains unresolved.

Where can all that unresolved information hide? Fix a prime cutoff P . For any pair of our ordinary sampled integers,

$$\mathbb{P}(\text{the pair shares a prime larger than } P) \leq \sum_{p>P} \frac{1}{p^2} \longrightarrow 0.$$

Thus the large primes can be ignored at a small cost in pair accuracy. This remains true when “correct” requires the *whole* shared-prime set: a pair with no common prime above P has no omitted part of its answer. Letting the cutoff grow slowly is enough to cover almost all pairs.

Uncertainty adds up differently. In the independent-prime model, the entropy of one box’s divisibility bits through P grows like $\log P$; the prime sum behind this is $\sum_{p \leq P} (\log p)/p \sim \log P$. There are many rare choices of large prime, and the numerical identity of each choice carries information. Their accumulated uncertainty can be substantial even when their effects occupy few pairs. Appendix F makes this comparison quantitative for ordinary integers, including the dependence among their prime divisibilities.

Nor has the surviving information become small in absolute terms. Parity recovery alone implies

$$I(G_N; Y_N) \geq N \log 2 - o(N) \quad \text{whenever } T_N \rightarrow \infty. \quad (8.6)$$

Indeed, the hidden parities are independent and asymptotically fair. If their mean decoding error is $e_N \rightarrow 0$, their conditional entropy is at most $Nh(e_N) = o(N)$, where $h(t) = -t \log t - (1-t) \log(1-t)$. The data-processing inequality then gives (8.6). In the paradox regime we therefore retain at least order N information, while the clean table contains order $N \log N$. The vanishing *fraction* and successful pair reconstruction fit together without losing this extensive amount of recoverable arithmetic.

9 Returning to the digits without their addresses

We have learned to identify prime divisibility without opening the boxes. Put that knowledge back into the original digit construction. This time a box contains a *digit position*, and its visible mark is the digit at that position. Shuffle the sampled positions out of sight, corrupt the marks, and also corrupt the coprimality answers between positions. The questions stored in the constant can still be read.

The distinction between a box number and a digit position is essential. Box i remains identifiable throughout the experiment; its hidden position X_i does not. We observe which noisy mark belongs to which row of the table. No ordering of the sampled digit positions is given.

Recall what a lane does. If its message bit is zero, every digit in that lane is zero. If its bit is one, the Liouville parity rule makes asymptotically half its digits equal to one. Once the graph has identified the lane, this difference can be read by averaging its marks. The individual numerical positions need never be reconstructed.

For a fixed mask g with $g_1 = 1$, recall

$$x_g(1) = 0, \quad x_g(n) = \frac{1 - \lambda(n)}{2} g_{\ell(n)} \quad (n \geq 2), \quad C_g = \sum_{n \geq 1} x_g(n) 2^{-n}.$$

In addition to (8.1), observe

$$Z_i = x_g(X_i) \oplus B'_i, \quad B'_i \sim \text{Bernoulli}(\eta_N), \quad b_N = 1 - 2\eta_N.$$

All digit flips are mutually independent and independent of the integers and edge flips. For the following finite theorem the digit-noise probability η_N is supplied; the edge-noise probability remains unknown.

The two channels have different jobs. The edge signal a_N lets us sort boxes into prime lanes. The digit signal b_N lets us distinguish an enabled lane from a disabled one after that sorting. Neither channel has to stay a fixed distance from a fair coin.

Theorem 9.1 (Recovering the arithmetic message). If $Na_N^2 \rightarrow \infty$ and $Nb_N^2 \rightarrow \infty$, a single procedure recovers each fixed bit g_j with probability tending to one, while recovering the exact common-prime sets on a fraction of pairs tending to one. It gives an estimator $\hat{C}_N \rightarrow C_g$ in probability. The two signal conditions are jointly necessary for one procedure to have these joint consistency properties for every fixed mask with $g_1 = 1$.

Reading a message bit. To read g_j , select the boxes whose contents are divisible by p_j and by no smaller prime. Corollary 8.2 identifies this set with a vanishing fraction of mistakes. It contains asymptotically $\delta_j N$ boxes, where $\delta_j > 0$ is the familiar lane density. After correcting the known digit noise, average their signs $(1 - 2Z_i)/b_N$. The average tends to $1 - g_j$: zero for an enabled lane, one for a disabled lane. Rounding recovers the message bit.

For example, the lane of 5 is recognized by divisibility by 5 and nondivisibility by 2 and 3. Before noise, an enabled lane has one-frequency $1/2$, whereas a disabled lane has

one-frequency 0. After noise these become $1/2$ and η_N . The signal condition says that the separation remains detectable by pooling the growing lane. For the actual Goldbach message, the bit carried by 5 is already known from $8 = 3 + 5$; the same reading procedure applies to every fixed lane, however far along the prime list it lies.

Recovering each fixed finite collection of message bits recovers each fixed initial block of the constant. The total contribution of binary digits beyond position M is at most 2^{-M} , so the estimates converge to the entire real number. Appendix G gives the full construction and proves necessity of both signals for the joint task in the theorem.

9.1 Knowing the Liouville signs does not solve the game

The digit construction uses the parity of the number of prime factors. It is natural to ask how much that extra arithmetic helps to identify shared primes. Suppose an oracle now reveals every exact Liouville sign. The recovery threshold is unchanged.

Corollary 9.2. Supplying all exact Liouville signs $\lambda(X_i)$ does not lower the common-prime recovery threshold. More generally, supplying all $\Omega(X_i) \bmod r$ for any fixed integer $r \geq 2$ leaves the threshold $Na_N^2 \rightarrow \infty$ unchanged.

Proof. For the first assertion use u and $4u$ as above. For the second use u and $2^r u$, with u odd and $u \leq N^2/2^r$. The paired event has limiting probability $2^{-r} > 0$, the residue of Ω is the same, and presence of the prime 2 differs. The identical testing argument applies. Sufficiency already holds without the extra data. $\square$

The small witness in this proof is worth keeping in view. Multiplying an odd integer by 4 preserves its Liouville sign while adding the prime 2 to its prime support. Multiplying by 2^r does the same while preserving Ω modulo r . These two kinds of parity—how many prime factors there are, and whether 2 occurs at all—encode different information.

The word “joint” in Theorem 9.1 matters. Its necessary edge threshold concerns a task that also reconstructs common-prime sets. At the level of limiting digit frequencies, the balance law already distinguishes the all-one message from a message with a disabled lane. Naming the individual disabled lanes demands more information.

Recovering just the message could require less edge information. Question 15.3 isolates this task while keeping the digit-noise rate supplied to the procedure.

9.2 Goldbach becomes a question of independence

One more question brings us directly back to Goldbach. Imagine looking at a single visible digit mark. Can knowing the entire graph change the probability that this mark is one? In the limiting experiment, Goldbach says precisely that it cannot.

The reason is the lane mechanism. When every lane is enabled, every mark is a fair bit independently of the prime profile. A disabled lane gives its marks a different distribution. The graph can identify that lane, so it can predict something about the marks. A

single Goldbach exception would create statistical dependence between the digits and the relationships among their hidden addresses.

Here is the exact experiment, with fixed noise rates. Let the profiles S_i be as in Section 7, let J_i be their least-prime index, and let V_i be independent fair bits independent of every profile. Give vertex i the clean label $V_i g_{J_i}$ and flip it with a fixed probability $\eta \neq 1/2$. Independently flip graph edges with a fixed $\varepsilon \neq 1/2$. Both rates may be unknown.

This is the limit of every fixed finite ordinary-integer marked experiment. For a fixed prime-divisibility pattern, expand its indicator by inclusion–exclusion. Each Liouville-weighted term is $\lambda(d)L(X/d) = o(X)$ for a fixed integer d . Thus the Liouville sign becomes a fair bit independent of the finite profile. The least-prime tail has probability at most $\prod_{p \leq R} (1 - 1/p) \rightarrow 0$, and the shared-prime tail is controlled as before. Taking the range limit and then the prime cutoff proves the assertion. This does not identify consecutive-digit frequencies.

Proposition 9.3. In this countable limiting model, the noisy marked graph determines the whole mask almost surely. For the Goldbach mask, one observed label is independent of the entire observed graph if and only if Goldbach holds.

Proof. Equations (7.5)–(7.6) and Theorem 7.2 recover every profile. Each lane has positive probability δ_j , so its observed label frequency is almost surely $1/2$ when $g_j = 1$ and η when $g_j = 0$. Since $\eta \neq 1/2$, these exact limiting frequencies recover all bits, even if η is unknown.

Let $\mathcal{G}$ be the sigma field of the observed infinite graph, put $S = \sum_{j: g_j=0} \delta_j$ and $b = 1 - 2\eta$. The recovered lane of a fixed vertex is $\mathcal{G}$ -measurable, and its observed label Z_i satisfies

$$\mathbb{P}(Z_i = 1 \mid \mathcal{G}) = \begin{cases} 1/2, & g_{J_i} = 1, \\ \eta, & g_{J_i} = 0. \end{cases}$$

Its conditional entropy is therefore $\log 2$ in an enabled lane and $h(\eta)$ in a blocked lane. Consequently,

$$I(Z_i; \mathcal{G}) = h\left(\frac{1 - bS}{2}\right) - (1 - S) \log 2 - Sh(\eta). \quad (9.1)$$

The enabled 2-lane makes $1 - S > 0$. Strict concavity of h shows that (9.1) is zero precisely when $S = 0$, because $\eta \neq 1/2$. For the Goldbach mask, $S = 0$ is equivalent to Goldbach. $\square$

When Goldbach holds, the entire label sequence is independent fair bits, independent of the graph, and the label-noise rate is unidentifiable in this limiting model. Fixed-pattern convergence does not transfer that exact unidentifiability statement to growing N -vertex experiments with range N^2 .

The message was repeated throughout its least-prime lanes. Coprimality lets us find those repetitions without their numerical addresses, and averaging lets us read them through noise. The same lane structure therefore appears in three concrete forms: a balance law for the constant, a recoverable message on hidden integers, and an independence criterion for Goldbach in the limiting experiment.

Remark 9.4 (Relation to other reconstruction problems). Feature reconstruction in random intersection graphs and statistical information/recovery comparisons are established subjects. In particular, Göbel, Ruff and Schiller [14] study efficient, robust recovery in a homogeneous growing-feature model. Our features have probabilities $1/p$, are numerically identified, and arise from ordinary sampled integers in the main theorems. The precise target here is recovery of shared primes on a fraction of pairs tending to one, under independent common-rate flips.

10 Choosing integers that do not share primes

Put the integers $2, 3, \dots, X$ on cards. We may keep as many cards as we like, provided no two kept cards share a prime factor. Thus 8 and 9 can sit together; 6 and 10 cannot. The prime 2 is a resource that the second pair tries to spend twice.

Now choose an admissible collection uniformly at random. How many cards does it contain? The cards interact in a complicated way, but the answer is almost the size of a pile of independent fair coin flips. Better still, an independent binomial part can be extracted *exactly*, before taking any limit.

10.1 A small example with a coin inside

For the five cards 2, 3, 4, 5, 6, the admissible collections have generating polynomial

$$1 + 5z + 6z^2 + 2z^3 = (1 + z)(1 + 4z + 2z^2).$$

The coefficient of z^j counts the collections of size j . There are fourteen collections in all, including the empty one. The factor $1 + z$ has an immediate explanation: the card 5 shares no prime with any other card, so it can be added or removed freely. A uniformly chosen collection therefore contains this card according to an independent fair coin.

For a larger interval the free coins need not have fixed names. Choosing a composite card may use up prime cards that would otherwise be available. Nevertheless, a large binomial component survives in the *distribution of the number of cards kept*.

To include later deletion experiments, let B be any set of allowed primes. Write $\mathcal{N}_B(X)$ for the integers in $[2, X]$ whose prime divisors all belong to B , and define

$$P_{B,X}(z) = \sum_{\substack{S \subseteq \mathcal{N}_B(X) \\ S \text{ pairwise coprime}}} z^{|S|}.$$

Here $N_B = \#(B \cap [2, X])$ counts available prime cards, $r_B = \#(B \cap [2, \sqrt{X}])$ counts the small ones, and

$$m_B = \max(0, N_B - 2r_B), \quad d_B = N_B - m_B.$$

Theorem 10.1 (The binomial component). There is a polynomial $Q_{B,X}$ with nonnegative integer coefficients, of degree d_B , such that

$$P_{B,X}(z) = (1+z)^{m_B} Q_{B,X}(z).$$

Give each admissible subset probability proportional to $\lambda^{|S|}$, where $\lambda > 0$, and put $\theta = \lambda/(1+\lambda)$. Its size has the exact distribution

$$|S| \stackrel{d}{=} Z + R, \quad Z \sim \text{Binomial}(m_B, \theta), \quad 0 \leq R \leq d_B, \quad Z \text{ independent of } R. \quad (10.1)$$

Proof. First choose the composite members T of an admissible subset. Let $u(T)$ count the distinct primes occurring in them. Every remaining prime through X can then be included or excluded freely, giving

$$P_{B,X}(z) = \sum_T z^{|T|} (1+z)^{N_B - u(T)}. \quad (10.2)$$

Every composite through X uses a prime at most $\sqrt{X}$ and at most one distinct prime greater than $\sqrt{X}$. Since the members of T have disjoint prime supports,

$$|T| \leq r_B, \quad |T| \leq u(T) \leq r_B + |T| \leq 2r_B.$$

Factor $(1+z)^{m_B}$ out of each summand. The remaining terms have nonnegative coefficients and degree at most $N_B - m_B$. Choosing all available primes shows that $P_{B,X}$ has degree N_B , so the quotient has degree exactly d_B . Finally,

$$\mathbb{E} t^{|S|} = \left(\frac{1+\lambda t}{1+\lambda} \right)^{m_B} \frac{Q_{B,X}(\lambda t)}{Q_{B,X}(\lambda)}$$

is the product of two probability generating functions, which proves the claimed independent representation. $\square$

The proof explains the abundance of coins. Every composite spends a small prime, and different selected composites must spend different small primes. There are too few small primes to entangle most of the available choices.

Taking $\lambda = 1$ makes all admissible collections equally likely. The following stronger approximation says that *every* event concerning the selected size has almost the same probability as in the binomial model. The distance d_{TV} is the largest difference between such event probabilities.

Corollary 10.2. If $N_B(X) \sim \pi(X)$, then, for fixed $\theta \in (0, 1)$,

$$d_{TV}(\mathcal{L}(|S|), \text{Binomial}(N_B(X), \theta)) = O_\theta((\log X)^{-1/2}).$$

The proportion of roots of $P_{B,X}$ that equal -1 , counted with multiplicity, tends to one. The integer $P_{B,X}(1)$ is divisible by 2^{m_B} .

The short smoothing estimate is in Appendix [H.1](#).

For the full prime set, the trial count is $\pi(X)$. A large, highly constrained family of subsets therefore has the size profile of $\pi(X)$ fair coins. The factorization leaves two further fingerprints: almost every polynomial root is exactly -1 , and the total number of admissible collections has a large power of two as a divisor. These are three readings of the same free choices.

10.2 An order that almost ceases to matter

A greedy choice can depend strongly on its order. From the cards 6, 10, 9, taking 6 first leaves a collection of size one; taking 10 and then 9 gives size two. The next experiment has a surprising large-sample answer: nearly all of a maximal collection is forced before we choose an order.

Sample N labelled integers independently and uniformly from $[1, [N^\beta]]$, with fixed $\beta > 1$. Repeated values keep separate labels. Let α_N be the largest pairwise-coprime selection size. Let $\mathcal{U}_N$ be the labels coprime to every other sampled label. Such a label can be added to any admissible collection, so every inclusion-maximal collection must contain it.

The answer involves Buchstab's function ω , which measures the frequency of integers with no small prime factor. For $1 < \beta \leq 2$, $\omega(\beta) = 1/\beta$; its full definition is given in Appendix [H.2](#).

Theorem 10.3 (A compulsory part of every answer). In the preceding random-input model,

$$|\mathcal{U}_N| \sim \alpha_N \sim \omega(\beta) \frac{N}{\log N} \quad \text{in probability.}$$

Uniformly over every pair S, T of inclusion-maximal coprime selections,

$$|S \triangle T| = o_{\mathbb{P}}(N/\log N).$$

Consequently every greedy ordering is asymptotically optimal, even when the order is chosen after the entire sample is seen.

For example, sampling from $[1, N^2]$ gives a largest collection of size asymptotic to $N/(2 \log N)$. Even an adversary who sees the entire sample before choosing the greedy order cannot lose a positive fraction of that answer.

The reason is concrete. Numbers made only from primes above N rarely share any of those primes with another sampled number. They supply the compulsory set $\mathcal{U}_N$. Meanwhile every selected number using a small prime must spend a different small prime. Counting those resources bounds what can be added beyond $\mathcal{U}_N$. The two bounds meet; the estimates are in Appendix [H.2](#).

10.3 A crowd can avoid collisions and leave more holes

Keep the prime-feature model, but reverse the experiment. Choose n independent integers from $[1, X]$ and condition them to be pairwise coprime. We first let $X \rightarrow \infty$ with n fixed, and only then let the crowd grow.

A prime can now belong to at most one crowd member. One might expect this economical use of primes to produce especially thorough coverage. In fact the first missing prime arrives on a much smaller scale.

The effect is visible before any asymptotics. An ordinary crowd is all odd with probability 2^{-n} . A pairwise-coprime crowd is all odd with probability $1/(n+1)$. With three members the probabilities are $1/8$ and $1/4$, respectively.

At each prime p , the limiting probabilities are

$$\mathbb{P}(p \text{ absent}) = \frac{p-1}{n+p-1}, \quad \mathbb{P}(p \text{ belongs to a specified member}) = \frac{1}{n+p-1}.$$

Different primes are independent in this limit. For finitely many primes this follows from the Chinese remainder theorem; the chance of a shared prime above a cutoff z is at most $\binom{n}{2} \sum_{p>z} p^{-2}$, which justifies removing the cutoff.

Proposition 10.4 (The first missing prime). Let L_n be the smallest prime absent from the crowd. In the coprime experiment,

$$\frac{L_n}{\sqrt{n \log n}} \Rightarrow R, \quad \mathbb{P}(R > t) = e^{-t^2} \quad (t \geq 0).$$

For an ordinary unconditioned crowd, $L_n \sim n/\log n$ in probability. More precisely,

$$\frac{n}{L_n} - \log n + 3 \log \log n \Rightarrow G, \quad \mathbb{P}(G \leq t) = e^{-e^{-t}}.$$

Here L_n records the first hole in the crowd's prime coverage. The coprime crowd finds its first hole around $\sqrt{n \log n}$; the ordinary crowd reaches the much larger scale $n/\log n$. The limit laws describe the fluctuations around these scales (Appendix H.3). Avoiding repeated ownership does not make ownership more complete.

10.4 One probability, two experiments

Inspect the crowd already assembled. Does its product contain any repeated prime power? Now perform a different experiment: invite one fresh integer, with no conditioning on the newcomer. Is it coprime to everyone present?

These events have exactly the same probability. The equality joins an internal property of the old crowd to the admission chance of a new member.

Proposition 10.5 (Old powers and a new member). In the same iterated limit,

$$\mathbb{P}(\text{crowd product squarefree}) = \mathbb{P}(\text{a fresh integer is coprime to the crowd}) = \frac{K(n+1)}{K(n)}.$$

For fixed $r \geq 3$, as $n \rightarrow \infty$,

$$\mathbb{P}(\text{crowd product is } r\text{-free}) \longrightarrow \frac{1}{\zeta(r-1)}.$$

In particular the cube-free probability tends to $6/\pi^2$, whereas the squarefree probability tends to zero.

Proof. Given that p is present, its unique owner's valuation has distribution $\mathbb{P}(v_p = j) = (1 - 1/p)p^{-(j-1)}$ for $j \geq 1$. Thus the probability that p^r divides the product is $n/[p^{r-1}(n + p - 1)]$. For $r = 2$, multiplying the complementary probabilities gives $K(n+1)/K(n)$, as does the conditional probability of admitting a new member. For $r \geq 3$, dominated convergence of the Euler product gives the zeta value. The squarefree probability tends to zero since, for any fixed cutoff, its limiting product is at most $\prod_{p \leq z} (1 - 1/p)$, which tends to zero as $z \rightarrow \infty$. $\square$

There is a second surprise in the last limit. As the crowd grows, being squarefree becomes unlikely, but being cube-free retains the positive probability $6/\pi^2$. A crowd can accumulate repeated primes while still usually avoiding high repetitions.

We can also keep a record of precisely what was repeated. For crowd product M , put

$$D_n = \frac{M}{\text{rad}(M)},$$

where $\text{rad}(M)$ keeps one copy of each prime. For example, the pairwise-coprime crowd 72, 25, 7 has $M = 2^3 3^2 5^2 7$ and $D_n = 2^2 \cdot 3 \cdot 5 = 60$. The exact limiting law is

$$\mathbb{P}(D_n = d) = \frac{K(n+1)}{K(n)} \frac{1}{d} \prod_{p|d} \frac{n}{n+p}. \quad (10.3)$$

Thus the excess powers have a probability law of their own, although the integers in the original experiment grow without bound.

On a logarithmic scale this remainder meets the classical Dickman law.

Corollary 10.6 (A small remainder). Let Y have Laplace transform

$$\mathbb{E}e^{-sY} = \exp\left(\int_0^1 \frac{e^{-st} - 1}{t} dt\right).$$

Then $\log D_n / \log n \implies Y$, and $\mathbb{P}(D_n \leq n) \rightarrow e^{-\gamma}$.

The local valuation calculation and the limit proof are in Appendix [H.3](#).

In particular, the chance that all repeated material fits below the crowd size tends to $e^{-\gamma} \approx 0.56146$:

$$\mathbb{P}(D_n \leq n) \longrightarrow 0.56146 \dots$$

The crowd size, rather than the much larger sampled integers, sets the natural scale for its repeated prime material.

11 What an average can hide

There are two ways to average a spectrum. We can find the eigenvalues in every experiment and average their histograms. Or we can average the characteristic polynomials first and then find the roots. These operations can tell different stories.

A two-dimensional example makes the distinction visible. With equal probability take

$$A_1 = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

The averaged eigenvalue histogram puts mass $1/4, 1/2, 1/4$ at $0, 2, 4$. But the average characteristic polynomial is

$$\frac{1}{2}z(z-2) + \frac{1}{2}(z-2)(z-4) = (z-2)^2.$$

Its two roots agree. Averaging the polynomial has hidden the spread.

Our coprimality probabilities produce an exact, large-dimensional version of this distinction, tied to the prime–composite factorization from the earlier chapters.

11.1 A polynomial made from chances of coprimality

Recall that $K(k)$ is the limiting probability that k random integers are pairwise coprime. Form

$$J_N(z) = \sum_{k=0}^N \binom{N}{k} K(k) z^k.$$

If we sample N labelled integers and count their admissible subcollections by size, J_N is the limiting *expected* counting polynomial. The factor $\binom{N}{k}$ chooses the labels, and $K(k)$ is the chance that those labels are compatible.

This is a different object from $P_{B,X}$, which counts choices in a fixed interval of integers. All roots of J_N are negative and real; write them as $-R_{N,1}, \dots, -R_{N,N}$. The prime-by-prime real-root argument is given in Appendix H.4.

Let W_c be the composite product from the exponential factorization, so $\mathbb{E}W_c^s = \Gamma(s+1)K(s)$ for $s > -1$. To build the matrix experiment, favor larger values of W_c by multiplying the probability weight of a value c by c^N and renormalizing. This gives the scalar $W_{c,N}$ below.

Theorem 11.1 (Roots of the average versus typical eigenvalues). Let Z_N have independent standard complex Gaussian entries ($\mathbb{E}|Z_{ij}|^2 = 1$), and independently let $W_{c,N}$ have distribution $c^N d\mathbb{P}_{W_c}(c)/(N!K(N))$. Put $V_N = Z_N Z_N^* / W_{c,N}$. Then

$$\mathbb{E} \det(zI - V_N) = (-1)^N \frac{J_N(-z)}{K(N)} = \prod_{j=1}^N (z - R_{N,j}).$$

With $a_N = e^\gamma \log N$,

$$\frac{1}{N} \sum_j \delta_{R_{N,j}/a_N} \implies \delta_1, \quad \frac{1}{N} \sum_j \delta_{\lambda_j(V_N)/a_N} \implies \text{MP}_1 \quad \text{in probability,}$$

where MP_1 has density $(2\pi)^{-1}\sqrt{(4-x)/x}$ on $(0, 4)$.

Read the first limit as follows: after dividing by $e^\gamma \log N$, almost all roots of the averaged polynomial gather near 1. The second limit says that a typical matrix has a whole spread of eigenvalues across $(0, 4)$ on that very same scale.

The distribution of $W_{c,N}$ uses the composite factor already paired with our prime product, and the identity follows from its moments. Expanding Gaussian minors gives factorials; the identity $\mathbb{E}W_c^k = k!K(k)$ supplies precisely the arithmetic coefficients. Appendix H.4 gives the calculation and both limits.

11.2 The hidden spread is a triangular spectrum

The roots of J_N have just collapsed to one point after division by $e^\gamma \log N$. That does not mean they have become equal. It means we have chosen a scale on which their interesting differences are hard to see.

A finer view reveals a familiar shape from an apparently distant experiment. Fill the region above a matrix diagonal with independent complex Gaussian entries of variance $1/N$, and put zeros elsewhere:

$$T_N = \begin{pmatrix} 0 & * & * & \cdots & * \\ 0 & 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & * \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

All eigenvalues of this triangular matrix are zero. Its singular values still measure how strongly it stretches vectors. Their squares are the eigenvalues of $T_N^* T_N$; these have a nontrivial limiting histogram.

That histogram is the Dykema–Haagerup law DH_1 , supported on $[0, e]$ [10, 7]. Its moments are

$$\int y^k dDH_1(y) = \frac{k^k}{(k+1)!} \quad (k = 1, 2, \dots),$$

so its mean is $1/2$. No triangular matrices occur in the definition of J_N . Nevertheless its roots reproduce this same law.

Theorem 11.2 (The triangular spectrum). Write $-R_{N,1}, \dots, -R_{N,N}$ for the roots of J_N , with multiplicity. Center their positive magnitudes by setting

$$t_{N,j} = e^{-\gamma} R_{N,j} - \log N - \log \log N.$$

Then

$$\boxed{\frac{1}{N} \sum_{j=1}^N \delta_{\exp(t_{N,j})} \Rightarrow DH_1.} \quad (11.1)$$

Equivalently, if T is the logarithm of a random variable with law DH_1 , then

$$\frac{1}{N} \sum_{j=1}^N \delta_{t_{N,j}} \implies \mathcal{L}(T). \quad (11.2)$$

The law of T has support $(-\infty, 1]$; its explicit density is given in (I.10).

Here is a practical reading of the theorem. List the positive magnitudes of the arithmetic roots. Multiply them by $e^{-\gamma}$, subtract $\log N + \log \log N$, and make a histogram. The result approaches a fixed asymmetric shape. Exponentiate the centered entries, and the shape becomes the squared-singular-value histogram of triangular Gaussian noise.

The common centering matters. Dividing by $\log N$ erases fluctuations on a constant scale; subtracting both $\log N$ and $\log \log N$ exposes them. The extra $\log \log N$ is part of the arithmetic calculation.

Where the bridge comes from. The proof has three steps, all supplied in Appendix I. First form the entire function

$$\mathcal{E}(z) = \sum_{k \geq 0} K(k) \frac{z^k}{k!}.$$

Its zeros are negative. If their magnitudes are ρ_j , the prime number theorem leads to the counting law

$$\#\{j : \rho_j \leq r\} \sim e^{-\gamma} \frac{r}{\log^2 r}.$$

Second, Campbell and Jalowy's transfer theorem [5, Corollary 2.3] turns this growth of the entire function's zeros into a limiting law for the polynomial roots. Finally, the precise arithmetic centering identifies that law as the logarithm of DH_1 , using the identification of Arizmendi and Hasebe [1, Proposition 4.4].

The connection therefore has a definite route: coprimality probabilities determine an entire function; its supply of zeros determines a free-stable law; exponentiating that law gives the triangular spectrum. The external transfer theorem supplies the bridge. The prime estimates determine which law and which scale cross it.

The histogram alone does not locate the largest arithmetic root. Question 15.5 asks whether it also reaches the edge of the triangular spectrum.

12 A bell curve inside a rare factorization

So far a prime factor has been a resource: present or absent, available or spent. Its size gives a different way to read the same integer.

Compare $72 = 8 \cdot 9$ with $2^{100} \cdot 3$. Both have two distinct prime factors. In the first, the two prime-power parts contribute almost equally to the logarithm. In the second, the factor 3 contributes very little. Counting both examples as *two factors* misses that difference.

For $n = \prod p^{a_p}$, give its prime-power parts logarithmic shares

$$w_p(n) = \frac{a_p \log p}{\log n}.$$

These shares add to one. Their entropy and effective factor count are

$$H(n) = - \sum_{p|n} w_p(n) \log w_p(n), \quad D(n) = e^{H(n)}.$$

At $n = 1$ we set $H(1) = 0$ and $D(1) = 1$. If k shares are equal, then $D(n) = k$. Unequal shares reduce the effective count:

$$D(72) \approx 1.99924, \quad D(2^{100} \cdot 3) \approx 1.08371.$$

Every prime power has effective count one, however large its exponent.

12.1 Infinitely many small factors, about three effective ones

Choose an integer uniformly from $[1, N]$ and let N grow. The expected number of distinct prime factors tends to infinity: divisibility counting and Mertens' theorem give

$$\mathbb{E} \#\{p : p \mid U_N\} = \sum_{p \leq N} \frac{\lfloor N/p \rfloor}{N} \sim \log \log N.$$

Yet its mean effective factor count approaches a finite number,

$$\boxed{\mathbb{E} D(U_N) \longrightarrow \mathfrak{P} = 2.9621 \dots}$$

An integer gains more and more prime factors without gaining an unbounded number of equally important logarithmic contributions.

The limiting shares have the classical Poisson–Dirichlet law $\text{PD}(0, 1)$. One way to make them is to break off a uniform fraction of a stick, then a uniform fraction of the remainder, and continue. Sort the resulting pieces into decreasing order. The same law describes the limiting normalized cycle lengths of a uniform random permutation [16, remarks after Theorem 1, part (d)]. This is a classical prime-factor and permutation connection, used here to ask about the shapes of unusually balanced factorizations.

Write $P = (P_i)$ for this random partition and put

$$H = - \sum_i P_i \log P_i, \quad D = e^H.$$

Its mean entropy is exactly 1, so $e^{\mathbb{E}H} = e$. Averaging the effective count instead gives $\mathbb{E}e^H = \mathfrak{P}$. The two numbers differ because exponentiating and averaging do not commute.

These limiting statements also respect our choice to group repeated copies of a prime into one prime-power part. Lemma H.1 makes that step quantitative. Appendix H.5 proves it and gives a positive recursion for $\mathfrak{P}$ with an explicit truncation error. Thus the displayed constant is computationally accessible as well as probabilistically defined.

12.2 What does an unusually large effective count cost?

The event $D > x$ asks for the equivalent of more than x substantial, well-balanced factors. It is much rarer than merely having more than x distinct prime divisors. Sawin [35, Lemma 9.5] bounded this same limiting entropy tail by a constant times $x^{9/2}e^{2x}x^{-x}$. Our expansion identifies its next logarithmic terms.

Theorem 12.1 (A sharp entropy tail). As $x \rightarrow \infty$,

$$-\log \mathbb{P}(D > x) = x \left(\log x + \frac{1}{2} \log \log x - 1 - \frac{1}{2} \log(2\pi) + o(1) \right).$$

Equivalently,

$$\mathbb{P}(D > x) = \left(\frac{e\sqrt{2\pi}}{x\sqrt{\log x}} \right)^x e^{o(x)}.$$

The main rarity cost is x^{-x} . Balancing the factors adds the further factor $(\log x)^{-x/2}$. The appearance of $\sqrt{2\pi}$ hints at the geometry behind this balancing penalty.

Here is the idea of the proof. To have effective count just above x , a partition can use about x pieces of size close to $1/x$. Near equal shares, the entropy loss is approximately a quadratic sum of their deviations. Counting a thin region around this balanced configuration gives a Gaussian volume. A matching exponential bound shows that configurations elsewhere cannot improve the leading cost. The full upper and lower estimates are in Appendix H.6.

12.3 The rare factors arrange themselves into a Gaussian

The tail estimate says how unlikely balance is. The next theorem says what balance usually looks like once we have insisted on it.

Take a factor share P_i , compare it with $1/x$, and magnify the relative difference by $\sqrt{\log x}$:

$$y_i = \sqrt{\log x} (xP_i - 1).$$

Give the point y_i weight P_i . The resulting histogram has total mass one. Conditioned on $D > x$, this histogram approaches the familiar bell curve.

Theorem 12.2 (A Gaussian profile). Given $D > x$, the random probability measure

$$\nu_x = \sum_i P_i \delta_{\sqrt{\log x}(xP_i - 1)}$$

converges weakly in probability to the standard Gaussian law. For each bounded interval (a, b) ,

$$\frac{1}{x} \#\{i : a < \sqrt{\log x}(xP_i - 1) < b\} \longrightarrow \Phi_G(b) - \Phi_G(a)$$

in the same conditional probability, where Φ_G is the standard Gaussian distribution function.

For example, asymptotically about 68.3% of the logarithmic mass lies in shares between

$$\frac{1 - 1/\sqrt{\log x}}{x} \quad \text{and} \quad \frac{1 + 1/\sqrt{\log x}}{x}.$$

The number of shares in this window, divided by x , approaches the same Gaussian proportion. The bell curve describes fluctuations *inside one rare factorization*, rather than fluctuations between independently sampled integers.

This conclusion is stronger than the symmetric construction used for the tail lower bound. That construction showed one economical way to be unusually balanced. The conditional theorem says that the Gaussian shape is what the whole rare event typically selects. Its proof, including exponential control of deviations, is in Appendix H.7.

12.4 How far can we return to ordinary integers?

For an actual integer n , a share near $1/x$ means that its corresponding prime-power part is near $n^{1/x}$ on a logarithmic scale. The limiting picture therefore suggests many comparably important prime-power parts, with the preceding Gaussian pattern in their logarithmic sizes.

A coupling with ordinary integers lets the threshold x grow with the sampling range. The following is the range justified here.

Corollary 12.3. Fix $\epsilon > 0$. If $x = x(N) \rightarrow \infty$ and

$$x \leq (1 - \epsilon) \frac{\log \log N}{\log \log \log N},$$

then the tail formula of Theorem 12.1 holds with $D(U_N)$ in place of D . Given $D(U_N) > x$, the measure

$$\sum_{p|U_N} w_p(U_N) \delta_{\sqrt{\log x}(xw_p(U_N)-1)}$$

converges weakly in probability to the standard Gaussian law.

The transfer proof is in Appendix H.7. This range is part of the assertion: the limiting partition theorem alone would not justify arbitrarily rare events among finite integers.

13 Return to the constant: delete seven

The information paradox concerned what survives a change of observation. For the finale, change the arithmetic message itself. Remove the prime 7, and ask every other prime to justify its place.

The rule comes straight from the original constant. Prime p_i carries the answer to the Goldbach question for $2(i+1)$. It may remain only if that even number is still the sum of two remaining primes. Apply this test simultaneously, delete the failures, and repeat. A deleted prime never returns.

The first two rounds can be done by hand. The prime $11 = p_5$ is responsible for 12. Its only prime certificate is $12 = 5 + 7$, so it falls in round one. Then $13 = p_6$ is responsible for 14. Both its certificates, $3 + 11$ and $7 + 7$, have failed, so it falls in round two. The cascade has begun.

In symbols the surviving prime sets are

$$B_0 = \mathcal{P} \setminus \{7\}, \quad B_{k+1} = \{p_i \in B_k : 2(i+1) = q + r \text{ for some } q, r \in B_k\}. \quad (13.1)$$

The same prime may be used twice in a certificate, as in $14 = 7 + 7$.

Theorem 13.1 (Abundance at every finite depth). For every fixed k and every $A > 0$,

$$\#\{p \leq X : p \notin B_k\} \ll_{k,A} \frac{X}{(\log X)^A}.$$

Nevertheless

$$\bigcap_{k \geq 0} B_k = \{2, 3, 5\}.$$

For every fixed k , asymptotically all ordered representations of a large odd integer as three primes use primes in B_k . The same conclusions hold if B_0 is the original Goldbach-enabled prime set with 7 deleted.

There are two clocks in this statement. Fix the number of rounds and look far enough along the prime sequence: almost all primes are still present. Fix any prime above 5 and keep applying the rule: that prime eventually disappears. Neither assertion cancels the other.

13.1 Why the collapse is inevitable

The permanent survivors carry their own certificates:

$$2 : 4 = 2 + 2, \quad 3 : 6 = 3 + 3, \quad 5 : 8 = 3 + 5.$$

For every later prime p_i with $i \geq 5$,

$$q + r = 2(i+1) \implies q, r < p_i.$$

Indeed the summands are odd and at most $2i - 1$, whereas $p_i \geq 2i + 1$. Every certificate therefore points backward to earlier primes.

Once 7 has gone, this backward dependence forces the rest of the collapse. Suppose all earlier primes except 2, 3, 5 have disappeared. Those three can make sums no larger than 10, so they cannot certify the target $2(i+1) \geq 12$. The next prime must disappear as well. Induction proves that only 2, 3, 5 survive forever.

The abundance assertion needs a different idea. Almost every sufficiently large even target has many Goldbach representations. Removing a sufficiently sparse set of primes cannot destroy all those representations. This remains true at each fixed number of rounds.

Appendix H.8 supplies the robust almost-all Goldbach estimate and the induction; the constants are allowed to depend on the depth.

The backward dependence also makes a finite calculation exact. Primes beyond the displayed window cannot rescue a prime inside it. Through 127, the deletion rounds are

0	7
1	11
2	13
3	17, 19, 43
4	23, 29, 31, 61, 83
5	37, 41, 47, 53, 59, 101, 109
6	67, 71, 73, 79, 89, 97, 103, 107, 113, 127.

The cascade does not simply march from left to right. For example, 43 falls before 23.

13.2 The cascade has a clock of its own

How long can a prime survive? Let $\tau(p)$ be the first round in which it is absent, with $\tau(7) = 0$ and $\tau(2) = \tau(3) = \tau(5) = \infty$. Let $M(x)$ be the latest deletion time among the other primes through x .

A second elementary consequence gives a surprisingly explicit answer.

Proposition 13.2 (The last deletion in a finite window).

$$M(x) := \max_{5 < p \leq x} \tau(p) \sim \frac{\log x}{\log \log x}.$$

Proof. Every certificate for a target at least 12 uses a prime above 5. All its primes are at most $2(\pi(p) + 1)$. Consequently

$$M(x) \leq 1 + M(F(x)), \quad F(x) = 2(\pi(x) + 1),$$

for large x . The prime number theorem gives $F(x) \sim 2x/\log x$. If $h(x) = \log x / \log \log x$, direct subtraction gives $h(x) - h(F(x)) \rightarrow 1$. For any $\epsilon > 0$, each sufficiently late backward step therefore spends at least $1 - \epsilon$ units of this clock. Iterating to a fixed cutoff proves $\limsup M(x)/h(x) \leq 1$.

For the reverse inequality, the certificate $2q = q + q$ guarantees

$$\tau(p_{q-1}) \geq \tau(q) + 1 \quad (q \geq 7 \text{ prime}).$$

Start at $q_0 = 7$ and set $q_{r+1} = p_{q_r-1}$. Then $\tau(q_r) \geq r$, while the prime number theorem gives $q_{r+1} \sim q_r \log q_r$ and $h(q_{r+1}) - h(q_r) \rightarrow 1$. Summing these increments gives $h(q_r) \sim r$. For $q_r \leq x < q_{r+1}$, we have $M(x) \geq r$ and $h(x) \sim r$. This proves the matching lower bound. $\square$

The lower bound has an explicit chain of witnesses:

$$7 \mapsto 13 \mapsto 37 \mapsto 151 \mapsto 863 \mapsto 6689 \mapsto 67139 \mapsto 843377 \mapsto \dots$$

Every arrow buys at least one more round by using the preceding prime twice. The upper bound says that no strategy buys asymptotically more time than this clock allows.

An exact computation through 10^6 gives

x	10^2	10^3	10^4	10^5	10^6
$M(x)$	6	8	10	12	13.

The leading asymptotic is not yet numerically accurate at these scales. The calculation also shows a concentration worth investigating: among the 78 495 primes above 5 through 10^6 , about 94.1% disappear in rounds 11, 12 or 13.

This concentration motivates Conjecture 15.1: nearly the whole population may follow the last survivor's clock. The experiment and its limitations are revisited with the conjecture in Section 15.

13.3 Three different readings of the three survivors

The surviving primes can now be returned to each of our constructions. They give a digit frequency, a selection law and a graph rank. These are exact endpoints of the same deletion process.

First, restore the surviving lanes to the number:

$$x_n^{(k)} = \frac{1 - \lambda(n)}{2} \mathbf{1}_{\{\text{lpf}(n) \in B_k\}}, \quad X_k = \sum_{n \geq 2} x_n^{(k)} 2^{-n}.$$

The opening lane calculation gives the frequency of ones

$$\rho_k = \frac{1}{2} \sum_{p_i \in B_k} \frac{1}{p_i} \prod_{j < i} (1 - 1/p_j).$$

Numbers with least prime factor 2, 3 or 5 have densities $1/2$, $1/6$ and $1/15$. Half the digits in each such lane are ones. Therefore

$$\boxed{\rho_k \downarrow \frac{1}{2} \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{15} \right) = \frac{11}{30}.}$$

The positive lane masses sum to one, so dominated convergence justifies passage to the surviving lanes. The decrease is strict: equality at a stage would mean no further prime had disappeared, hence permanent stabilization, contrary to finite-depth abundance and the three-prime intersection.

The numbers X_k themselves decrease to the three-lane constant. That endpoint is still irrational. Its 2-lane gives $x_{2m} + x_{4m} = 1$. An eventual period T would make these two digits equal for every large multiple m of T , a contradiction. A finite list of surviving primes has produced a nonperiodic infinite digit sequence.

Next choose a uniformly random pairwise-coprime collection of the surviving smooth numbers in $[2, X]$. At every fixed finite depth, its size is close in total variation to

$$\text{Binomial}(N_k(X), 1/2), \quad N_k(X) = \#(B_k \cap [2, X]).$$

This is Theorem 10.1, with error $O_k((\log X)^{-1/2})$. Although $N_k(X) \sim \pi(X)$, its exact value must be retained: an error negligible compared with $\pi(X)$ need not be negligible compared with the binomial fluctuations.

Take infinite deletion first, however, and at most three cards can be kept. The earlier cloud of coin flips is replaced by a three-point law.

Proposition 13.3 (The last selection law). Take infinite deletion first, then choose a uniform pairwise-coprime subset of the $\{2, 3, 5\}$ -smooth integers in $[2, X]$. As $X \rightarrow \infty$, its size H_X satisfies

$$(\mathbb{P}(H_X = 1), \mathbb{P}(H_X = 2), \mathbb{P}(H_X = 3)) \longrightarrow (1, 9, 6)/16.$$

Proof. Put $L = \log X$. For any nonempty $S \subseteq \{2, 3, 5\}$, the number of integers with exact prime support S is

$$\frac{L^{|S|}}{|S|! \prod_{p \in S} \log p} + O_S(L^{|S|-1}).$$

This follows by counting positive integer lattice points in $\sum_{p \in S} a_p \log p \leq L$: bounding unit boxes above and below changes the simplex boundary by a fixed distance and contributes $O(L^{|S|-1})$.

Distinct members of a coprime selection have disjoint supports. Terms of order L^3 must use all three primes, whose supports therefore form a set partition. After removing the common factor $L^3/(\log 2 \log 3 \log 5)$, the coefficients are $1/3!$ for one block, $3/(1!2!)$ for two blocks, and 1 for three blocks. These unordered support partitions already count each selection once. Their sum is $16/6$, giving the three probabilities. Selections omitting a prime have lower order, as does the empty set. $\square$

The weights are easy to remember through the three possible ways to group the available prime resources:

$$\{2, 3, 5\}; \quad \{2\} \mid \{3, 5\} \text{ and its two companions}; \quad \{2\} \mid \{3\} \mid \{5\}.$$

A card may use all three primes, two cards may split them into two blocks, or three cards may use one each. Counting the possible powers within each block produces the weights $1 : 9 : 6$.

The integer 1 was excluded throughout. Including it would add one independent fair coin, since it is coprime to everything. For any fixed set of b allowed primes, the same argument gives a limiting selection law depending only on b : the individual prime logarithms cancel when probabilities are normalized.

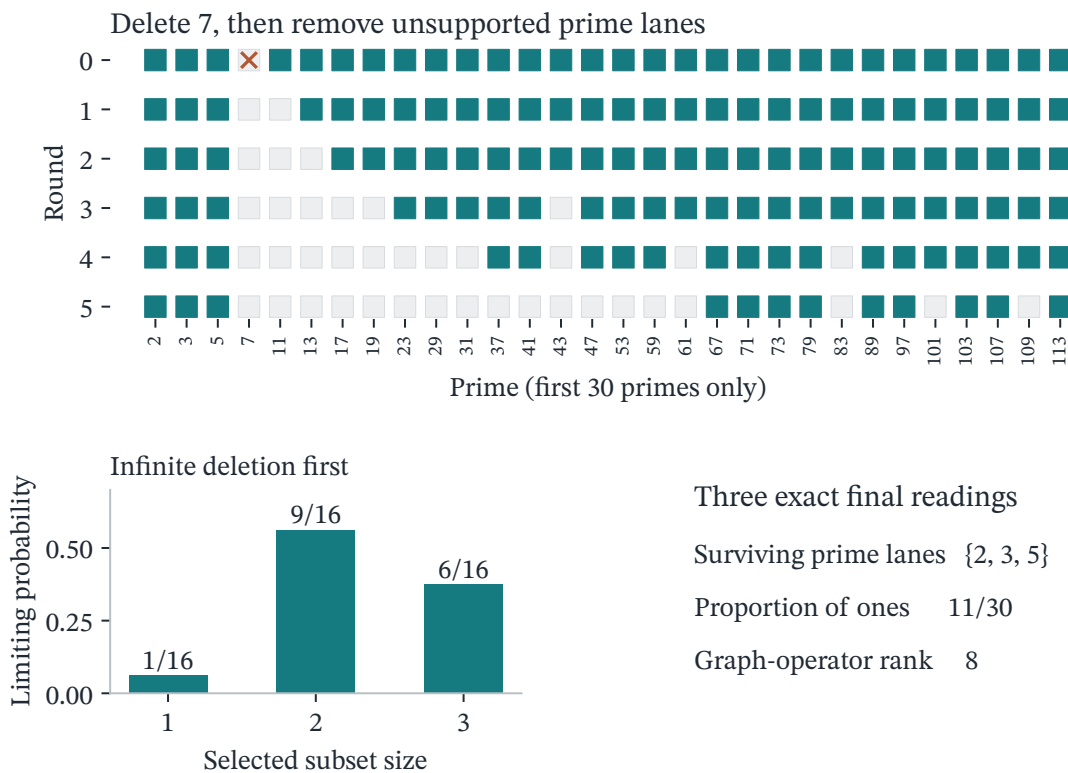


Figure 5: The upper panel follows the exact permanent-deletion rule for the first thirty primes; it is not evidence for the infinite density assertion. The lower bars give the limiting sizes of a uniformly chosen coprime subset of the $\{2, 3, 5\}$ -smooth integers in $[2, X]$, as $X \rightarrow \infty$ after infinite deletion. The integer 1 is excluded.

13.4 An infinite spectrum becomes eight numbers

For the last reading, give every graph vertex an independent prime profile $\omega = (\omega_p)_{p \in \mathcal{P}}$: it carries p , meaning $\omega_p = 1$, with probability $1/p$. Join two vertices when they share no prime in the retained set B . The graph kernel and its integral operator are

$$G_B(\omega, \omega') = \prod_{p \in B} (1 - \omega_p \omega'_p), \quad (T_B u)(\omega) = \int G_B(\omega, \omega') u(\omega') d\mathbb{P}(\omega').$$

This operator records the graph's spectral patterns. It is self-adjoint and Hilbert–Schmidt, and it has eigenvalues of both signs. It is distinct from the positive mask operator A_g used in the RH criterion.

With only 2, 3, 5 retained, a vertex has eight possible relevant profiles:

$$(\omega_2, \omega_3, \omega_5) \in \{0, 1\}^3.$$

Nothing else in its prime profile can affect an edge. The final graph therefore has a finite matrix description with eight types. Each of the three local two-state matrices is invertible, so all eight dimensions really occur.

Every finite round has a very different spectrum. Let $N_{k,+}(u)$ and $N_{k,-}(u)$ count its positive and negative eigenvalues of magnitude at least u . Lemma H.3 proves

$$N_{k,+}(u) \sim N_{k,-}(u) \sim \frac{A_k}{2u} \quad (u \downarrow 0).$$

Here $A_k > 0$ measures the supply of small spectral values. Write $K_B(4)$ for the probability that four independent profiles form a clique when the retained restrictions are B .

Theorem 13.4 (Infinite spectra at every finite depth). Let $T_k = T_{B_k}$. At each fixed finite depth it has infinitely many positive and negative eigenvalues. Their counting coefficient A_k from Lemma H.3 decreases to zero. Nevertheless T_k converges in Hilbert–Schmidt norm to an operator with exactly eight nonzero eigenvalues. At the same time

$$K_{B_k}(4) \uparrow K_{\{2,3,5\}}(4) = \frac{512}{3375}.$$

Removing prime restrictions makes such a clique more likely, so its probability rises to 512/3375. At the same time A_k tends to zero and the infinite spectral tail disappears. A fixed four-vertex experiment and the full spectrum see different aspects of the same changing graph.

The exact tensor spectrum, the small-eigenvalue estimate, and the norm convergence are proved in Appendix H.9. The full-prime tensor formula is due to Zagier [28, Appendix C]; here it is applied along the deletion process.

13.5 The initial edit can be arbitrarily late

Deleting 7 made the cascade easy to watch. One can also begin arbitrarily far out in the original constant.

Let $S = \{p_i : 2(i+1) \text{ has a Goldbach representation}\}$. Choose $J \geq 4$, put $r = p_J$, and initially delete the finite block

$$\{p_i : J < i < r\}.$$

The same robust argument leaves density one at each finite round. For $i \geq r$, the target $2(i+1) > 2r$ cannot be made from the first J primes. Backward dependence therefore leaves an eventual set contained in $\{p_1, \dots, p_J\}$. For example, $J = 4$ deletes 11 and 13 and leaves exactly 2, 3, 5, 7.

The edited lanes begin at position p_{J+1} . Consequently the initial real number changes by at most $2^{1-p_{J+1}}$. An arbitrarily small change to the starting number can lead, under this explicit repeated rule, to a finite eventual prime supply.

We began by hiding arithmetic answers in prime lanes. We have now read those lanes as digits, choices and spectra. After deleting 7 and waiting forever, the three readings are

$$\frac{11}{30}, \quad \frac{1}{16}(1, 9, 6), \quad 8.$$

At every finite round, almost every sufficiently large prime remains. The endpoint becomes visible only when we change the order of the two limits.

14 Random digits, complex heights and echoes

The same constructions behave differently under three familiar tests. Binary normality asks for every finite word; complex value distribution asks which heights a curve reaches; Fourier decay asks whether a measure forgets high frequencies. The next observations connect those tests to arithmetic already built into our objects.

14.1 What would genuinely random digits mean?

The original constant began by repeating Goldbach answers. If every answer is yes, its digits become $(1 - \lambda(n))/2$. It is tempting to call these digits random. Ordinary normality gives that word a precise meaning: every binary word of length k must occur with frequency 2^{-k} , for every k .

The relevant classical assertion is the ordinary, naturally averaged Chowla conjecture for the Liouville function. In its sign-pattern form it predicts precisely these frequencies; see [25, Conjecture 1.3]. Its correlation form says that for every nonempty finite set J of distinct nonnegative integers,

$$\frac{1}{N} \sum_{n \leq N} \prod_{j \in J} \lambda(n+j) \longrightarrow 0. \quad (14.1)$$

In particular, this is a fixed-step assertion. Averaging over the step in the sampling theorem does not establish it.

Proposition 14.1. The original Goldbach constant is normal in base two if and only if Goldbach's conjecture and the Liouville Chowla conjecture (14.1) both hold.

Proof. Normality implies simple normality, which already implies Goldbach by Theorem 1.2. Under Goldbach the digit at n is $(1 - \lambda(n))/2$. For a word $w = (w_0, \dots, w_{k-1})$, its indicator at n is

$$2^{-k} \prod_{j=0}^{k-1} (1 + (-1)^{w_j} \lambda(n + j)).$$

Expanding this product shows that (14.1) gives the frequency 2^{-k} . Conversely, uniform frequencies of all words give zero average to every nonconstant product of their signs. Every finite J lies inside a consecutive block, so this recovers (14.1). $\square$

This equivalence helps locate the difficulty. Normality of this one real number simultaneously asks that no Goldbach answer be missing and that every finite pattern of Liouville signs have its expected frequency. It is a meeting point of two existing conjectures, rather than a newly available shortcut to either one. The primitive sampling theorem supplies a useful contrast: its fair-bit law is unconditional because it averages over an additional variable.

14.2 Every complex height returns

On the real line our prime density is a positive decreasing curve. Away from that line, its entire continuation behaves very differently. Put

$$m = f(0) = \frac{1}{2\mathfrak{C}_2}.$$

The reflection law $f(z) + f(-z) = 2m$ has a surprisingly strong consequence when combined with Picard's theorem.

Proposition 14.2. For every $a \in \mathbb{C}$ other than possibly m , the equation $f(z) = a$ has infinitely many solutions. In particular f has infinitely many complex zeros, but none on the real or imaginary axis.

Proof. The nonzero coefficients at every prime position show that f is transcendental entire. Picard's theorem says that such a function can have at most one complex value with only finitely many preimages [31, Chapter II, Section 9]. Reflection pairs the solutions of $f(z) = a$ with those of $f(z) = 2m - a$. If a had only finitely many preimages, so would $2m - a$; hence these values must coincide, giving $a = m$.

The real-axis limits and monotonicity in Corollary 5.4 imply $0 \leq f(x) \leq 2m$ there. Equality at either endpoint would make f constant on a real half-line, hence constant everywhere by the identity theorem. Thus $0 < f(x) < 2m$ for every real x . Finally the Taylor coefficients are real, so $f(-iy) = \overline{f(iy)}$. Reflection therefore gives $\Re f(iy) = m > 0$, excluding imaginary-axis zeros as well. $\square$

The conclusion uses a classical theorem, but it sharpens the picture of this particular density: a curve positive everywhere on the real line has infinitely many zeros off both coordinate axes. Exactly one height escapes the argument, its central height m .

Reflection alone cannot settle the middle height: an odd entire function can have only one zero. We initially approached it as a question about how the prime gaps constrain an entire function. But a simpler feature of the primes settles it, and more: the only prime divisible by 3 is 3. Among the exponents $p - 2$, therefore, only the exponent 1 is congruent to 1 modulo 3. With $\omega = e^{2\pi i/3}$, rotating the argument through three equally spaced angles and taking a weighted sum gives

$$f(z) + \omega^2 f(\omega z) + \omega f(\omega^2 z) = \frac{3z}{K'(-2)}.$$

Every term except the linear one cancels. This finite identity contains information that the size of the prime gaps alone misses. Using larger primes in the same way gives the following conclusion.

Theorem 14.3 (Every polynomial target is met infinitely often). For every polynomial $P \in \mathbb{C}[z]$, the equation

$$f(z) = P(z)$$

has infinitely many distinct solutions in $\mathbb{C}$. In particular every complex value, including the middle height $m = f(0)$, is attained infinitely often.

Proof. Choose an odd prime $q > \deg P + 2$; when $P = 0$, any odd prime will do. Put $\zeta = e^{2\pi i/q}$ and $h = f - P$. The root-of-unity filter gives

$$\sum_{j=0}^{q-1} \zeta^{-j(q-2)} h(\zeta^j z) = \frac{qz^{q-2}}{K'(1-q)}. \quad (14.2)$$

Indeed, the weighted sum kills a monomial z^n unless $n \equiv q - 2 \pmod{q}$. It kills all of P , while a surviving exponent $p - 2$ of f would require $q \mid p$, hence $p = q$. The coefficient on the right is nonzero because the zero of K at $1 - q$ is simple.

Suppose h had finitely many zeros. Factoring them out, with multiplicity, gives $h = De^H$, where D is a nonzero polynomial and H is entire: the remaining zero-free entire function has an entire logarithm. The function H is nonconstant, since h is transcendental. Substitution into (14.2) gives a polynomial-coefficient relation among

$$e^{H(z)}, e^{H(\zeta z)}, \dots, e^{H(\zeta^{q-1}z)}, e^0.$$

Narasimhan's theorem says that exponentials of entire functions whose pairwise differences are nonconstant are linearly independent over $\mathbb{C}[z]$ [29, Theorem 1]. To apply it, group the rotated exponents whenever their difference is constant, absorbing the corresponding constant exponential factors into their polynomial coefficients. Discard any group whose combined coefficient vanishes. The exponent 0 remains in its own group, since every $H(\zeta^j z)$ is nonconstant. Its coefficient is $-qz^{q-2}/K'(1-q)$, which is nonzero. The resulting relation contradicts Narasimhan's theorem. $\square$

Thus the bounded real curve does more than revisit every complex height. It meets every prescribed polynomial target infinitely often. The proof uses a classical independence theorem; its arithmetic input is the elementary fact that the only prime divisible by a prime q is q itself. No estimate for the distribution of primes is needed in this step.

There is a small geometric consequence of the same cancellation. Every complex height occurs infinitely often, yet it cannot occur at all three vertices of an equilateral triangle centred at the origin. More generally, no nondegenerate regular polygon with an odd prime number q of vertices, centred at the origin, has all its vertices at one height of f . Indeed, if $f(\zeta^j z) = a$ for every j and $z \neq 0$, the left side of (14.2), with $P = a$, is zero and the right side is not.

The infinitude conclusion also survives any number of derivatives: for every integer $r \geq 0$ and every polynomial P , the equation $f^{(r)}(z) = P(z)$ has infinitely many solutions. The same proof chooses a prime $q > \deg P + r + 2$ and filters the exponent $q - 2 - r$, whose coefficient is $(q - 2)! / ((q - 2 - r)! K'(1 - q)) \neq 0$; for $P = 0$, take any $q > r + 2$.

There is nevertheless a useful connection with the gap-series literature. Erdős recorded a conjecture of Fejér and Pólya that every entire series $\sum_k a_k z^{n_k}$ with all $a_k \neq 0$ and $n_k/k \rightarrow \infty$ attains every complex value infinitely often [11, Section IV.8, p. 250]. For $f - m$ the exponents are $n_k = p_{k+1} - 2 \sim k \log k$. The stronger summability condition $\sum_k 1/n_k < \infty$ in the proved Fejér-gap theorem does not hold here [27, Section 1]. Our conclusion uses the extra residue-class structure of these particular exponents; it makes no assertion about the general Fejér–Pólya conjecture.

Before finding this argument, we explored the middle height numerically. Truncate the Euler product defining K at primes at most Q , call it K_Q , and use the odd polynomial

$$P_Q(z) = \sum_{3 \leq p \leq 251} \frac{z^{p-2}}{K'_Q(1-p)}.$$

Numerical root finding gives the following nonzero roots in the first quadrant.

Euler cutoff Q	Root of P_Q , rounded
2 000	$4.202432 + 3.410403 i$
8 000	$4.200425 + 3.410428 i$
32 000	$4.199995 + 3.410432 i$

At each fixed Euler cutoff, replacing the Taylor cutoff 251 by 149 leaves the displayed digits unchanged. These are roots of finite approximations, not certified roots of $f - m$; no bound on the remaining Euler-product error is included. The theorem now proves infinitely many roots without locating them. Their positions and rate of accumulation are taken up in Question 15.6.

The origin is the only real solution: a nonconstant analytic function that is nonincreasing on the real line cannot take the same value at two distinct real points. Oddness of $f - m$ and its real coefficients put every root off the axes into a quartet $z, -z, \bar{z}, -\bar{z}$. The infinitude theorem alone does not determine how many roots belong to such quartets.

14.3 A full-dimensional staircase that keeps an echo

The primitive sampling law provides another small surprise. Its distribution function may be singular, strictly increasing and supported on an interval. Yet a frequency present in that measure reappears unchanged at every doubling of its pitch.

Proposition 14.4 (Dyadic echoes). Under the hypotheses of Theorem 3.1, the measure ν_g is invariant under $T(y) = 2y \pmod{1}$. If any lane is disabled, its Fourier transform does not tend to zero at infinity. More precisely, with

$$\hat{\nu}_g(t) = \int_0^1 e^{2\pi i t y} d\nu_g(y),$$

there is an integer $k \neq 0$ such that $\hat{\nu}_g(2^j k) = \hat{\nu}_g(k) \neq 0$ for every $j \geq 0$.

Proof. In the sampling model (B.1), deleting the first binary digit replaces (U, V) by $(U + V, V)$ and shifts the sequence of fair bits. At every prime the map $(U, V) \mapsto (U + V, V)$ permutes the nonzero residue pairs; it preserves their uniform distribution and their independence between primes. The shifted fair bits have the same independent law. This proves invariance. The binary expansion ambiguity has measure zero because ν_g is nonatomic.

Invariance gives $\hat{\nu}_g(2k) = \hat{\nu}_g(k)$ for integer k . If all coefficients with $k \neq 0$ vanished, integration against every trigonometric polynomial would agree with Lebesgue measure. Their density among continuous functions on the circle would force $\nu_g = \text{Leb}$. This is impossible when a lane is disabled, because (3.3) then has a nonzero singular part. Choose a nonzero coefficient and iterate. $\square$

In particular, the explicit singular law from (3.4) has Hausdorff dimension one but no Fourier decay along an infinite sequence. Its Fourier dimension is zero. The mechanism is the familiar one for a nonuniform dilation-invariant measure; the sampling construction supplies a specific arithmetic instance.

There is also more than one way for dimension to vary inside a single sampling law. As a small exact example, disable only the lane of 3. Depending on the residues of the step V , the proportion of free bits takes three values:

Condition on the primitive step	Probability	Free-bit proportion
$3 \mid V$	$1/4$	1
$3 \nmid V, 2 \nmid V$	$1/2$	$5/6$
$3 \nmid V, 2 \mid V$	$1/4$	$2/3$

Indeed, $\mathbb{P}(p \mid V) = 1/(p+1)$ independently at distinct primes. If $3 \mid V$, the disabled lane is avoided. If V is odd and not divisible by 3, exactly one index in every six has least prime factor 3. If V is even and not divisible by 3, primitivity forces all progression values to be odd, and one index in every three is closed.

For a fixed one of these periodic patterns, every compatible n -digit cylinder has probability $2^{-dn+O(1)}$, where d is its free-bit proportion. Distinct patterns having incompatible forced-zero sets are almost surely eventually distinguished by their fair bits. In the finite mixture, compatible patterns with larger d contribute exponentially less. Consequently the dyadic local dimension of this law is 1, 5/6 or 2/3 with the respective probabilities in the table. Here dyadic local dimension means

$$d_\nu(y) = \lim_{n \rightarrow \infty} \frac{-\log_2 \nu(I_n(y))}{n},$$

when the limit exists, where $I_n(y)$ is the dyadic interval of length 2^{-n} containing y . This example shows why the full-dimension statement in the sampling theorem does not assert one local dimension almost everywhere.

The three values give an exact starting point for the infinite-mask problem in Question 15.7. The geometry already varies inside one law, independently of which Goldbach answers occur.

15 Conjectures and open questions

The proved statements leave a small set of next steps. We distinguish the one prediction stated as a conjecture from six questions whose answers are not established here. The definitions are recalled so that each can be read independently. The connection to classical conjectures remains visible in Proposition 14.1: normality of the original constant is equivalent to Goldbach together with the naturally averaged Liouville Chowla conjecture.

15.1 A conjecture about the typical deletion time

In the deletion experiment of Section 13, begin with all primes except 7. At every round retain p_i only when $2(i + 1)$ is the sum of two currently retained primes; the same prime may be used twice, and all deletions happen simultaneously. Write $\tau(p)$ for the first round in which p is absent, with $\tau(7) = 0$ and $\tau(2) = \tau(3) = \tau(5) = \infty$, and put

$$M(x) = \max_{5 < p \leq x} \tau(p),$$

where the maximum is over primes. Proposition 13.2 proves $M(x) \sim \log x / \log \log x$. This is the clock of the last survivor in a finite window. We conjecture that it also gives the typical deletion time.

Conjecture 15.1 (Most primes share the last survivor's clock). Let P_x be a uniformly chosen prime in $(5, x]$. Then

$$\frac{\tau(P_x)}{M(x)} \longrightarrow 1 \quad \text{in probability.}$$

Here the randomness is only the uniform choice of P_x ; the deletion process itself is deterministic. The exact computation through 10^6 found that about 94.1% of the 78 495 primes above 5 disappear in rounds 11, 12 or 13, with $M(10^6) = 13$. The proposition proves the clock for the last surviving prime. The conjecture asks whether nearly the whole population follows it. The computation is a finite experiment, not evidence for a specific error term; fixed-depth abundance alone does not settle this question.

15.2 How much data does arithmetic recovery need?

Both questions in this subsection use the ordinary-integer experiment of (8.1). There are N labelled boxes, with hidden contents $X_1, \dots, X_N$ independently uniform on $\{1, \dots, N^2\}$. For each unordered pair we observe one bit saying whether the contents are coprime, independently flipped with common probability ε_N . The flips are independent of the integers; the decoder is not told ε_N . Theorem 8.1 identifies the consistency threshold

$$T_N = N(1 - 2\varepsilon_N)^2 \longrightarrow \infty.$$

It names the shared primes on a fraction of pairs tending to one. Corollary 8.2 then recovers divisibility by each fixed prime at a fixed labelled box. These are asymptotic results; the following asks for a useful finite guarantee. Take $N^2 \geq p$ so that conditioning on either divisibility outcome is meaningful.

Question 15.2 (How many boxes are needed to name a prime?). In the finite experiment, fix a prime p , a constant edge-flip rate $\varepsilon_N = \varepsilon \neq 1/2$, and $0 < \delta < 1/2$. How large must N be to decide whether $p \mid X_i$, with conditional error at most δ both for a multiple and for a nonmultiple of p ? Require a computationally feasible procedure that is not supplied the error rate. How does the answer grow with p and with the closeness of the channel to a fair coin?

The second question concerns the original message rather than the full collection of shared-prime sets. Fix an unknown mask $g = (g_j)$ with $g_1 = 1$, and set

$$x_g(1) = 0, \quad x_g(n) = \frac{1 - \lambda(n)}{2} g_{\ell(n)} \quad (n \geq 2),$$

where $\ell(n)$ is the index of the least prime factor of n . Alongside the noisy graph, observe aligned digit marks

$$Z_i = x_g(X_i) \oplus B'_i, \quad B'_i \sim \text{Bernoulli}(\eta_N).$$

These digit flips are independent of one another, the integers and the edge flips. As in Theorem 9.1, the digit-noise probability η_N is supplied; the edge-noise probability is not. Writing $a_N = 1 - 2\varepsilon_N$ and $b_N = 1 - 2\eta_N$, that theorem recovers both the message and almost every shared-prime set when $Na_N^2 \rightarrow \infty$ and $Nb_N^2 \rightarrow \infty$. Here recovering the message means recovering each fixed finite prefix of g with probability tending to one as N grows.

Question 15.3 (How much graph does the message need?). If the only required output is the arithmetic message, what conditions on the two noise sequences suffice? In particular, can the edge signal be weaker than the threshold needed to recover shared-prime sets?

This asks for a separate decoding theorem. The joint-task result supplies the digit-noise rate to the procedure, and makes no assertion of adaptation to an arbitrary unknown rate approaching one half.

15.3 Can one hear every disabled prime?

The exact bias can confuse a single disabled lane with infinitely many of them, as Proposition 4.5 shows. The whole function distinguishes that example immediately: one function is linear and the other is transcendental. How much farther does this recovery go?

For this question only, write $\Phi_{\mathcal{F}}$ for the normalized entire function in (4.4) whose disabled-prime set is $\mathcal{F}$. Its Taylor coefficients are $M_k/k!$, where M_k are the conditional correlations of Section 4. The counting exponent of $\mathcal{F}$ is

$$\limsup_{R \rightarrow \infty} \frac{\log \max(1, \#(\mathcal{F} \cap [2, R]))}{\log R}.$$

Restrict to masks with counting exponent strictly below one, so that their zero multisets determine the normalized entire functions by Corollary 4.4.

Question 15.4 (Hearing the mask). Let $\mathcal{F}$ and $\mathcal{H}$ be sets of odd primes whose counting exponents are strictly below one. Does

$$\Phi_{\mathcal{F}} \equiv \Phi_{\mathcal{H}} \implies \mathcal{F} = \mathcal{H}?$$

Equivalently, can two different such masks have exactly the same complex zeros, including multiplicities?

There are three equivalent ways to supply the data: all conditional correlations M_k , the entire function Φ , or the nonzero eigenvalues of the positive mask operator A_g . Indeed, Ψ is obtained from Φ by multiplying its k th Taylor coefficient by the fixed nonzero number $K(k)$, and $\Psi(z) = \prod_j (1 + \alpha_j z)$ records those eigenvalues. Thus this is also an inverse spectral question for a restricted arithmetic family of operators.

For finite masks the answer is yes: even the first derivative $\Phi'(0) = S_g$ determines the mask. An infinite mask cannot agree with a finite one because the corresponding functions have different degrees. A possible failure must therefore involve two infinite masks. We already know that identical functions would force the same counting exponent. The question is whether the higher correlations retain the individual prime locations as well.

15.4 Does the triangular match include the last root?

Recall

$$J_N(z) = \sum_{k=0}^N \binom{N}{k} K(k) z^k,$$

where $K(k)$ is the limiting probability that k random integers are pairwise coprime. Its roots are negative; write them as $-R_{N,1}, \dots, -R_{N,N}$, with $R_{N,j} > 0$ and multiplicities included. Theorem 11.2 shows that the empirical distribution of

$$\frac{\exp(e^{-\gamma} R_{N,j})}{N \log N}$$

approaches the Dykema–Haagerup law DH_1 , whose support is $[0, e]$; γ is Euler’s constant. This is a statement about the whole histogram.

Question 15.5 (Does the last root find the same edge?). Do the largest arithmetic roots satisfy

$$e^{-\gamma} \max_j R_{N,j} - \log N - \log \log N \longrightarrow 1,$$

or, equivalently,

$$\max_j \frac{\exp(e^{-\gamma} R_{N,j})}{N \log N} \longrightarrow e?$$

This asks whether the match reaches the edge as well as the histogram. For the Gaussian triangular matrix, the largest squared singular value does tend to e [7, Theorem 2]. The empirical convergence proved here leaves room for a small number of arithmetic outliers. It implies the corresponding lower limit is at least e , since the limiting law has mass arbitrarily close to that edge; excluding larger outliers needs more. The same distinction leaves convergence of transformed empirical moments unproved.

15.5 Locating the returns to the middle height

Let f be the entire prime density of Theorem 5.3, and set $m = f(0) = 1/(2\mathfrak{C}_2)$. Theorem 14.3 proves that every polynomial target is met infinitely often. For the target m , the origin is the only real solution; reflection and conjugation place roots off the axes in quartets. The theorem does not locate those roots.

Question 15.6 (Where does the middle height return?). Are there infinitely many solutions of $f(z) = m$ on the imaginary axis? How fast does

$$N_m(R) = \#\{z \in \mathbb{C} : |z| \leq R, f(z) = m\},$$

with multiplicities counted, grow as $R \rightarrow \infty$?

The first-quadrant calculation in Section 14 is an experiment with finite approximations, not a certification of a root. The imaginary-axis question asks about a different part of the geometry, while the counting question measures how often the middle height returns in the whole plane.

15.6 The dimensions inside a singular staircase

Let p_j be the j th prime. Disable exactly the lanes in $\mathcal{F}_* = \{p_{2j} : j \geq 2\}$, the explicit mask (3.4), and write ν_* for its primitive sampling measure from Theorem 3.1. This measure is singular continuous, has support $[0, 1]$, and has Hausdorff dimension one. The local dimension describes a different feature: how quickly the probability around an individual point shrinks.

For $y \in [0, 1]$, let $I_n(y)$ be its dyadic interval of length 2^{-n} ; endpoints may be assigned consistently to either adjacent interval since the measure is nonatomic. Choose Y with law ν_* . Then the following local dimension is itself a random variable.

Question 15.7 (The dimensions inside the staircase). For the explicit infinite mask $\mathcal{F}_* = \{p_{2j} : j \geq 2\}$, determine the distribution of

$$\underline{d}_*(Y) = \liminf_{n \rightarrow \infty} \frac{-\log_2 \nu_*(I_n(Y))}{n}, \quad Y \sim \nu_*.$$

In particular, does this distribution have any atoms?

The example with only the lane of 3 disabled, in Section 14, gives an exact comparison: its local dimension equals 1, 5/6 or 2/3, with probabilities 1/4, 1/2 and 1/4. Infinitely many prime lanes may produce a richer distribution than these three values. The question concerns the geometry of the construction independently of which Goldbach answers occur.

16 Back to the constant

The first construction fitted a list of arithmetic questions into one real number. Each prime carried a question; its lane repeated the answer. That repetition let a single missing answer alter a limiting percentage, break a self-similarity, and turn Euler's convergent sum into infinity.

Reading the positions through coprimality changed the questions we could ask. Correlations counted missing lanes. Their generating functions turned that count into degrees, zeros and spectral directions. A random scale connected two ways of averaging, and its density became the curve whose derivatives recognize primes. The prime and composite parts could even be put back together to recover an ordinary exponential law.

Then we hid the numbers. The relationships still revealed prime names. The cancellation of three observed frequencies explained why; the noisy experiment quantified how far the idea could go. Its paradox survives the precise rules of the game: the exact shared primes of a fraction of pairs tending to one can be recovered while the retained fraction of information about the complete table tends to zero. The surviving information grows, but the uncertainty about the whole table grows faster. Both facts belong in the picture.

The later experiments asked what arithmetic arrangements look like. Coprime crowds could leave earlier holes in the list of covered primes. Averaging a matrix and averaging its characteristic polynomial revealed different spectra. Conditioning a factorization to be unusually balanced produced a Gaussian profile. These conclusions make sense on their own; the shared constructions gave us a route from one question to the next.

Finally we returned to the original assignment and removed seven. Almost all primes survived every fixed number of rounds. Only three survived forever. The same arithmetic could appear abundant or almost empty, depending on when we looked.

A last change of view sent the curve into the complex plane. Its prime coefficients made it meet every polynomial infinitely often, yet prevented any three points with the same value from forming an equilateral triangle centred at the origin. A rotation that keeps just one prime explained both conclusions.

Section 15 gathers the conjecture and questions left by these experiments: their hidden messages, complex values, spectra and long-term behavior. The earlier connections to classical arithmetic questions also remain invitations to further work. Goldbach, RH and twin primes remain open, and a new formulation needs new input before it can settle them.

The unifying question is simpler: what can a number remember after we change the way we observe it? Here it remembered arithmetic in a digit frequency, in a curve's missing derivatives, in the signs of a matrix, and in noisy relationships between closed boxes. Each observation revealed something the previous one had made difficult to see.

A Proofs behind the constant and the curve

The main text keeps the elementary digit arguments beside their conclusions. This appendix supplies the longer operator and density arguments.

A.1 The finite part of the heat trace

Proof of Proposition 2.4. Summing a Dirichlet series on one least-prime lane gives

$$\begin{aligned} \sum_{\ell(n)=j} n^{-s} &= p_j^{-s} \zeta(s) \prod_{r < j} (1 - p_r^{-s}), \\ \sum_{\ell(n)=j} \lambda(n) n^{-s} &= -p_j^{-s} \frac{\zeta(2s)}{\zeta(s)} \prod_{r < j} (1 + p_r^{-s}). \end{aligned}$$

Initially for $\Re s > 1$, convolution with the constant-one function therefore gives

$$Z_{\mathcal{G}}(s) := \sum_{m \geq 1} c_m m^{-s} = \zeta(2s)(1 + C(s)) + \zeta(s)^2 A(s). \quad (\text{A.1})$$

This identity supplies meromorphic continuation. Mellin inversion of the exponential, followed by a contour shift from $\Re s > 1$ to $\Re s = -1/2$, gives

$$\Theta_{\mathcal{G}}(t) = 1 + \frac{2}{2\pi i} \int \Gamma(s) Z_{\mathcal{G}}(s) (\pi t)^{-s} ds$$

with residues at $1, 1/2, 0$. The finite Dirichlet polynomials are bounded on each vertical line, zeta has polynomial growth there, and gamma decays exponentially. The shifted integral is therefore $O_F(\sqrt{t})$; these are the standard Mellin contour estimates in [30, Section 2.5].

At $s = 1$, the double pole of $\zeta(s)^2 A(s)$, together with $\Gamma'(1) = -\gamma$, gives the first line of (2.6). The residue of $\zeta(2s)$ at $1/2$ gives its second divergent term. Finally $C(0) = B$, $A(0) = 0$ because the lane of 2 is enabled, and $\zeta(0) = -1/2$. Hence $Z_{\mathcal{G}}(0) = -(1 + B)/2$. The residue at zero and the separate zero-energy state combine to give $1 + 2Z_{\mathcal{G}}(0) = -B$. $\square$

A.2 A positive operator for the message

Let $\mathcal{F} = \{p_j : g_j = 0\}$ and $D = z d/dz$. For a prime cutoff R , start with $U = 1$ and apply successively

$$U \longleftarrow (D + p - 1 + \mathbf{1}_{\{p \in \mathcal{F}\}} z) U, \quad p \leq R. \quad (\text{A.2})$$

Call the result U_R . For the mask in which only $\mathcal{F} \cap [2, R]$ is disabled, denote the first function in (4.4) by $\Phi_{g,R}$. Then

$$[z^k] \Phi_{g,R}(z) = \frac{[z^k] U_R(z)}{\prod_{p \leq R} (p + k - 1)}. \quad (\text{A.3})$$

Indeed, let v_j be the probability that exactly j coordinates have received a first label, all such labels were disabled, and no coordinate has received a successful first label. At prime p ,

$$v_j^{\text{new}} = \frac{(p + j - 1)v_j + \mathbf{1}_{\{p \in \mathcal{F}\}}(k - j + 1)v_{j-1}}{p + k - 1}.$$

Divide v_j by the falling factorial $(k)_j$ and clear the common denominators. This is exactly the coefficient recurrence in (A.2). At $j = k$, the probability is $M_k(g, R)$, proving (A.3).

The all-active form of this differential recurrence is a generalized Bell-polynomial recurrence studied in [9]. We give the masked normalization and positive construction explicitly, because they are what connect the recurrence to the digits.

Proof of Theorem 4.2. Set

$$\eta_R = \prod_{p \leq R} \frac{p-1}{p}, \quad \mathcal{P}_R(z) = \frac{U_R(\eta_R z)}{\prod_{p \leq R} (p-1)}.$$

Its constant term is one. If $K_R(k) = \prod_{p \leq R} (1 - 1/p)^k (1 + k/(p-1))$, then (A.3) gives

$$[z^k] \mathcal{P}_R(z) = K_R(k) \frac{M_k(g, R)}{k!}. \quad (\text{A.4})$$

These coefficients are between zero and $1/k!$ and tend to $K(k)M_k(g)/k!$. It follows that $\mathcal{P}_R \rightarrow \Psi_g$ locally uniformly.

We construct positive matrices with determinant $\mathcal{P}_R$. Suppose the previous matrix has eigenvalues $\alpha_1, \dots, \alpha_m > 0$. At prime p , put $c = p - 1$, $u = c/p$ and $\eta_{\text{new}} = u\eta_{\text{old}}$. Begin with

$$B = \text{diag}(u\alpha_1, \dots, u\alpha_m), \quad w = (\sqrt{u\alpha_1}, \dots, \sqrt{u\alpha_m})^\top.$$

If $p \in \mathcal{F}$, adjoin one zero coordinate to B and the component $\sqrt{\eta_{\text{new}}}$ to w . In either case set $A_R = B + c^{-1}ww^\top$. Start with the empty matrix. The matrix determinant lemma gives

$$\det(I + zA_R) = \left(1 + \frac{D}{c} + \frac{\mathbf{1}_{\{p \in \mathcal{F}\}} \eta_{\text{new}} z}{c}\right) \mathcal{P}_{<p}(uz),$$

which is the normalization of (A.2). The matrices are positive definite whenever nonempty: a newly adjoined null coordinate is filled by the nonzero last component of w .

The old diagonal entries of A_R are exactly the previous α_i , because $u(1 + 1/c) = 1$. At a disabled prime the new diagonal entry is $\eta_{\text{old}}/p = \delta_{\ell(p)}$. Therefore

$$\text{tr } A_R = S_R := \sum_{\substack{p \leq R \\ p \in \mathcal{F}}} \delta_{\ell(p)} \longrightarrow S_g. \quad (\text{A.5})$$

Sort each eigenvalue list decreasingly and pad it with zeros. For every fixed j , the sum of its first j entries is nondecreasing through the updates: the variational characterization of that sum bounds it below by the trace on the old leading j coordinate directions. These sums are bounded by one and hence converge. Their consecutive differences define a decreasing summable list $\alpha_1, \alpha_2, \dots \geq 0$.

No trace disappears in this passage. Given a cutoff with m disabled primes, every later sum of the first m eigenvalues is at least S_R . Passing to the limit and then increasing the cutoff shows $\sum_j \alpha_j \geq S_g$. The reverse inequality follows from (A.5). Thus the padded lists converge in ℓ^1 , and their determinant products converge locally uniformly to

$$\Psi_g(z) = \prod_j (1 + \alpha_j z), \quad \sum_j \alpha_j = S_g.$$

Take A_g to be the diagonal operator with these positive eigenvalues. In the finite-mask case its rank equals the degree, which is $\#\mathcal{F}$ by Proposition 4.1. In the infinite case every coefficient is positive, so the function cannot be a finite product. This proves the rank statement and the location of the zeros. $\square$

A.3 Why the entire series is the actual density

Proof of Theorem 5.3. For a cutoff R , write

$$L_R = \prod_{p \leq R} \frac{p}{p-1}, \quad Y_R = \sum_{p \leq R} \frac{E_p}{p-1}, \quad W_R = L_R e^{-Y_R}.$$

The summands of Y_R have distinct exponential rates $r_p = p-1$. Partial fractions of $\prod_{p \leq R} r_p/(r_p + s)$ show that the density of Y_R is

$$\sum_{p \leq R} r_p \prod_{\substack{q \leq R \\ q \neq p}} \frac{r_q}{r_q - r_p} e^{-r_p y}, \quad y > 0.$$

After the change of variable $w = L_R e^{-y}$ this becomes

$$f_R(w) = \sum_{p \leq R} c_{p,R} w^{p-2} \quad (0 < w < L_R), \quad c_{p,R} = \frac{1}{K'_R(1-p)}. \quad (\text{A.6})$$

It is zero for $w > L_R$. One can check the coefficient without transform notation: the change of variable multiplies the coefficient of $e^{-r_p y}$ by $L_R^{-r_p}$, which is exactly the reciprocal derivative in (A.6).

For fixed p , the derivative $K'(1-p)$ is nonzero. Its factors from primes below p are negative, and all remaining factors apart from the differentiated one are positive, proving the claimed sign. Moreover

$$|c_{p,R}| \leq |c_p| \quad (R \geq p), \quad c_{p,R} \longrightarrow c_p. \quad (\text{A.7})$$

Indeed, adding a prime $q > p$ to the derivative multiplies it by

$$\left(1 + \frac{1}{q-1}\right)^{p-1} \left(1 - \frac{p-1}{q-1}\right),$$

a number strictly between zero and one. Taking logarithms and using $\log(1+u) < u$ and $\log(1-v) < -v$ proves the upper bound of one. The infinite tail converges to a nonzero product.

We next bound the infinite coefficient. Put $m = p-1$. Separating the prime factors at $2p$ gives the exact expression

$$\begin{aligned} \log |K'(-m)| &= m \sum_{q \leq 2p} \log \frac{q}{q-1} - \log m \\ &\quad + \sum_{\substack{q \leq 2p \\ q \neq p}} \log \frac{|q-p|}{q-1} \\ &\quad + \sum_{q > 2p} \left[m \log \frac{q}{q-1} + \log \left(1 - \frac{m}{q-1}\right) \right]. \end{aligned}$$

Mertens' product estimate makes the first line $p \log \log p + O(p)$; a statement of the classical product estimate is recalled in [22, Section 2]. Each ratio in the middle line lies between $1/(2p)$ and $2p$. Chebyshev's bound $\pi(2p) = O(p/\log p)$ therefore bounds that line in absolute value by $O(p)$. For $q > 2p$, cancellation of the linear terms leaves an error $O(p^2/q^2)$, whose sum is $O(p)$ even if all integers $q > 2p$ are included. This proves (5.7). It follows that the series in (5.5) converges absolutely and uniformly on every compact subset of $\mathbb{C}$.

Finally, $L_R \rightarrow \infty$ by Mertens' theorem. On any fixed interval $[0, B]$, the polynomial formula (A.6) therefore applies for large R , with values at zero supplied by continuity. Equations (A.7) and (5.7) imply uniform convergence $f_R \rightarrow f$ there, by dominated convergence of the power series. On the other hand, $W_R \rightarrow W$ almost surely. For every continuous compactly supported test function in $(0, \infty)$, uniform convergence of densities and weak convergence of the random variables give the same integral. Hence the restriction of f to $(0, \infty)$ is exactly the law's density. Since W is positive and finite almost surely, no missing mass occurs at an endpoint. Its integral is one. Each c_p is nonzero, proving (5.6). $\square$

A.4 Why signed correlations become probabilities

Proof of Proposition 4.1. Write $a_g = h_g + (1 - h_g)\lambda$ as in (1.3). Truncate the lane mask to primes at most R , and truncate the coprimality condition at the same cutoff. The mean error is at most a constant depending on k times

$$\prod_{p \leq R} (1 - 1/p) + \sum_{p > R} p^{-2} + o_N(1).$$

The first term bounds coordinates with no prime divisor at most R ; the second bounds a large-prime collision.

For fixed R , expand the product of the truncated signed digits. Every term containing a Liouville factor in at least one coordinate has average zero. To see this, partition the finite prime divisibility patterns among coordinates. In each coordinate, inclusion–exclusion reduces the required sum to finitely many expressions $\lambda(d)L(N/d)$ with fixed d , which are $o(N)$. The other coordinates contribute bounded normalized sums. Thus only the term $\prod_i h_g(n_i)$ remains in the limit. Letting $R \rightarrow \infty$ and dividing by the positive limit $K(k)$ proves existence and identifies the conditional moment as a probability.

Conditioning the local independent divisibility model on at most one divisible coordinate gives precisely the probabilities in the statement. Conditioning at finitely many primes preserves independence across those primes; passage to all primes is valid because the no-collision probability tends to $K(k) > 0$. Each coordinate receives a first prime almost surely, since $\prod_p (1 - 1/(p+k-1)) = 0$. The first labels are distinct. There can therefore be no successful outcome with fewer than k disabled primes. Given k such primes, prescribe them as the first labels of the coordinates and prescribe no incompatible choices at the finitely many earlier primes. This event has positive probability. The case $k = 1$ is the lane-density formula. $\square$

B Proof of the primitive sampling theorem

This appendix proves Theorem 3.1. The main text gave its probabilistic mechanism; here we justify the limiting distribution, the exact Lebesgue decomposition, and the dimension statement.

Proof. We first obtain the limiting probability model. Take a pair of Haar-uniform profinite integers (U, V) , conditioned on having no common prime divisor. This event has probability $\prod_p (1 - p^{-2}) = 6/\pi^2$. At each prime p , the residues (U, V) are therefore uniform among the $p^2 - 1$ pairs other than $(0, 0)$; these choices are independent at distinct primes. Independently choose fair

bits $B_0, B_1, \dots$. The candidate limit is the law of

$$Z_g = \sum_{j \geq 0} (1 - h_g(U + jV)) B_j 2^{-j-1}. \quad (\text{B.1})$$

The least prime of each $U + jV$ exists almost surely. For example, under this local law $\mathbb{P}(p \mid U + jV) = 1/(p + 1)$, and independence over primes shows that the probability of avoiding every prime is zero.

For completeness, the analytic input that produces the fair bits is the following variable-step cancellation: for any nonempty finite $J \subset \{0, 1, \dots\}$ and any fixed function $P(a, d)$ periodic in both variables,

$$\frac{1}{N^2} \sum_{a, d \leq N} P(a, d) \prod_{j \in J} \lambda(a + jd) \longrightarrow 0. \quad (\text{B.2})$$

Theorem 2.5 of Frantzikinakis–Host [13] gives vanishing Gowers norms of every fixed order for the aperiodic function λ ; their Lemma 9.6 controls products on independent linear forms. To include P , expand it in its finite Fourier series. For $j_0 \in J$ a phase $e((ra + sd)/Q)$ factors as $e(r(a + j_0d)/Q)e((s - rj_0)d/Q)$, where $e(t) = e^{2\pi it}$. Absorb the first factor into $\lambda(a + j_0d)$ and the second into the additional form d . Multiplication by a linear phase preserves Gowers norms of order at least two. Extend the first function evenly from positive integers, as required by that lemma. The forms $a + jd$ and d are pairwise independent, proving (B.2). Thus no assertion about correlations at a fixed step is being used.

Fix a prefix length k . Truncate both the mask and the coprimality condition at primes at most R . All mask coefficients and restrictions are now periodic in (a, d) . Expand a prefix indicator using $1 - 2x_g = h_g + (1 - h_g)\lambda$. Every term containing a Liouville factor vanishes by (B.2); the other terms converge by the Chinese remainder theorem. These are exactly the prefix probabilities in (B.1), with the same truncations. The proportion of pairs missed by truncated coprimality is at most $\sum_{p > R} p^{-2}$, since $[N/p]^2 \leq N^2/p^2$. A mask error at $a + jd$ requires that integer to avoid all primes at most R . For fixed R , its limiting proportion in the square is $\prod_{p \leq R} (1 - 1/p)$, which tends to zero. Summing this error over the k positions and then removing both truncations proves prefix convergence. A continuous function of the sampled real is uniformly approximated by a function of its first k digits, since the remaining binary tail is at most 2^{-k} . This proves weak convergence to the law of Z_g .

Next fix a primitive profinite pair (U, V) . A lane p is visited by $U + jV$, $j \geq 0$, if and only if $p \nmid V$. If $p \mid V$, primitivity gives $p \nmid U$, so the lane is avoided. Otherwise prescribe the unique root modulo p and avoid the roots modulo each smaller prime q with $q \nmid V$. There is an allowed residue at each such q ; primes dividing V already divide no term. The Chinese remainder theorem gives an index with least prime p , and the same event repeats periodically in j .

All digits of the progression are open precisely when $p \mid V$ for every $p \in \mathcal{F}$. Locally this has probability $(p - 1)/(p^2 - 1) = 1/(p + 1)$, proving that the all-open event has probability L_g . On it, Z_g is uniform on $[0, 1]$. An all-closed progression would require $p \mid V$ for every successful prime. There are infinitely many such primes by (3.1), so that event has probability zero. Almost every fiber therefore has an open lane, hence infinitely many independent fair bits. Its binary law is nonatomic, and so is its image on $[0, 1]$.

If a fiber has any closed position, that position repeats along an infinite arithmetic progression of indices. All the corresponding bits are forced to zero. The set of binary sequences with zeros on some infinite arithmetic progression is a countable union of fair-coin null sets. Its image has

Lebesgue measure zero. Every fiber with a closed position is carried by this same null set. This common support, together with non-atomicity, proves the exact decomposition (3.3).

To prove full support, choose any k distinct successful primes $s_0, \dots, s_{k-1}$, and take $R > \max s_j$. At disabled primes $q \leq R$ impose $V = 0$, $U \neq 0$ modulo q . At successful primes $q \leq R$ impose $V \neq 0$; at s_j also impose $U + jV = 0$. These compatible local conditions have positive probability. They make each of the first k positions open: it has a successful prime divisor below R and no disabled divisor there. The independent fair bits can then realize any prescribed word. Every dyadic interval has positive mass.

Finally, impose the same conditions below R , without prescribing the first k positions. Call this positive-probability event D_R . For any fixed local residues on D_R , every closed index belongs to the periodic set avoiding the single roots at all successful primes $q \leq R$. This set has density

$$\varepsilon_R = \prod_{\substack{q \leq R \\ q \notin \mathcal{F}}} (1 - 1/q).$$

With $Q_R = \prod_{q \leq R} q$, at least $(1 - \varepsilon_R)n - Q_R$ of the first n bits are consequently open. The submeasure $\mu_R(B) = \mathbb{P}(Z_B \in B, D_R)$ satisfies, for every dyadic interval I of length 2^{-n} ,

$$\mu_R(I) \leq 2^{Q_R} 2^{-(1 - \varepsilon_R)n}.$$

The same bound with a different constant holds for arbitrary intervals, by covering one interval with at most two dyadic intervals of comparable length. Any full- ν_g -measure Borel set has positive μ_R mass. Covering it by intervals and summing this bound shows that its Hausdorff dimension is at least $1 - \varepsilon_R$. Hypothesis (3.1) gives $\varepsilon_R \rightarrow 0$, proving dimension one. A nonatomic full-support probability has a continuous strictly increasing distribution function; differentiation of its Lebesgue decomposition gives the final assertion. $\square$

C Growth estimates for the mask function

The proof of Theorem 4.3 matches an upper bound from all possible assignments with a lower bound from one carefully chosen set of assignments.

Proof. For every $\eta_g < \beta < 1$, $B_g(u) \ll_\beta u^\beta$; this also ensures convergence of the reciprocal sum. Integration by parts in (4.8) yields

$$\log P_{\mathcal{F}}(r) = r \int_2^\infty \frac{B_g(u)}{u(u+r)} du \ll_\beta r^\beta, \quad (\text{C.1})$$

since $\int_0^\infty v^{\beta-1}/(1+v) dv < \infty$. This proves the upper bound for the order.

For the lower bound suppose $\mathcal{F}$ is infinite and list it as $q_1 < q_2 < \dots$. Require $q_1, \dots, q_k$ to select distinct labels and every other prime at most q_k to select no label. The $k!$ possible assignments are disjoint. Therefore

$$\begin{aligned} a_k &\geq \prod_{j=1}^k \frac{1}{q_j + k - 1} \prod_{\substack{p \leq q_k \\ p \notin \{q_1, \dots, q_k\}}} \frac{p-1}{p+k-1} \\ &\geq (q_k + k)^{-k} \prod_{p \leq q_k} (1 + k/(p-1))^{-1} \geq \left(\frac{c}{(q_k + k) \log q_k} \right)^k. \end{aligned} \quad (\text{C.2})$$

The last inequality uses $\log(1+t) \leq t$ and the classical Mertens estimate $\sum_{p \leq y} (p-1)^{-1} \leq \log \log y + O(1)$; see also [22, Section 2]. Since $q_k \geq k+1$, taking $r_k = (2e/c)q_k \log q_k$ gives $\log \Phi_g(r_k) \geq k$. Hence the order is at least

$$\limsup_{k \rightarrow \infty} \frac{\log k}{\log q_k + \log \log q_k + O(1)} = \eta_g.$$

For a finite mask the degree is its cardinality, so both exponents are zero. This completes the general claim.

For the Goldbach mask, $p_j \sim j \log j$ shows that the exponent of B_g equals the exponent of E . Apply the established bound (3.2) in (C.1). The part $u \leq \sqrt{r}$ is $O(r^{9/25})$. For $u > \sqrt{r}$, replace $(\log u)^{-18/25}$ by $O((\log r)^{-18/25})$ and scale $u = rv$ in the remaining integral. This proves (4.10). Finally $M_k \leq k! \Phi_g(r) r^{-k}$; choose $r = k^{25/18} \log(k+1)$ and use $k! \leq k^k$ to obtain (4.11). $\square$

D Proofs for the variance and twin-prime criteria

D.1 The RH variance criterion

Lemma D.1 (The smoothed prime discrepancy). Fix a real $m > 1$, and set

$$H_m(v) = \sum_p (p+v)^{-m} - \int_2^\infty \frac{du}{(u+v)^m \log u}, \quad v > 0.$$

Put

$$B_m = \frac{\Gamma(1/2)\Gamma(m-1/2)}{2\Gamma(m)}, \quad S = 1 + \frac{\gamma_E}{2} - \frac{1}{2} \log(4\pi).$$

On RH, with nontrivial zeros counted with multiplicity,

$$v^{m-1/2} \log v H_m(v) = -B_m - \sum_\rho \frac{\Gamma(\rho)\Gamma(m-\rho)}{\Gamma(m)} v^{i\Im \rho} + o(1). \quad (\text{D.1})$$

The series is absolutely convergent, uniformly in $\log v$, and its absolute value is at most $2SB_m < B_m$. If RH fails, the expression on the left of (D.1) has $\limsup +\infty$ and $\liminf -\infty$.

Proof. We spell out the removal of the logarithmic prime weight. The classical ingredients are the logarithmic derivative of zeta, its explicit formula, the zero-counting bound and the RH estimate for the Chebyshev function; see [39, Chapters II, III, IX and XIV]. For $a \geq 0$, define

$$Q_{m,a}(v) = \sum_p \frac{(\log p)p^{-a}}{(p+v)^m} - \int_2^\infty \frac{u^{-a}}{(u+v)^m} du.$$

Since $\int_0^\infty u^{-a} da = 1/\log u$, applying Tonelli separately to the two nonnegative terms gives

$$H_m(v) = \int_0^\infty Q_{m,a}(v) da. \quad (\text{D.2})$$

Assume RH and take $0 \leq a \leq a_0 = 1/8$. For $1 < c < m$, Mellin inversion is

$$\sum_{n \geq 2} \frac{\Lambda(n)n^{-a}}{(n+v)^m} = \frac{1}{2\pi i} \int_{(c)} \frac{\Gamma(s)\Gamma(m-s)}{\Gamma(m)} \left(-\frac{\zeta'(s+a)}{\zeta(s+a)} \right) v^{s-m} ds.$$

Move the line to $\Re s = 1/4$. The pole at $1 - a$ gives the continuous integral over $(0, \infty)$; a zero ρ gives the residue

$$-\frac{\Gamma(\rho - a)\Gamma(m - \rho + a)}{\Gamma(m)}v^{\rho - a - m}.$$

The remaining integral is $O_m(v^{1/4 - m})$, uniformly in a . Indeed, the new zeta arguments have real parts in $[1/4, 3/8]$, separated from all zeros on RH. Polynomial bounds for the logarithmic derivative and exponential decay of the gamma factors control the vertical integral; the horizontal segments may be chosen at the usual heights avoiding zeros. Changing the continuous lower endpoint from 0 to 2 costs $O_m(v^{-m})$.

We must remove the prime powers from the von Mangoldt sum. The squares contribute

$$\sum_p \frac{(\log p)p^{-2a}}{(p^2 + v)^m} = v^{1/2 - a - m} \left\{ \frac{\Gamma(1/2 - a)\Gamma(m - 1/2 + a)}{2\Gamma(m)} + o_m(1) \right\}.$$

This is the prime number theorem applied as a Stieltjes integral against $\vartheta(u)$, followed by $u = \sqrt{v}y$. The rescaled kernels and their derivatives have common integrable majorants for $0 \leq a \leq 1/8$, so the error is uniform. Powers of exponent at least three contribute $O_m(v^{1/3 - m} \log^2 v)$, by Chebyshev's bound and dyadic summation. Consequently

$$Q_{m,a}(v) = v^{1/2 - a - m} \left\{ -B_m(a) - \sum_{\rho} C_{m,\rho}(a)v^{i\Im \rho} + o_m(1) \right\},$$

uniformly on the parameter interval, where

$$B_m(a) = \frac{\Gamma(1/2 - a)\Gamma(m - 1/2 + a)}{2\Gamma(m)}, \quad C_{m,\rho}(a) = \frac{\Gamma(\rho - a)\Gamma(m - \rho + a)}{\Gamma(m)}.$$

The zero sums of the coefficients and their first a -derivatives converge uniformly. The elementary endpoint estimate

$$L \int_0^{a_0} e^{-aL} g(a) da = g(0) + o(1), \quad L \longrightarrow \infty,$$

therefore gives (D.1) from this part of (D.2).

The discarded parameter tail also needs a bound. With $E(u) = \vartheta(u) - u + 2$, interpreted from 2^- , that tail equals

$$\int_{[2, \infty)} \frac{u^{-a_0}}{\log u (u + v)^m} dE(u).$$

On RH, $E(u) = O(\sqrt{u} \log^2 u)$. Integration by parts bounds the last display by

$$O_m \left(\int_2^\infty \frac{u^{-a_0 - 1/2} \log u}{(u + v)^m} du + \int_2^\infty \frac{u^{1/2 - a_0} \log u}{(u + v)^{m+1}} du \right) = O_m(v^{1/2 - a_0 - m} \log v).$$

This is $o(v^{1/2 - m} / \log v)$, as required.

Next we compare the square term with the zeros. The Hadamard product gives, on RH,

$$\sum_{\gamma > 0} \frac{1}{\gamma^2 + 1/4} = S = 0.0230957 \dots$$

The gamma integral, reflection formula and $\cosh(\pi\gamma) \geq 4(\gamma^2 + 1/4)^2$ imply

$$\sum_{\rho} \left| \frac{\Gamma(\rho)\Gamma(m - \rho)}{\Gamma(m)} \right| \leq 2B_m S.$$

For the elementary hyperbolic bound it suffices to retain the constant, quadratic and quartic terms of the series for cosh. This proves the claimed strict negative margin on RH.

For the converse, initially in $1 < \Re s < m$, Mellin transformation gives

$$\int_0^\infty v^{m-s-1} H_m(v) dv = \frac{\Gamma(s)\Gamma(m-s)}{\Gamma(m)} F(s), \quad (\text{D.3})$$

where

$$F(s) = P(s) - J(s), \quad P(s) = \sum_p p^{-s}, \quad J(s) = \int_2^\infty \frac{u^{-s}}{\log u} du.$$

Write $R(s) = \sum_{k \geq 2} P(ks)/k$. This is analytic on $\Re s > 1/2$, and continuation of the derivative of F gives

$$F'(s) = \frac{\zeta'(s)}{\zeta(s)} - R'(s) + \frac{2^{1-s}}{s-1}. \quad (\text{D.4})$$

The singularities at the real point 1 cancel. There are no other real singularities above $1/2$. An off-line zero ρ with $\Re \rho > 1/2$, however, gives a nonzero pole of F' , hence a nonremovable logarithmic singularity of F .

Suppose the normalized discrepancy were eventually bounded below by $-C$. For a sufficiently large $V > 1$, the function

$$h(v) = v^m H_m(v) + C v^{1/2} / \log v$$

would be nonnegative on $[V, \infty)$. Its Mellin integral $\int_V^\infty h(v) v^{-s-1} dv$ has an abscissa of convergence $\sigma_c \leq 1$. The off-line singularity implies $\sigma_c \geq \Re \rho > 1/2$: the added term is analytic to the right of $1/2$, and the short integral omitted from (D.3) is analytic for $\Re s < m$.

Landau's positivity argument forces a singularity at the real point σ_c , contradicting (D.4). Here is the argument in the required form. If a Mellin transform of a nonnegative function were analytic at its finite real abscissa, expand at a slightly larger real point. The Taylor coefficients evaluated in the leftward direction are integrals of nonnegative powers of $\log v$. Tonelli then turns convergence of the Taylor series just to the left of the abscissa into convergence of the defining integral there, a contradiction. Applying the same reasoning to $-H_m$ rules out an eventual upper bound. This proves both unbounded oscillations. $\square$

Theorem 6.1 follows by taking $m = 2$ and $v = t - 1$. There is also a useful warning about the order of quantifiers. At any fixed $v > 0$, the atom at 2 eventually dominates as m grows:

$$\sum_p (p+v)^{-m} \geq (v+2)^{-m}, \quad \int_2^\infty \frac{du}{(u+v)^m \log u} \leq \frac{(v+2)^{1-m}}{(m-1) \log 2}.$$

Thus $H_m(v) > 0$ whenever $(m-1) \log 2 > v+2$. Each fixed order can eventually have the RH sign; one cannot require that sign for every order on one common tail.

D.2 Counting twin pairs with negative directions

Proof of Theorem 6.2. The finite quotients are

$$F_y(t) = \prod_{2 < p \leq y} (1 - p^{-1})^3 \left(1 + \frac{3}{t + p - 1} \right).$$

Each factor is completely monotone on $t > -2$: its successive derivatives alternate in sign. The products converge locally uniformly near that real interval, because the terms of order $1/p$ in their logarithms cancel. Their derivatives converge too. Leibniz's rule proves complete monotonicity of the limit; separating the factor for $p = 3$ makes all the inequalities strict. This proves entrywise positivity.

The residue at the pole $1 - p$ is

$$b_p = \frac{K_o(4-p)}{K'_o(1-p)} = 3(1-p^{-1})^3 \prod_{\substack{q>2 \\ q \neq p}} (1-q^{-1})^3 \left(1 + \frac{3}{q-p}\right).$$

For fixed p , the tail factors are $1 + O_p(q^{-2})$. Among the factors with $q < p$, exactly one can be negative: the factor with $q = p - 2$, if that prime exists. No factor vanishes, since odd-prime differences cannot equal three. Hence

$$b_p < 0 \iff p - 2 \text{ is prime.} \quad (\text{D.5})$$

To pass from residues to derivatives we need a uniform bound. Let $b_{p,y}$ be the finite-product residue. For $q < p$, each factor in absolute value is at most one. For $p < q \leq 2p$, comparison with all even distances gives

$$\prod_{p < q \leq 2p} \left(1 + \frac{3}{q-p}\right) \leq \prod_{1 \leq j \leq p/2} \left(1 + \frac{3}{2j}\right) \ll p^{3/2}.$$

For $q > 2p$, the logarithm of the combined factor is at most

$$-\frac{3}{q} + \frac{3}{q-p} \leq \frac{6p}{q^2}.$$

Even its sum over all integers is bounded. Thus, uniformly for $y \geq p$,

$$|b_{p,y}| \leq Cp^{3/2}.$$

Differentiate the finite partial fractions at least twice and use dominated convergence:

$$\frac{(-1)^k F^{(k)}(a)}{k!} = \sum_{p>2} \frac{b_p}{(a+p-1)^{k+1}}, \quad k \geq 2. \quad (\text{D.6})$$

The absolute majorant for $k = 2$ is already bounded by $C \sum_{n \geq 1} n^{-3/2}$. This explains why the matrix starts with the second derivative.

Define the finite signed measure

$$\sigma_a = \sum_{p>2} \frac{b_p}{(a+p-1)^3} \delta_{1/(a+p-1)}.$$

Its support is compact, with its only possible accumulation point at zero. Equation (D.6) identifies the matrix quadratic form as

$$c^T H_N(a) c = \int \left(\sum_{j=0}^{N-1} c_j u^j \right)^2 d\sigma_a(u).$$

If σ_a has J negative atoms, the negative part of this form has rank at most J , giving the upper bound on its negative index.

Conversely, choose any k negative atoms. They are isolated, so disjoint continuous bump functions can select them and vanish at all other support points. Normalize the functions so that their signed Gram matrix is $-I_k$. Uniform polynomial approximation preserves this strict negative definiteness, because σ_a has finite total variation. The resulting polynomials have some common finite degree and span a k -dimensional negative subspace of a finite matrix. This proves the reverse bound, whether there are finitely or infinitely many negative atoms. Finally use (D.5) and the fact that successive matrices are nested principal submatrices. $\square$

E Why the recovery threshold is sharp

The construction has three stages. Estimate the common error rate; denoise the table using its finite-prime approximations; then turn approximate frequencies into exact rationals. The last stage can be very slow. The theorem asserts consistency and gives no useful practical sample bound for the particular implementation below.

E.1 Calibrating the noise without knowing its sign

Write J for the all-ones matrix and I_N for the identity. All data matrices have zero diagonal. Set

$$\hat{q} = \frac{1}{N(N-1)} \sum_{i \neq j} Y_{ij}, \quad \hat{a} = \text{clip}_{[-1,1]} \left(\frac{2\hat{q} - 1}{2\kappa - 1} \right).$$

Here $\text{clip}_{[u,v]}$ means projection onto $[u, v]$. If q_0 is the clean empirical edge density, conditioning on E gives

$$\hat{q} = \frac{1}{2} + a_N(q_0 - \frac{1}{2}) + O_{\mathbb{P}}(N^{-1}).$$

Changing one integer changes q_0 by at most $2/N$, whence $q_0 - \mathbb{E}q_0 = O_{\mathbb{P}}(N^{-1/2})$. With $X = N^2$, Möbius inversion gives

$$\mathbb{E}q_0 = \frac{1}{X^2} \sum_{d \leq X} \mu(d) [X/d]^2 = \kappa + O(\log X/X). \quad (\text{E.1})$$

Replacing squared floors costs $O(X^{-1} \sum_{d \leq X} d^{-1})$; the omitted tail of $\sum_d \mu(d)/d^2$ costs $O(X^{-1})$. Thus, for nonzero a_N , clipping cannot increase the absolute error and

$$\frac{\hat{a}}{a_N} - 1 = O_{\mathbb{P}} \left(N^{-1/2} + \frac{1}{N|a_N|} + \frac{\log N}{N^2} \right). \quad (\text{E.2})$$

When $T_N \rightarrow \infty$, this tends to zero, the sign is eventually correct with probability tending to one, and $N\hat{a}^2/T_N \rightarrow 1$ in probability. In the independent-prime model the finite-range bias in (E.1) is absent.

E.2 The matrix estimate

For a matrix U , denote its operator and Frobenius norms by $\|U\|_{\text{op}}$ and $\|U\|_F$. We record the exact deterministic estimate needed, so no uniformity condition is hidden in an external denoising theorem.

Lemma E.1 (Spectral truncation). Suppose $B = A + Z$ are symmetric and $\|Z\|_{\text{op}} \leq t/2$. Retain the eigenvalues of B with absolute value at least t , with their signs, to obtain $\hat{A}$. For every matrix R of rank at most r ,

$$\|\hat{A} - A\|_F^2 \leq C(rt^2 + \|A - R\|_F^2)$$

for an absolute constant C .

Proof. Let s_i be the decreasing singular values of A and let r_0 count those at least $t/2$. Weyl's inequality bounds the rank of $\hat{A}$ by r_0 . If A_0 is the rank- r_0 truncation of A , then

$$\|\hat{A} - A_0\|_{\text{op}} \leq \|\hat{A} - B\|_{\text{op}} + \|Z\|_{\text{op}} + \|A - A_0\|_{\text{op}} \leq 2t.$$

The difference has rank at most $2r_0$, giving $\|\hat{A} - A\|_F^2 \leq 16r_0t^2 + 2\sum_{i>r_0}s_i^2$. For every r ,

$$r_0t^2 \leq rt^2 + 4\sum_{i>r}s_i^2, \quad \sum_{i>r_0}s_i^2 \leq \sum_{i>r}s_i^2 + rt^2/4.$$

The best-rank approximation inequality $\sum_{i>r}s_i^2 \leq \|A - R\|_F^2$ completes the proof. $\square$

Apply this to

$$B = Y - \frac{1}{2}(J - I_N), \quad A = a_N M, \quad M = E - \frac{1}{2}(J - I_N),$$

with threshold $t = 10\sqrt{N}$. Conditional on the integers, the entries of $Z = B - A$ above the diagonal are independent, centered, and supported on intervals of length one. A $1/4$ -net of the unit sphere has at most 9^N points. For a fixed unit vector v , Hoeffding's inequality gives $\mathbb{P}(|v^T Z v| > u \mid X_1, \dots, X_N) \leq 2e^{-u^2}$, because the sum of squared interval lengths in $2\sum_{i<j} v_i v_j Z_{ij}$ is at most two. The net inequality therefore yields

$$\mathbb{P}(\|Z\|_{\text{op}} > 4\sqrt{N} \mid X_1, \dots, X_N) \leq 2\exp(-(4 - \log 9)N). \quad (\text{E.3})$$

The arithmetic input is a low-rank approximation. Let $F_{ij}^{(m)}$, including the diagonal, indicate that i, j share none of the first m primes. There are only 2^m truncated vertex profiles, so

$$\text{rank}(F^{(m)} - J/2) \leq 2^m + 1.$$

Off the diagonal this differs from M only if a larger prime is shared. The probability of sharing p is at most p^{-2} , also for ordinary integers, and $p_m \geq m + 1$. Consequently

$$\mathbb{E} \frac{\|M - (F^{(m)} - J/2)\|_F^2}{N^2} \leq \frac{1}{m} + \frac{1}{4N}. \quad (\text{E.4})$$

The last term accounts for the diagonal, whose approximation error is exactly $1/2$ in absolute value.

For the proof choose $m = \lfloor \frac{1}{2} \log_2 T_N \rfloor$. This choice is not used by the algorithm. Lemma E.1, (E.3), (E.4), and Markov's inequality give

$$\frac{\|\hat{A} - a_N M\|_F^2}{a_N^2 N^2} = O_{\mathbb{P}} \left(T_N^{-1/2} + \frac{1}{\log T_N} \right).$$

For $i \neq j$, define $\hat{E}_{ij} = \text{clip}_{[0,1]}(1/2 + \hat{A}_{ij}/\hat{a})$, and set the diagonal to zero. At $\hat{a} = 0$ use an arbitrary fixed table. Equation (E.2) and clipping show

$$Q_N := \frac{1}{N^2} \|\hat{E} - E\|_F^2 = O_{\mathbb{P}}((\log T_N)^{-1}). \quad (\text{E.5})$$

E.3 Approximate frequencies become exact primes

The algorithm commutes with vertex permutations. Thus (E.5) controls every fixed row in probability, even though it is initially a statement about the whole matrix. To see the rate carefully, let $G_C = \{Q_N \leq C/\log T_N\}$. This event is permutation invariant. On it, the expected squared error of any fixed row, averaged across its columns and multiplied by $\mathbf{1}_{G_C}$, is $\mathbb{E}[Q_N \mathbf{1}_{G_C}] \leq C/\log T_N$. Choose C to make G_C likely, then use Markov's inequality on G_C . Hence for fixed i ,

$$\frac{1}{N} \sum_k (\hat{E}_{ik} - E_{ik})^2 = O_{\mathbb{P}}((\log T_N)^{-1}). \quad (\text{E.6})$$

Form $\hat{d}_i = N^{-1} \sum_k \hat{E}_{ik}$ and $\hat{c}_{ij} = N^{-1} \sum_k \hat{E}_{ik} \hat{E}_{jk}$. Cauchy-Schwarz bounds their denoising errors by $O_{\mathbb{P}}((\log T_N)^{-1/2})$. For the second assertion use

$$|uv - u'v'| \leq |u - u'| + |v - v'| \quad (0 \leq u, v, u', v' \leq 1).$$

For ordinary integers put $D(n) = \varphi(n)/n$ and $L_{ij} = \text{lcm}(X_i, X_j)$. Inclusion-exclusion gives

$$|\mathbb{P}_{U \leq X}((U, n) = 1) - D(n)| \leq \frac{2^{\omega(n)}}{X}. \quad (\text{E.7})$$

For every fixed $\delta > 0$, $2^{\omega(n)} \leq C_\delta n^\delta$: in the product $2^{\omega(n)}/n^\delta$ only finitely many primes can give a factor above one. Take $\delta = 1/4$ and note $L_{ij} \leq X^2$, $X = N^2$. The bias in (E.7) is uniformly $O(N^{-1})$. Conditional on X_i, X_j , the other vertices are independent, so their empirical clean degree and codegree errors are $O_{\mathbb{P}}(N^{-1/2})$. Their targets are $D(X_i), D(X_j), D(L_{ij})$.

Division by the last target is controlled uniformly, since

$$\mathbb{E}[-\log D(L_{ij})] \leq 2 \sum_p \frac{-\log(1 - 1/p)}{p} < \infty. \quad (\text{E.8})$$

Thus $D(L_{ij})^{-1} = O_{\mathbb{P}}(1)$. The corresponding infinite-model products satisfy the same estimate. With clipping and a default at a zero estimated denominator, we have in both models

$$\hat{r}_{ij} := \text{clip}_{[0,1]} \left(\frac{\hat{d}_i \hat{d}_j}{\hat{c}_{ij}} \right) = \frac{\varphi(R_{ij})}{R_{ij}} + O_{\mathbb{P}}((\log T_N)^{-1/2}). \quad (\text{E.9})$$

Set $\hat{T} = N\hat{a}^2$ and

$$H_N = \max \left\{ 1, \left\lfloor [\log(\max\{e, \hat{T}\})]^{1/16} \right\rfloor \right\}.$$

Choose the nearest reduced rational to $\hat{r}_{ij}$ in $[0, 1]$ with denominator at most H_N , using a fixed rule for ties. If it is a finite totient ratio, apply Lemma 7.1; otherwise return a fixed default. This is a finite test: reject zero, reject a largest denominator prime of exponent exceeding one, remove that prime as prescribed, and reject if the ratio leaves $(0, 1]$ or the largest denominator prime fails to decrease. Accept only on reaching one.

The true denominator is tight. For ordinary integers it is at most $\text{gcd}(X_i, X_j)$ and, for $H \geq 1$,

$$\mathbb{P}(\text{gcd}(X_i, X_j) > H) \leq \sum_{d>H} \mathbb{P}(d \mid X_i, d \mid X_j) \leq \sum_{d>H} d^{-2} = O(H^{-1}). \quad (\text{E.10})$$

In the independent-prime model R_{ij} is a finite random integer, which gives the same tightness requirement without a rate. Calibration implies $H_N \asymp (\log T_N)^{1/16}$ in probability. Distinct candidate rationals differ by at least H_N^{-2} , while

$$H_N^2 \left| \hat{r}_{ij} - \frac{\varphi(R_{ij})}{R_{ij}} \right| = O_{\mathbb{P}}((\log T_N)^{-3/8}) \longrightarrow 0.$$

The nearest rational is therefore correct with probability tending to one. Permutation equivariance makes this the expected fraction of incorrect pair outputs; Markov's inequality proves the fraction assertion in Theorem 8.1.

E.4 A single hidden parity bit proves necessity

Suppose along a subsequence that $T_N \leq C$, so eventually $|a_N| \leq 1/2$. Giving a decoder additional information can only help it. We give almost everything away and retain one fair bit that changes whether a pair shares the prime 2.

With $X = N^2$, the disjoint pairs $\{u, 2u\}$, $u \leq X/2$ odd, cover a fraction $1/2 + o(1)$ of the possible values of X_i . On this event reveal the odd base u , leaving a fair choice between u and $2u$. Reveal every other integer and restrict also to X_j being even. This joint event has probability $1/4 + o(1)$. The correct R_{ij} contains 2 precisely in the second case.

Only the at most $N - 1$ edges incident to i can distinguish the cases. Each differing observation exchanges Bernoulli parameters $(1 - a_N)/2$ and $(1 + a_N)/2$, whose relative entropy is

$$a_N \log \frac{1 + a_N}{1 - a_N} \leq 4a_N^2 \quad (|a_N| \leq 1/2).$$

Conditional independence bounds the total relative entropy by $4T_N$. For two laws P, Q with equal priors, the testing error is at least $e^{-D(P\|Q)}/4$. Indeed, their Hellinger affinity ρ satisfies $\rho \geq e^{-D(P\|Q)/2}$ by Jensen, while $\text{TV}(P, Q) \leq \sqrt{1 - \rho^2}$ by Cauchy-Schwarz. The error $(1 - \text{TV})/2$ is consequently at least $\rho^2/4$. Every pair estimator thus has error at least

$$(1/16 + o(1))e^{-4T_N}. \quad (\text{E.11})$$

In the independent-prime model, hide only $B_{i,2}$ and require $B_{j,2} = 1$; the same argument gives $e^{-4T_N}/8$. The bounds are uniform in the choice of pair. Their positive lower bound on a bounded- T_N subsequence also bounds the expected pair-error fraction. A random variable in $[0, 1]$ tending to zero in probability must have expectation tending to zero. Necessity follows.

F Counting the information that survives

We prove Theorem 8.3 directly for ordinary integers. This avoids approximating a growing array of divisibility bits by independent prime bits. The upper bound asks how much information small primes can carry and how much the remaining primes can transmit through the noise. The lower bound temporarily reveals a small set of reference integers and uses them to identify many prime features of the others.

F.1 The prime sums and the conditioning principle

Write $h(t) = -t \log t - (1 - t) \log(1 - t)$, with its endpoint values defined by continuity, and put

$$S(y) = \sum_{p \leq y} h(1/p), \quad \rho_y = \sum_{p > y} p^{-2}.$$

The classical prime estimates give, for $y \geq 2$,

$$\begin{aligned} S(y) &= \log y + \log \log y + O(1), \\ \sum_{p \leq y} \frac{\log p}{p} &= \log y + O(1), \quad \sum_{p \leq y} \log p = O(y), \quad \rho_y \ll \frac{1}{y \log y}. \end{aligned} \quad (\text{F.1})$$

The first follows by expanding $h(1/p) = (\log p)/p + 1/p + O(p^{-2})$ and using Mertens' sums. The last follows by partial summation from $\pi(y) \ll y/\log y$. These are classical estimates, with stronger explicit versions in Rosser and Schoenfeld [34]; see also Mertens [26] for the prime sums.

We shall use one elementary information inequality repeatedly. If A and Z are any data determined by the hidden integers, then the channel noise is independent of them given $G = G_N$, so

$$\begin{aligned} I(G; Y) &= I(G, A; Y) = I(A; Y) + I(G; Y | A) \\ &\geq I(G; Y | A) \geq I(Z; Y | A). \end{aligned} \quad (\text{F.2})$$

The last step is conditional data processing. Thus reference values may be revealed *in a proof of a lower bound* without claiming they were given to the actual decoder. This particular inequality depends on the conditional independence of the observation channel.

F.2 Upper bound: truncate the prime features

Reveal the array B of prime-divisibility bits for all vertices and all $p \leq y$. The bits of one integer are dependent, but entropy subadditivity still gives

$$H(B) \leq N \sum_{p \leq y} h\left(\frac{|X/p|}{X}\right) \leq NS(y), \quad X = N^2. \quad (\text{F.3})$$

Here h is increasing on $[0, 1/2]$ and $|X/p|/X \leq 1/p \leq 1/2$.

Given B , let F_{ij} be the answer that uses only primes at most y . If $F_{ij} = 0$, the clean answer is determined. If $F_{ij} = 1$, the clean answer differs only if a prime above y divides both hidden integers. The average probability of this event is at most ρ_y .

First assume $|a_N| \leq 1/2$. The divergence between the two output laws for an edge is at most $4a_N^2$, as in the testing proof. For a binary input that differs from a fixed baseline with probability r , its mutual information through this channel is at most $4a_N^2 r$. To verify this, compare the two output distributions with the baseline output law: the average divergence is rD , and subtracting the nonnegative divergence of the mixture from the baseline gives mutual information. Conditional entropy subadditivity across edges, together with independent noise given G , now yields

$$I(G; Y) \leq H(B) + I(G; Y | B) \leq NS(y) + 4a_N^2 \binom{N}{2} \rho_y. \quad (\text{F.4})$$

Only the averaged residual probability is used here; independence of prime divisibility for ordinary integers is unnecessary.

If $T_N \geq 2$, choose $y = T_N$. Equations (F.1) and (F.4) give

$$I(G; Y) \leq N \log T_N + N \log \log T_N + O(N).$$

For bounded T_N , choosing $y = 2$ instead gives $I(G; Y) = O(N)$, uniformly on each bounded range of T_N .

If $|a_N| > 1/2$, then $T_N > N/4$. We use a clean entropy bound. Reveal prime bits through N and then record which edges differ from the truncated graph. Entropy subadditivity and concavity of h give

$$H(G) \leq NS(N) + \binom{N}{2} h(\rho_N) \leq N \log N + N \log \log N + O(N), \quad (\text{F.5})$$

because $h(\rho_N) = O(N^{-1})$. Since $I(G; Y) \leq H(G)$ and $\log(1 + T_N) = \log N + O(1)$ in this regime, these estimates prove the upper half of (8.3) for every noise rate.

F.3 Lower bound: reference integers reveal many small primes

Put $L = \log N$, $m = \lfloor N/L \rfloor$, and

$$y = T_N/L^{12}. \quad (\text{F.6})$$

If $y < 2$, then $\log(1 + T_N) = O(\log \log N)$, and the required lower bound follows from $I(G; Y) \geq 0$. Suppose henceforth $y \geq 2$. Reveal the actual values of the first m sampled integers, calling this reference data A . The remaining $N - m$ vertices are targets.

For a target value n , call it good when $2^{\omega(n)} \leq L^3$. The divisor identity $2^{\omega(n)} = \sum_{d|n} \mu(d)^2$ gives

$$\frac{1}{X} \sum_{n \leq X} 2^{\omega(n)} \leq \sum_{d \leq X} \frac{1}{d} \leq 1 + \log X.$$

Thus a uniformly sampled target is bad with probability $O(L^{-2})$. For each prime $p \leq y$, use all reference integers divisible by p . Let their number be L_p . It is binomial with mean at least $m/(2p)$. A Chernoff bound gives

$$\mathbb{P}(L_p < m/(4p)) \leq e^{-cm/p} \quad (\text{F.7})$$

for an absolute $c > 0$ and all sufficiently large N .

If $p \mid n$, every clean answer from this target to these references is zero. If $p \nmid n$ and n is good, inclusion–exclusion gives

$$\begin{aligned} \#\{b \leq X : p \mid b, (b, n) = 1\} &= \frac{X}{p} D(n) + O(2^{\omega(n)}) \\ &\geq \frac{X}{pL^3} - L^3 \geq \frac{X}{2pL^3}. \end{aligned} \quad (\text{F.8})$$

We used $D(n) \geq 2^{-\omega(n)}$ and $X/p \geq NL^{12}$, since $p \leq T_N/L^{12} \leq N/L^{12}$. Consequently a random reference conditioned to be divisible by p is coprime to this good n with probability at least $1/(2L^3)$.

Conditional on n and on which reference indices are divisible by p , their values are independent and uniform over the multiples of p in $[1, X]$. Their noisy answers are therefore independent. When $p \mid n$ their mean is ε_N ; otherwise it is $\varepsilon_N + a_N \theta$, with $\theta \geq 1/(2L^3)$ on good targets. Use the threshold $\varepsilon_N + a_N/(4L^3)$, reversing the comparison when $a_N < 0$. The parameter is fixed in the probability law and may be used by this auxiliary estimator for the information lower bound. The reconstruction algorithm in Theorem 8.1 does not require it.

On the event in (F.7), Hoeffding's inequality bounds the bit error on a good target by

$$2 \exp\left(-c \frac{a_N^2 L_p}{L^6}\right) \leq 2 \exp\left(-c \frac{T_N}{pL^7}\right) \leq 2e^{-cL^5}. \quad (\text{F.9})$$

The count failure probability is smaller after modifying c . The independence calculation conditions on the reference divisibility indicators, *not* on their full revealed values. We average over those values to obtain the displayed error bound. The resulting estimator is nevertheless a function of (Y, A) , as required in (F.2).

Let $D_i^* = \prod_{p \leq y, p | X_i} p$ be a target's truncated squarefree mask. Union-bound over the at most N candidate primes. The probability of failing to recover the whole mask is

$$e_N = O(L^{-2}) + O(Ne^{-cL^5}). \quad (\text{F.10})$$

There are at most X possible masks, since each is an integer divisor of some number at most X . Replace an impossible estimate by 1. The usual error-flag proof of Fano's inequality gives

$$H(D_i^* | Y, A) \leq h(e_N) + e_N \log X = O(L^{-1}). \quad (\text{F.11})$$

For completeness, condition additionally on the indicator that the estimated mask is wrong. On a correct estimate there is no uncertainty; on an incorrect one there are at most X choices. The entropy of the indicator is at most $h(e_N)$.

There is a simple lower bound for the mask's own entropy. For every possible value d , $\mathbb{P}(D_i^* = d) \leq \mathbb{P}(d | X_i) \leq 1/d$. Therefore, using (F.1),

$$\begin{aligned} H(D_i^*) &\geq \mathbb{E} \log D_i^* = \sum_{p \leq y} \log p \frac{|X/p|}{X} \\ &\geq \sum_{p \leq y} \frac{\log p}{p} - \frac{1}{X} \sum_{p \leq y} \log p \geq \log y - O(1). \end{aligned} \quad (\text{F.12})$$

The target masks are mutually independent and independent of A . Apply (F.2) to their collection and subtract their conditional entropy using (F.11). We obtain

$$\begin{aligned} I(G; Y) &\geq (N - m)(\log y - O(1)) - O(N/L) \\ &\geq N \log T_N - 12N \log \log N - O(N). \end{aligned} \quad (\text{F.13})$$

Here $m \log y = O(N)$ because $y \leq N$. Replacing T_N by $1 + T_N$ costs $O(N)$ when $T_N \geq 1$; for smaller T_N the lower bound was already trivial. This proves (8.3). Finally take $\varepsilon_N = 0$, so $Y = G$ and $T_N = N$, to obtain (8.4). This derivation of the clean entropy uses no earlier graph-entropy theorem.

The proof shows why the leading scale is logarithmic in T_N . The entropy in prime bits through y is roughly $\log y$ per vertex, and reference testing accesses these bits up to T_N with only logarithmic losses. The coefficient comes from arithmetic prime sums. General entropy behavior of random-free graph models is studied by Hatami and Norine [18]; graph mutual information and signal-dependent information laws have broader precedents in Skeja–Olhede [37] and Deshpande–Abbe–Montanari [8]. The assertion here is the specific finite, labelled coprimality law.

G Proof of reconstruction without digit addresses

Proof of Theorem 9.1. By Corollary 8.2, for each fixed j we can form an estimated lane $\hat{L}_j$ differing from $L_j = \{i : \text{lpf}(X_i) = p_j\}$ at $o_{\mathbb{P}}(N)$ vertices. The exceptional value $X_i = 1$ belongs to no lane. Ordinary residue counting and the law of large numbers give

$$\frac{|\hat{L}_j|}{N} \longrightarrow \delta_j = \frac{1}{p_j} \prod_{q < p_j} (1 - 1/q) > 0. \quad (\text{G.1})$$

Use

$$\hat{h}_j = \frac{1}{|\hat{L}_j|} \sum_{i \in \hat{L}_j} \frac{1 - 2Z_i}{b_N}, \quad (\text{G.2})$$

with a fixed default when the denominator vanishes.

Conditional on the integers and edge noise, the reconstructed lane is fixed and the digit errors remain independent. Each summand in (G.2) has conditional mean $1 - 2x_g(X_i)$, whose absolute value is at most one, and conditional variance at most b_N^{-2} . Equations (G.1) and $Nb_N^2 \rightarrow \infty$ make the centered error $o_{\mathbb{P}}(1)$. Replacing the estimated lane by the true lane in the *conditional mean* costs only $o_{\mathbb{P}}(1)$. No division of this replacement error by $|b_N|$ is needed.

The average Liouville sign in each fixed lane tends to zero. To spell out the classical input, write $Q_j = \prod_{q < p_j} q$ and $L(t) = \sum_{n \leq t} \lambda(n) = o(t)$, the prime number theorem in its Liouville form. Inclusion–exclusion and complete multiplicativity give

$$\sum_{m \leq t, (m, Q_j)=1} \lambda(m) = \sum_{d|Q_j} \mu(d) \lambda(d) L(t/d) = \sum_{d|Q_j} L(t/d) = o(t).$$

The least-prime lane consists of $p_j m$ with $(m, Q_j) = 1$; multiplication by p_j reverses the sign. This proves that its finite-range mean of $1 - 2x_g(n)$ tends to $1 - g_j$. The independently sampled lane average has the same limit by bounded-variance concentration and (G.1). Thus $\hat{h}_j \rightarrow 1 - g_j$ in probability. Rounding at $1/2$ recovers the bit. A union bound recovers any fixed finite prefix of mask bits.

Define, with arbitrary binary defaults where needed,

$$\hat{C}_N = \sum_{n=2}^N \frac{1 - \lambda(n)}{2} \hat{g}_{\ell(n), N} 2^{-n}.$$

For every fixed M , the first M digits are correct with probability tending to one; the remaining tail is at most 2^{-M} . First taking $N \rightarrow \infty$ and then $M \rightarrow \infty$ proves convergence to C_g .

For necessity of the digit condition, compare two fixed masks differing only at some $j \geq 2$. Supply an oracle with the exact integers, every other mask bit, and both noise probabilities. The two hypotheses differ at at most N digit observations. On a subsequence where Nb_N^2 is bounded, their total conditional relative entropy is at most $4Nb_N^2$ eventually. The testing bound used in (E.11) leaves positive error under at least one mask. Thus one procedure cannot be consistent for both.

For the edge condition, use the all-one mask. The earlier pair $u, 2u$ has opposite Liouville signs, so it would be distinguished by a clean digit label. Instead use the disjoint pairs $\{u, 4u\}$ with u odd and $u \leq N^2/4$. Their union has probability $1/4 + o(1)$, and the two values have exactly the same Liouville sign. Reveal the base and every other hidden integer and require the partner to be even. This event has probability $1/8 + o(1)$ and leaves a fair bit that changes membership of 2 in R_{ij} but changes no digit observation law. The incident edges again have total divergence at most $4Na_N^2$. On a bounded- Na_N^2 subsequence the fixed-pair error is bounded away from zero, uniformly over pairs, which also excludes a vanishing error fraction. This proves the joint necessity statement. $\square$

H Proofs for selections, crowds, entropy and deletion

The main text keeps the short prime-resource arguments beside their conclusions. This appendix supplies the quantitative estimates and limit transitions used there.

H.1 The binomial approximation

Proof of Corollary 10.2. Write $M_m = \max_j \mathbb{P}(\text{Binomial}(m, \theta) = j)$. Unimodality gives $d_{\text{TV}}(Z + a, Z + b) \leq |a - b|M_m$ for integer shifts. Couple R with an independent $\text{Binomial}(d_B, \theta)$ and retain the common independent Z . The distance is at most $d_B M_{m_B} = O_\theta(d_B/\sqrt{m_B})$. The prime number theorem gives $r_B \leq \pi(\sqrt{X}) = O(\sqrt{X}/\log X)$ and $N_B \sim X/\log X$, proving the bound. The other assertions follow from the polynomial factorization. $\square$

H.2 The compulsory part of a random selection

We use Buchstab's function ω : it satisfies $u\omega(u) = 1$ for $1 \leq u \leq 2$ and $(u\omega(u))' = \omega(u - 1)$ for $u > 2$. The classical rough-number estimate is

$$\Phi(x, y) := \#\{n \leq x : \text{lpf}(n) > y\} \sim \omega(u) \frac{x}{\log y}, \quad u = \frac{\log x}{\log y} > 1 \quad (\text{H.1})$$

when u stays near a fixed number greater than one; 1 is counted. We also use $\Phi(x, y) \ll x/\log y$ for $x \geq y \geq 2$. These forms follow, for example, from the estimates reviewed and made explicit in [12].

Proof of Theorem 10.3. Put $h = \log N$, $X = \lfloor N^\beta \rfloor$, and let R count sampled integers with no prime divisor at most N . By (H.1) and binomial concentration, $R \sim \omega(\beta)N/h$ in probability.

A rough label fails to belong to $\mathcal{U}_N$ only by sharing a prime $p > N$ with another label. If r_p is the probability that the first label is rough and divisible by p , the expected number of such failures is at most $N^2 \sum_{p > N} r_p/p$. For $p \leq X/N$, the rough-number bound gives $r_p \ll 1/(p \log N)$; for $p > X/N$, the only rough cofactor is 1, so $r_p = 1/X$. Thus the expected number is

$$O_\beta \left(\frac{N}{h^2} + \frac{N^2}{X} \right) = o(N/h).$$

Here $\sum_{p > N} p^{-2} \ll 1/(N \log N)$ and $\sum_{N < p \leq X} p^{-1} = O_\beta(1)$. It follows that $|\mathcal{U}_N| \geq R - o_\mathbb{P}(N/h)$.

For the matching upper bound, put $y = N/h^2$. In any coprime selection, each non- y -rough member uses a distinct prime at most y . Its size is therefore at most $R_y + \pi(y)$, where R_y is the number of y -rough sampled labels. Again (H.1) and binomial concentration give $R_y \sim \omega(\beta)N/h$, while $\pi(y) = o(N/h)$. Since $|\mathcal{U}_N| \leq \alpha_N$, both asymptotics follow.

Finally every maximal selection contains $\mathcal{U}_N$, so

$$|S \triangle T| \leq 2(\alpha_N - |\mathcal{U}_N|).$$

A greedy scan always produces a maximal selection. All conclusions hold simultaneously over its possible orders. $\square$

H.3 Crowds, missing primes and repeated powers

Proof of Proposition 10.4. In the conditioned experiment,

$$\mathbb{P}(L_n > y) = \prod_{p \leq y} \frac{n}{n + p - 1}.$$

For $y = t\sqrt{n \log n}$, the negative logarithm equals $n^{-1} \sum_{p \leq y} (p-1) + O(n^{-2} \sum_{p \leq y} p^2) \rightarrow t^2$ by the prime number theorem.

In the ordinary experiment the survival probability is $\prod_{p \leq y} (1 - (1 - 1/p)^n)$. Put $a = \log n - 3 \log \log n + t$ and $y = n/a$. The quantitative prime number theorem and partial summation give

$$\sum_{p \leq y} (1 - 1/p)^n \sim \int_2^y \frac{e^{-n/u}}{\log u} du = n \int_a^{n/2} \frac{e^{-v}}{v^2 \log(n/v)} dv \sim \frac{ne^{-a}}{a^2 \log(n/a)} \rightarrow e^{-t}.$$

Primes below $y/2$ contribute $o(1)$; on $[y/2, y]$ replacement by the integral is valid even on its effective width $y/\log n$, using the standard prime-number-theorem error $O(ye^{-c\sqrt{\log y}})$. The largest summand tends to zero, so the product tends to $e^{-e^{-t}}$. This is the asserted law after the change $y = n/a$. $\square$

The following local calculation also proves the remainder law (10.3). To verify it, the excess exponent at p has the representation $B_p G_p$, where B_p is Bernoulli with parameter $n/(n+p-1)$ and G_p is geometric with $\mathbb{P}(G_p = j) = (1 - 1/p)p^{-j}$, independently. The nonzero excesses occur at only finitely many primes almost surely. Taking the ratio of the mass at j to the mass at zero gives $(n/(n+p))p^{-j}$ and proves the formula.

Proof of Corollary 10.6. Couple the preceding excess exponents with $H_n = \prod_{p \leq n} p^{G_p}$. The chance that any exponent differs is at most $\pi(n)/n + n \sum_{p > n} p^{-2} = O(1/\log n)$. For H_n , logarithmic Laplace transforms expand into

$$\sum_{p \leq n} \sum_{j \geq 1} \frac{p^{-j}}{j} (e^{-sj \log p / \log n} - 1).$$

The terms $j \geq 2$ vanish by dominated convergence. Mertens' theorem turns the $j = 1$ terms into the asserted integral.

The limit is the sum of a Poisson process on $(0, 1]$ with intensity dt/t . Its largest point is uniform, so $Y \stackrel{d}{=} U(1 + Y')$, with U uniform and Y' an independent copy. Its density equals $\mathbb{E}[(1 + Y')^{-1} \mathbf{1}_{y < 1+Y'}]$, constant for $0 < y < 1$. The Laplace transform is asymptotic to $e^{-\gamma}/s$ as $s \rightarrow \infty$, identifying that constant as $e^{-\gamma}$. $\square$

H.4 Real roots and the averaged Wishart polynomial

For completeness, each prime acts on a polynomial by positive rescaling and $P \mapsto P + azP'$, $a > 0$. This preserves negative real roots: for $P(z) = \prod_j (z + r_j)$, $r_j > 0$, the imaginary part of $zP'(z)/P(z) = \sum_j z/(z + r_j)$ is strictly positive in the upper half-plane. The new polynomial has no roots there or in the lower half-plane, and its positive coefficients exclude nonnegative roots. Start with $(1 + z)^N$ and pass to the coefficient limit over primes.

Proof of Theorem 11.1. Cauchy–Binet expresses a principal minor of $Z_N Z_N^*$ of order j as a sum of $\binom{N}{j}$ squared Gaussian determinants, each with mean $j!$. Consequently, for the Laguerre polynomial $L_N(x) = \sum_k \binom{N}{k} (-x)^k / k!$,

$$\mathbb{E} \det(Z_N Z_N^* + tI) = N! L_N(-t).$$

Since $\mathbb{E} W_c^k = k! K(k)$, averaging at $t = W_c z$ gives $N! J_N(z)$. Power biasing by W_c^N proves the characteristic-polynomial identity.

Put $\kappa(t) = -\log K(t)$. Differentiating its Euler product and using the prime number theorem and Mertens' product theorem gives

$$\kappa'(t) = \log \log t + \gamma + o(1), \quad \kappa''(t) \sim \frac{1}{t \log t}.$$

For the second estimate, substitute $u = tv$ in $\int_2^\infty (u+t-1)^{-2} d\pi(u)$; for the first, split the logarithmic derivative at $p = t$. The correction to $-\sum_{p \leq t} \log(1 - 1/p)$ is $O(1/\log t)$ by partial summation. Write $b_N = K(N-1)/K(N) \sim a_N$. Coefficient comparison yields

$$\frac{1}{N} \sum_j R_{N,j} = b_N, \quad \frac{1}{N} \sum_j (R_{N,j} - b_N)^2 = (N-1) \left(b_N^2 - \frac{K(N-2)}{K(N)} \right) \sim \frac{b_N^2}{N}.$$

The last equivalence uses the second difference of κ , asymptotic to $1/(N \log N)$. Chebyshev's inequality gives the root limit.

The biased scalar satisfies

$$\mathbb{E} W_{c,N} = (N+1) \frac{K(N+1)}{K(N)} \sim e^{-\gamma} N / \log N, \quad \frac{\text{Var}(W_{c,N})}{(\mathbb{E} W_{c,N})^2} = \frac{N+2}{N+1} \frac{K(N+2)K(N)}{K(N+1)^2} - 1 \sim \frac{1}{N}.$$

It therefore concentrates at that scale. Apply the square Marchenko–Pastur theorem [23] to $Z_N Z_N^*/N$ and multiply by $N/(W_{c,N} a_N) \rightarrow 1$. $\square$

H.5 The prime-factor coupling and its moments

Write $P = (P_i)$ for the limiting partition, and set $H = -\sum_i P_i \log P_i$, $D = e^H$. The correlation formula [17, Theorem 2.1] is

$$\mathbb{E} \sum_{i_1, \dots, i_k \text{ distinct}} F(P_{i_1}, \dots, P_{i_k}) = \int_{\substack{u_j > 0 \\ \sum u_j < 1}} F(u_1, \dots, u_k) \prod_j \frac{du_j}{u_j}. \quad (\text{H.2})$$

One can see this formula directly from the gamma process: take Poisson jumps with intensity $e^{-v} dv/v$, divide by their exponential total, and use independence of that total and the normalized jumps. After choosing k jumps and writing the leftover total as $s(1 - \sum u_j)$, the Jacobian cancels their common power of s ; integrating $e^{-s} ds$ leaves (H.2). Taking $k = 1$ and $F(u) = u \log(1/u)$ also gives $\mathbb{E} H = \int_0^1 \log(1/u) du = 1$.

Lemma H.1 (Grouping repeated primes). For a uniform integer $U_N \in [1, N]$, let Q_N be its decreasing prime-power share sequence, with $Q_N = (1, 0, \dots)$ at $U_N = 1$. There is a coupling with P such that

$$\mathbb{E} \|Q_N - P\|_1 \ll (\log N)^{-1}, \quad \mathbb{E} |H(Q_N) - H(P)| \ll_\beta (\log N)^{-\beta} \quad (0 < \beta < 1).$$

Moreover $\sup_N \mathbb{E} e^{tH(Q_N)} < \infty$ for every fixed $t > 0$.

Proof. Haddad and Koukoulopoulos [16] couple the decreasing prime factors q_i with multiplicity to P so that $\mathbb{E} \sum_i |\log q_i - P_i \log N| = O(1)$. Changing the denominator from $\log N$ to $\log U_N$ costs $1 - \log U_N / \log N$ in ℓ^1 , with mean $O(1/\log N)$. The exceptional value $U_N = 1$ costs $O(1/N)$.

Grouping repeated occurrences of p changes the vector by at most twice their excess logarithmic mass

$$R(n) = \frac{1}{\log n} \sum_p \sum_{j \geq 2} (\log p) \mathbf{1}_{p^j | n}.$$

For $n \geq \sqrt{N}$, divisibility counting bounds its mean by $2(\log N)^{-1} \sum_p (\log p)/(p(p-1))$. The remaining integers contribute at most $N^{-1/2}$. Sorting minimizes matching distance and cannot increase this bound. This proves the first assertion.

We record a moment bound used twice. For $0 < \alpha < 1$ and a mass partition Q , write $S_\alpha(Q) = \sum_i Q_i^\alpha$. For every fixed positive integer r ,

$$\sup_N \mathbb{E} S_\alpha(Q_N)^r < \infty. \quad (\text{H.3})$$

On $n \geq \sqrt{N}$, bound S_α by $\sum_{p|n} (2 \log p / \log N)^\alpha$ and expand its r th power. A block of b equal primes has exponent sum

$$\sum_{j_1, \dots, j_b \geq 1} p^{-\max(j_1, \dots, j_b)} = \sum_{m \geq 1} \frac{m^b - (m-1)^b}{p^m} \ll_b \frac{1}{p}.$$

Then $\sum_{p \leq N} (\log p)^{\alpha b} / p \ll_{\alpha, b} (\log N)^{\alpha b}$ bounds each block, and there are only finitely many partitions of r occurrences. Integers below $\sqrt{N}$ contribute negligibly because $S_\alpha \leq (\log N / \log 2)^{1-\alpha}$.

For $h(u) = u \log(1/u)$, $\alpha = (1 - \beta)/2$, elementary differentiation and a split into $u \leq v/2$ and $v/2 < u \leq v$ give

$$|h(u) - h(v)| \ll_\beta |u - v|^\beta (u^\alpha + v^\alpha).$$

Hölder's inequality, first on coordinates and then on expectations, bounds the entropy difference by a constant times $(\mathbb{E} \|Q_N - P\|_1)^\beta$; the remaining factors are bounded by $\mathbb{E} S_{1/2}(Q_N)$ and $\mathbb{E} S_{1/2}(P) = 2$, the latter from (H.2). Finally Jensen's inequality gives $e^{(1-\alpha)H(Q)} \leq S_\alpha(Q)$, so (H.3) bounds every fixed exponential moment. $\square$

It follows that the limiting mean effective count is $\mathfrak{P} = \mathbb{E} e^H$, numerically about 2.9621. This is different from $e^{\mathbb{E} H} = e$. The following recursion makes that numerical constant reproducible. If $h_2(u) = -u \log u - (1-u) \log(1-u)$, stick breaking gives $H \stackrel{d}{=} h_2(U) + (1-U)H'$. For $a_r = \mathbb{E} H^r / r!$,

$$a_0 = 1, \quad a_r = \frac{r+1}{r} \sum_{j < r} a_j \int_0^1 \frac{h_2(u)^{r-j}}{(r-j)!} (1-u)^j du, \quad \mathfrak{P} = \sum_{r \geq 0} a_r.$$

All terms are nonnegative. Writing the entropy as a sum of stick-breaking contributions bounds $H \leq (\log 2)(1 + \sum_{j \geq 1} \prod_{\ell \leq j} (1 - U_\ell))$. The product terms are exponentials of a rate-one Poisson process's arrival times; its exponential formula gives $\mathbb{E} e^{tH} \leq 2^t \exp(2^t - 1)$. Thus truncating the last series at $r = R$ has error at most $2^{-R-1} \mathbb{E} e^{2H} < 81 2^{-R-1}$. Lemma H.1, with uniform integrability, also proves $\mathbb{E} D(U_N) \rightarrow \mathfrak{P}$.

H.6 The sharp entropy tail

Proof of Theorem 12.1. Let $M(t) = \mathbb{E} e^{tH}$ and $h(u) = u \log(1/u)$. Expanding a nonnegative product and using (H.2) gives, for every $a > 0$,

$$\log M(t) \leq a + I_t(a), \quad I_t(a) = \int_0^1 e^{-au} (e^{th(u)} - 1) \frac{du}{u}.$$

Here the simplex constraint was bounded by $e^{a(1-\sum u_j)}$. Put $L = \log t$, choose $b > 0$ from $b + \frac{1}{2} \log b + \frac{1}{2} \log(2\pi) = L$, and set $a = t(\log(t/b) - 1)$. Then

$$b = L - \frac{1}{2} \log L - \frac{1}{2} \log(2\pi) + o(1).$$

Under $z = tu/b$, the positive exponent becomes $b\chi(z)$ with $\chi(z) = z(1 - \log z)$, whose unique maximum is $\chi(1) = 1$ and whose second derivative there is -1 . Laplace's method yields

$$I_t(a) = e^b \sqrt{2\pi/b} (1 + o(1)) = \frac{t}{b} (1 + o(1)).$$

To control the endpoint, the cancellation in $e^{th(u)} - 1$ makes the integral over $0 < z < b^{-2}$ bounded. On $[b^{-2}, 1/2] \cup [2, e]$ the maximum of χ is strictly below one, and on $[e, t/b]$ the integral is $O(\log t)$. All these errors are $o(e^b/\sqrt{b})$. Consequently

$$\log M(t) \leq t(L - \log L - 1) + \frac{t}{L} \left(\frac{1}{2} \log L + 1 + \frac{1}{2} \log(2\pi) \right) + o(t/L).$$

Markov's inequality with $t = x \log x$ gives the required upper bound on the probability.

For the reverse bound write $X = \log x$ and choose

$$\eta = X^{-1/8}, \quad \epsilon = X^{-1/4}, \quad v = (1 - \eta)/X, \quad k = \lceil x \exp((1/2 + \eta)/X) \rceil.$$

In $\mathbb{R}^k$ consider the region

$$\mathcal{R} = \left\{ |z_i| \leq \epsilon, \quad -\frac{1}{2} \leq \sum_i z_i \leq 0, \quad (1 - \eta/4)kv \leq \sum_i z_i^2 \leq (1 + \eta/4)kv \right\}.$$

Set $u_i = (1 + z_i)/k$. Their total lies in $[1 - 1/(2k), 1]$. The unlisted mass is too small to contain another component in $[(1 - \epsilon)/k, (1 + \epsilon)/k]$. Thus the factorial count, divided by $k!$, is an event probability, rather than an overcount.

Taylor's inequality $(1 + z) \log(1 + z) \leq z + (\frac{1}{2} + C\epsilon)z^2$ shows that on this event

$$H \geq (1 - \frac{1}{2k}) \log k - (\frac{1}{2} + C\epsilon)(1 + \eta/4)v > \log x$$

for large x . Also $\prod_i (1 + z_i)^{-1} \geq 1$, so

$$\mathbb{P}(D > x) \geq \frac{\text{vol}(\mathcal{R})}{k!}.$$

We give the volume estimate to identify the constant. Take independent $N(0, v)$ variables, each conditioned to lie in $[-\epsilon, \epsilon]$. The conditioning costs $\exp(-o(k))$, changes the variance by a relative $o(\eta)$, and keeps the standardized third moments bounded. Berry-Esseen gives probability at least $c/\sqrt{kv}$ for their sum to lie in $[-1/2, 0]$: its $O(k^{-1/2})$ error is smaller than this slab probability. Chebyshev bounds failure of the quadratic shell by $O((k\eta^2)^{-1}) = o((kv)^{-1/2})$. Undoing the Gaussian density on the shell therefore gives

$$\log \text{vol}(\mathcal{R}) \geq \frac{k}{2} \log(2\pi ev) - o(k).$$

Stirling's formula now yields

$$-\log \mathbb{P}(D > x) \leq k \log k - k - \frac{k}{2} \log(2\pi ev) + o(k) = x \left(X + \frac{1}{2} \log X - 1 - \frac{1}{2} \log(2\pi) + o(1) \right).$$

This matches the first bound. □

H.7 The Gaussian profile and its transfer to integers

Proof of Theorem 12.2. Put $X = \log x$, $t = xX$, and use a, b from the preceding saddle calculation. Fix a continuous compactly supported function ψ , and write $T_x = \int \psi d\nu_x$. For fixed real s , the exponent

$$q(u) = tu \log(1/u) + sxu \psi(\sqrt{X}(xu - 1))$$

is nonnegative for large x . On the perturbation's support, $u = (1 + O(X^{-1/2}))/x$, so its first term is of order X^2 and dominates the bounded perturbation. The same nonnegative factorial expansion gives

$$\mathbb{E} e^{tH + sxT_x} \leq \exp(a + I_t(a) + \Delta_x(s)),$$

where

$$\Delta_x(s) = \int_0^1 e^{tu \log(1/u) - au} \left(e^{sxu \psi(\sqrt{X}(xu - 1))} - 1 \right) \frac{du}{u}.$$

Under $y = \sqrt{X}(xu - 1)$, uniformly on that fixed compact support, the exponential is $e^{b - y^2/2 + o(1)}$, and $du/u = (1 + o(1))dy/\sqrt{X}$. The saddle equation gives $e^b/(x\sqrt{X}) \rightarrow 1/\sqrt{2\pi}$. Hence, with Z standard Gaussian,

$$\frac{\Delta_x(s)}{x} \longrightarrow \Lambda_\psi(s) := \mathbb{E}(e^{s\psi(Z)} - 1).$$

The sharp lower and upper bounds just proved also give $\log \mathbb{P}(H > X) = a + I_t(a) - tX + o(x)$. For $s > 0$, exponential Markov therefore implies

$$\limsup \frac{1}{x} \log \mathbb{P}(T_x \geq \mathbb{E}\psi(Z) + \delta \mid H > X) \leq \Lambda_\psi(s) - s(\mathbb{E}\psi(Z) + \delta) < 0$$

when s is small enough. Negative s gives the lower deviation. Thus compactly supported continuous tests converge, with exponentially small conditional deviation probabilities. All measures have mass one and the limit has mass one, so this vague convergence is weak convergence. On each fixed standardized interval, $xP_i = 1 + O(X^{-1/2})$, allowing weights P_i to be replaced by $1/x$. This proves the count assertion. $\square$

Proof of Corollary 12.3. Let $L = \log N$, choose $\beta \in (1 - \epsilon/2, 1)$, and first take $\eta = (\log x)^{-2}$. Lemma H.1 bounds the probability of entropy error exceeding η by $O_\beta(\eta^{-1}L^{-\beta})$. Both comparison tails at xe^η and $xe^{-\eta}$ are at least $L^{-(1-\epsilon/2)+o(1)}$. The coupling error is negligible relative to them. Sandwiching the integer event between these two tails proves the tail formula because $\eta \log x \rightarrow 0$ changes the exponent by only $o(x)$.

For the profile, choose instead $\eta = x^{-2}(\log x)^{-2}$. For a compactly supported Lipschitz ψ , the function $u \mapsto u\psi(\sqrt{\log x}(xu - 1))$ is Lipschitz with constant $O_\psi(1 + x\sqrt{\log x})$. Thus its coupled sums differ in expectation by $O_\psi((1 + x\sqrt{\log x})/L)$. Both this Markov error and the entropy-error probability remain negligible relative to the integer conditioning probability. The bad-profile probability in the limiting model, on $H > \log x - \eta$, is at most $\exp(-c_\psi x + o(x))$ times its tail probability, by the proof of Theorem 12.2. The shift changes the saddle normalization by $o(x)$ and the standardized test by $o(1)$. The ratio of that tail to the integer tail is $\exp(o(x))$, by the already proved logarithmic asymptotics. Therefore its conditional bad-profile probability still tends to zero. A countable determining set of Lipschitz tests completes the proof. $\square$

H.8 Why every finite deletion depth is abundant

The iteration needs a lower bound on the *number* of Goldbach representations; mere almost-everywhere existence would not suffice.

Lemma H.2 (Robust almost-all Goldbach). Let $R_2(m)$ count ordered prime pairs summing to the even integer m . There is $c > 0$ such that, for every $A > 0$,

$$\#\{m \leq Y : m \text{ even, } R_2(m) < cm/(\log m)^2\} \ll_A Y/(\log Y)^A.$$

Proof. We use the weighted form explicitly obtained in the proof of [4, Theorem 4.3], the classical almost-all Goldbach argument. Outside a set with arbitrarily large logarithmic saving, it gives

$$\sum_{a+b=m} \Lambda(a)\Lambda(b) \geq \frac{1}{2}m\mathfrak{S}(m).$$

The singular series has a positive absolute lower bound for even m . Pairs involving a non-prime prime power contribute $O(\sqrt{m}(\log m)^3) = o(m)$: there are $O(\sqrt{m} \log m)$ possible proper prime powers and each weight product is at most $(\log m)^2$. Remove them and bound each remaining prime-pair weight by $(\log m)^2$. This gives the asserted lower bound, with a smaller c . One may apply the weighted estimate on dyadic intervals and sum them; the finitely many small targets are absorbed in the implicit constant. $\square$

Finite-depth assertions of Theorem 13.1. For abundance, write $D_k(t) = \#\{p \leq t : p \notin B_k\}$. The estimate at $k = 0$ is immediate when starting from all primes. For the Goldbach-enabled initial set it follows from Lemma H.2, since an index i with $p_i \leq X$ has target at most $2(\pi(X) + 1)$.

Assume the estimate at depth k . Each missing prime spoils at most two ordered pairs for a fixed target, so

$$R_{2,B_k}(m) \geq R_2(m) - 2D_k(m).$$

Taking the induction estimate with exponent 4 gives $D_k(m) = o_k(m/\log^2 m)$. Every sufficiently large robust target from Lemma H.2 therefore still has a certificate at depth k . The newly missing indices through $\pi(X)$ form a subset of the lemma's exceptional targets through $2(\pi(X)+1)$. Together with the previously missing primes and finitely many small targets this proves the estimate at depth $k + 1$, for every requested logarithmic power. Its constants can depend on k .

For the ternary statement, the same robust lemma already implies $R_3(n) \gg n^2/\log^3 n$ for odd n : take primes $q \in [n/3, n/2]$. The prime number theorem supplies $\gg n/\log n$ choices, and all but $O_A(n/\log^A n)$ have robust even target $n - q$. Each such target has $\gg n/\log^2 n$ representations. A triple using a missing prime can be counted by choosing its position, the missing prime, and one other prime. There are at most $3D_k(n)\pi(n) = o(R_3(n))$ such triples. Hence $R_{3,B_k}(n)/R_3(n) \rightarrow 1$. $\square$

H.9 The graph spectrum and its collapse

The two-dimensional local matrix and its eigenvalues are

$$L_p = \begin{pmatrix} 1 - 1/p & \sqrt{(1/p)(1 - 1/p)} \\ \sqrt{(1/p)(1 - 1/p)} & 0 \end{pmatrix}, \quad a_p^\pm = \frac{p - 1 \pm \sqrt{(p - 1)(p + 3)}}{2p}.$$

Put $b_p = -a_p^-/a_p^+$ and $\Lambda_B = \prod_{p \in B} a_p^+$. The product is positive since $a_p^+ = 1 + O(p^{-2})$. The nonzero spectrum, with multiplicity, is

$$\Lambda_B (-1)^{|F|} \prod_{p \in F} b_p, \quad F \subset B \text{ finite.} \quad (\text{H.4})$$

For the full prime set this is Zagier's spectrum [28, Appendix C]. The tensor calculation also proves the version for B . Indeed finite prime cutoffs have exactly these products. Their kernels converge in L^2 because $\sum_{p > z} 1/p^2 \rightarrow 0$; a product involving a moving prime tends to zero since $b_p \sim 1/p$. Thus every eigenvalue bounded away from zero comes from a fixed finite set F , establishing the entire nonzero list.

Lemma H.3 (The spectral tail). Let B omit a prime set D with $\sum_{p \in D} 1/p < \infty$. If $N_{B,+}(u)$ and $N_{B,-}(u)$ count eigenvalues of T_B of each sign and magnitude at least u , then

$$N_{B,+}(u) \sim N_{B,-}(u) \sim \frac{A_B}{2u},$$

where

$$A_B = \sqrt{K_B(4)} \prod_{p \in D} (1 - 1/p), \quad K_B(4) = \prod_{p \in B} (1 - 1/p)^4 (1 + 4/(p-1)).$$

Proof. For $B = \mathcal{P}$, the absolute and signed spectral Dirichlet series are

$$D_{\text{abs}}(s) = \Lambda_{\mathcal{P}}^s \prod_p (1 + b_p^s) = \zeta(s) \mathcal{A}(s), \quad D_{\text{sgn}}(s) = \Lambda_{\mathcal{P}}^s \prod_p (1 - b_p^s) = \mathcal{E}(s) \zeta(s).$$

Since $b_p = p^{-1} + O(p^{-2})$, the correction $\mathcal{A}$ is analytic for $\Re s > 1/2$, and $\mathcal{E}$ is analytic for $\Re s > 0$. These assertions follow from locally uniform product estimates $1 + O(p^{-\Re s - 1})$ and $1 + O(p^{-\Re s - 1})$, respectively. At $s = 1$,

$$\mathcal{A}(1) = \prod_p (1 - 1/p)(a_p^+ - a_p^-) = \sqrt{K(4)} =: A.$$

The identity follows by squaring each local factor. The two positive-coefficient series $(D_{\text{abs}} \pm D_{\text{sgn}})/2$ have a simple pole of residue $A/2$ at one and a continuous remainder on $\Re s = 1$; the prime number theorem supplies nonvanishing of ζ on that line. The Wiener-Ikehara theorem [20], in its Laplace-Stieltjes form, gives $N_{\mathcal{P},\pm}(u) \sim A/(2u)$.

For the omitted set D , divide out its local factors:

$$D_{\text{abs},B}(s) = D_{\text{abs}}(s) \Lambda_D^{-s} \prod_{p \in D} (1 + b_p^s)^{-1}, \quad D_{\text{sgn},B}(s) = D_{\text{sgn}}(s) \Lambda_D^{-s} \prod_{p \in D} (1 - b_p^s)^{-1}.$$

Expand each inverse factor into a geometric series. At exponent one the total absolute coefficient weight is bounded by $\Lambda_D^{-1} \prod_{p \in D} (1 - b_p)^{-1} < \infty$. The full-prime counting functions are $O(x)$ at reciprocal threshold x , so dominated convergence applies to these convolutions. The absolute count has coefficient

$$\frac{A}{\Lambda_D \prod_{p \in D} (1 + b_p)} = A \prod_{p \in D} \frac{p}{\sqrt{(p-1)(p+3)}},$$

while the signed count remains $o(x)$. The displayed coefficient equals the asserted A_B , again by the local identity for $K(4)$. $\square$

Proof of Theorem 13.4. Theorem 13.1 and partial summation imply $\sum_{p \notin B_k} 1/p < \infty$ for each fixed k , so the spectral tail lemma applies. Its equivalent product is

$$A_k = \sqrt{K(4)} \prod_{p \notin B_k} \frac{p}{\sqrt{(p-1)(p+3)}}.$$

Every factor is below one. Their logarithms are $-1/p + O(p^{-2})$, and the omitted sets increase to $\mathcal{P} \setminus \{2, 3, 5\}$. Divergence of the prime harmonic series proves $A_k \downarrow 0$.

Two independent prime profiles share only finitely many primes almost surely, since $\sum_p 1/p^2 < \infty$. All shared primes above 5 eventually disappear from B_k . Thus the kernels converge almost everywhere to $G_{\{2,3,5\}}$, and bounded convergence gives Hilbert–Schmidt convergence. Its three local matrices each have two nonzero eigenvalues, so the tensor has rank $2^3 = 8$. Finally the local four-clique factors are below one, so removing restrictions increases the product. Its final value is $(5/16)(16/27)(512/625) = 512/3375$. $\square$

I Proof of the triangular spectral limit

This appendix proves Theorem 11.2, retaining the zero-counting and centering estimates that determine its normalization. Throughout, $c = e^{-\gamma}$.

I.1 An entire function with a measurable supply of zeros

Put

$$\mathcal{E}(z) = \sum_{k \geq 0} K(k) \frac{z^k}{k!}, \quad \kappa(t) = -\log K(t) \quad (t \geq 0).$$

The more accurate prime estimates needed below are

$$\begin{aligned} \kappa'(t) &= \log \log t + \gamma + O((\log t)^{-2}), \\ \kappa''(t) &= \frac{1 + O((\log t)^{-1})}{t \log t}, \\ |\kappa'''(t)| &\ll (t^2 \log t)^{-1}. \end{aligned} \tag{I.1}$$

Here and below derivative asymptotics concern $t \rightarrow \infty$.

We explain the extra precision in the first line, since an error of order $1/\log t$ would change the final location. Differentiation gives

$$\kappa'(t) = \sum_p \left\{ -\log(1 - 1/p) - \frac{1}{p+t-1} \right\}.$$

After splitting at $p = t$, Mertens' product theorem supplies $\log \log t + \gamma$ with error $o((\log t)^{-2})$. Replacing $p+t-1$ by $p+t$, and the logarithm by $1/p$ above t , costs $O(1/(t \log t))$. The remaining prime sum is

$$-\sum_{p \leq t} \frac{1}{p+t} + \sum_{p > t} \frac{t}{p(p+t)}.$$

Partial summation with the classical prime-number-theorem error turns its part on $t^{1/2} \leq p \leq t^{3/2}$ into the integral of $h(v)/(\log t + \log v)$, where

$$h(v) = \begin{cases} -(1+v)^{-1}, & 0 < v \leq 1, \\ [v(1+v)]^{-1}, & v > 1. \end{cases}$$

The omitted ranges are $O(t^{-1/2}/\log t)$. Since $\int_0^\infty h(v) dv = 0$ and $\int_0^\infty |h(v) \log v| dv < \infty$, the correction is $O((\log t)^{-2})$. This proves the first line of (I.1). Applying the same partial summation to $\sum_p (p+t-1)^{-2}$ proves the second; differentiating once more and bounding $\sum_p (p+t-1)^{-3}$ proves the third.

In particular $K(k)^{1/k} \rightarrow 0$. For every $\epsilon > 0$,

$$|\mathcal{E}(z)| \leq C_\epsilon e^{\epsilon|z|}.$$

Also $J_N(z/N) \rightarrow \mathcal{E}(z)$ locally uniformly, since its k th coefficient tends to $K(k)/k!$ and is bounded by $1/k!$. The negative real roots of J_N therefore give

$$\mathcal{E}(z) = \prod_{j \geq 1} (1 + z/\rho_j), \quad \rho_j > 0, \quad \sum_j \rho_j^{-1} = 1. \quad (\text{I.2})$$

Here is a direct way to check the representation and its missing exponential factor. Write $J_N(z/N) = \prod_j (1 + \alpha_{N,j} z)$ with $\alpha_{N,j} \geq 0$ and $\sum_j \alpha_{N,j} = 1$. Sort these numbers and take coordinate limits along a subsequence. The small factors, using $\log(1 + \alpha z) = \alpha z + O_z(\alpha^2)$, contribute an exponential e^{az} with $a \geq 0$; the remaining factors give the product in (I.2). A positive a contradicts the zero exponential type bound on the positive axis. Hence $a = 0$, and $\mathcal{E}'(0) = K(1) = 1$ gives the reciprocal sum. There must be infinitely many factors, since every coefficient of $\mathcal{E}$ is positive. In particular $\mathcal{E}(-z)$ belongs to the Laguerre–Pólya class and has only positive zeros.

Lemma I.1 (Arithmetic zero density). With $n(r) = \#\{j : \rho_j \leq r\}$,

$$n(r) \sim c \frac{r}{\log^2 r}, \quad \rho_n \sim c^{-1} n \log^2 n.$$

Proof. For $r > 0$, define a random nonnegative integer L_r by

$$\mathbb{P}(L_r = k) = \frac{K(k)r^k/k!}{\mathcal{E}(r)}.$$

Choose $m > 0$ from $r = me^{\kappa'(m)}$; the right side is strictly increasing from zero to infinity. If P is Poisson with mean m , put $\Delta = P - m$ and

$$\beta(k) = \kappa(k) - \kappa(m) - \kappa'(m)(k - m) \geq 0.$$

Convexity of κ gives the inequality, and direct cancellation gives the exact change of measure

$$\mathbb{P}(L_r = k) = \frac{e^{-\beta(k)}}{\mathbb{E}e^{-\beta(P)}} \mathbb{P}(P = k). \quad (\text{I.3})$$

Write $A = \kappa''(m)$. For $|\Delta| \leq m/2$,

$$\beta(P) = \frac{1}{2}A\Delta^2 + O\left(\frac{|\Delta|^3}{m^2 \log m}\right).$$

The complementary Poisson tail is exponentially small; multiplication by any of the fixed powers below preserves that bound. For example, $\kappa(k) = O(k \log(k+2))$ controls $\beta(k)$ there. Using the Poisson central moments

$$\mathbb{E}\Delta^2 = m, \quad \mathbb{E}\Delta^3 = m, \quad \mathbb{E}\Delta^4 = 3m^2 + m, \quad \mathbb{E}\Delta^6 = 15m^3 + 25m^2 + m$$

and Cauchy–Schwarz for intermediate absolute moments yields

$$\begin{aligned}\mathbb{E}\beta(P) &= \frac{1}{2}Am + O(m^{-1/2}/\log m), & \mathbb{E}\beta(P)^2 &= O((\log m)^{-2}), \\ \mathbb{E}[\Delta^2\beta(P)] &= \frac{1}{2}A(3m^2 + m) + O(\sqrt{m}/\log m), & \mathbb{E}[\Delta^2\beta(P)^2] &= O(m/(\log m)^2).\end{aligned}$$

Insert $e^{-\beta} = 1 - \beta + O(\beta^2)$ in (I.3). The mean displacement is $O(\sqrt{m}/\log m)$, and the second moment about m is

$$\mathbb{E}(L_r - m)^2 = m - \frac{1}{2}A(2m^2 + m) + o(m/\log m) = m - \frac{m}{\log m} + o(m/\log m).$$

The square of the mean displacement is $O(m/(\log m)^2)$. Consequently

$$\text{Var}(L_r) - \mathbb{E}L_r = -\frac{m}{\log m}(1 + o(1)). \quad (\text{I.4})$$

For $g(r) = \log \mathcal{E}(r)$, differentiation of the probability weights and of (I.2) gives

$$g'(r) = \frac{\mathbb{E}L_r}{r}, \quad g''(r) = \frac{\text{Var}(L_r) - \mathbb{E}L_r}{r^2} = -\sum_j (r + \rho_j)^{-2}.$$

Thus

$$\sum_j (r + \rho_j)^{-2} \sim \frac{c}{r \log^2 r}. \quad (\text{I.5})$$

We will also need the more accurate first-derivative estimate

$$g'(r) = \frac{c}{\log m} + O((\log m)^{-3}), \quad r = me^{\kappa'(m)}, \quad (\text{I.6})$$

which follows from the mean bound and (I.1).

For clarity, we give the short Tauberian step that extracts $n(r)$. Form the positive measure

$$d\eta_r(u) = \frac{\log^2 r}{cr} \frac{dn(ru)}{(1+u)^2} \quad (u \geq 0)$$

and push it forward under $v = (1+u)^{-1}$ to a measure λ_r on $[0, 1]$. Its mass tends to one by (I.5), and for every $t > 0$,

$$\int_0^1 \frac{d\lambda_r(v)}{[1 + (t-1)v]^2} = \frac{r \log^2 r}{c} \sum_j (tr + \rho_j)^{-2} \longrightarrow \frac{1}{t}.$$

The uniform probability measure on $[0, 1]$ has these same integrals. Expanding near $t = 1$ determines all moments, which determine a measure on this compact interval. Every subsequential limit is therefore uniform. It follows that

$$\frac{\log^2 r}{cr} n(r) = \int_{1/2}^1 v^{-2} d\lambda_r(v) \longrightarrow 1.$$

The discontinuity at $1/2$ has zero limiting mass. Asymptotic inversion now gives the formula for ρ_n . $\square$

I.2 The transfer theorem and the location constant

We invoke Campbell and Jalowy [5, Corollary 2.3(2)]. For a Laguerre–Pólya function with zero-counting law regularly varying with index $\alpha \in (0, 2)$, their theorem gives a free-stable limit for suitably scaled and centered Appell roots. The one-sided index-one case is included.

Apply it to $F(z) = \mathcal{E}(-z)$, whose compensated product is

$$F(z) = e^{-z} \prod_j (1 - z/\rho_j) e^{z/\rho_j}.$$

Its linear parameter is 1, its Gaussian parameter is zero, and all its zeros are positive. Put

$$B_n = \rho_n, \quad S(B) = \sum_j \frac{\rho_j}{\rho_j^2 + B^2}.$$

In the convention of the cited theorem, take limiting location zero. Its prescribed exponential shift is

$$a_n = -\frac{B_n}{n} + \frac{1}{n} \sum_j \frac{(B_n/\rho_j)^3}{1 + (B_n/\rho_j)^2} = -\frac{B_n}{n} S(B_n),$$

where the equality uses $\sum_j 1/\rho_j = 1$. The Appell polynomial of F is

$$A_{n,F}(x) = \sum_{k=0}^n (-1)^k K(k) \binom{n}{k} x^{n-k} = x^n J_n(-1/x).$$

Its roots are $R_{n,j}^{-1}$. Replacing $F(z)$ by $e^{-a_n z} F(B_n z/n)$ multiplies these roots by B_n/n and adds a_n . The transfer theorem therefore gives

$$\frac{1}{n} \sum_j \delta_{X_{n,j}} \implies \mu_+, \quad X_{n,j} = \frac{B_n}{n} \{R_{n,j}^{-1} - S(B_n)\}. \quad (\text{I.7})$$

The limit has weighted Lévy measure $\mathbf{1}_{x>0}(1+x^2)^{-1}dx$ and zero location.

To identify its normalization, the free Lévy–Khintchine formula gives

$$R_{\mu_+}(u) = \int_0^\infty \frac{u+x}{(1-xu)(1+x^2)} dx = -\log(-u).$$

For instance, at $u = -s < 0$ the integrand equals $x/(1+x^2) - s/(1+sx)$, whose integral is $-\log s$. Analytic continuation fixes the branch on the lower half-plane.

The remaining step is arithmetic: replace $S(B_n)$ by an explicit elementary center. We claim

$$S(B) - g'(B) = o((\log B)^{-2}). \quad (\text{I.8})$$

Lemma I.1 makes the rescaled measure $(\log^2 B/B) dn(Bu)$ converge vaguely to $c du$, so

$$\begin{aligned} & \log^2 B \{S(B) - g'(B)\} \\ &= \int_0^\infty \left\{ \frac{u}{1+u^2} - \frac{1}{1+u} \right\} \frac{\log^2 B}{B} dn(Bu) \longrightarrow 0. \end{aligned}$$

Indeed the kernel is bounded at zero, is $O(u^{-2})$ at infinity, and has integral zero: its antiderivative is $\frac{1}{2} \log(1+u^2) - \log(1+u)$. To justify the tails, below $u = B^{-1/2}$ the rescaled mass is $O(B^{-1/2})$;

from there to a fixed small ϵ , $n(Bu) \ll Bu/\log^2 B$ gives $O(\epsilon)$. Above a fixed $A > 1$, the same bound and partial summation give $O(A^{-1})$. This proves (I.8).

Choose m_n with $B_n = m_n e^{\kappa'(m_n)}$. The zero density gives $m_n \sim n \log n$, and hence $\log m_n = \log n + \log \log n + o(1)$. Combining (I.6) and (I.8) yields

$$\frac{c}{S(B_n)} = \log n + \log \log n + o(1), \quad \frac{cn}{B_n S(B_n)^2} \rightarrow 1, \quad \frac{n}{B_n S(B_n)} \rightarrow 0. \quad (\text{I.9})$$

The exact identity

$$cR_{n,j} - \frac{c}{S(B_n)} = -\frac{cn}{B_n S(B_n)^2} \frac{X_{n,j}}{1 + \frac{n}{B_n S(B_n)} X_{n,j}}$$

now completes the transfer. The empirical laws of $X_{n,j}$ are tight by (I.7); on each fixed compact interval the right side converges uniformly to $-X_{n,j}$. Thus (11.2) has the reflected limit law of T , with R -transform $\log u$.

The limiting density has the following explicit parametrization:

$$x(\theta) = \log \frac{\sin \theta}{\theta} + \theta \cot \theta, \quad f_T(x(\theta)) = \frac{\sin^2 \theta}{\pi \theta}, \quad 0 < \theta < \pi. \quad (\text{I.10})$$

For completeness, it can be read from the Cauchy-transform equation $z = 1/G(z) + \log G(z)$. On the support write $G(x + i0) = re^{-i\theta}$. The imaginary part forces $r = \sin \theta/\theta$; the real part and Stieltjes inversion give exactly (I.10). The function $x(\theta)$ decreases from 1 to $-\infty$. Arizmendi and Hasebe [1, Proposition 4.4] identify the exponential of this precisely normalized law with DH_1 . Alternatively, applying the exponential change of variables to (I.10) gives the Dykema–Haagerup density directly. The continuous mapping theorem proves (11.1).

J A finite certificate for an infinite matrix

This appendix specifies the rational calculation cited after Theorem 6.2. The accompanying file `code/verify_twin_certificate.py` uses integer and rational arithmetic for every decisive comparison. Its decimal output is only a readable summary.

Let $Y = 100$, and write

$$A_Y(z) = \prod_{2 < p \leq Y} \frac{1 - z/(p+3)}{1 - z/p} = \sum_{n \geq 0} \alpha_n z^n, \quad \frac{F(1-z)}{F(1)} = A_Y(z) B_Y(z).$$

The tail estimate in the main text gives

$$B_Y(z) = \exp\left(\sum_{k \geq 1} \frac{\epsilon_k z^k}{k}\right) = \sum_{m \geq 0} \beta_m z^m, \quad 0 \leq \epsilon_k \leq 3Y^{-k}.$$

Coefficient comparison with $(1 - z/Y)^{-3}$ yields

$$\beta_0 = 1, \quad 0 \leq \beta_m \leq \binom{m+2}{2} Y^{-m}. \quad (\text{J.1})$$

Keeping one common sequence (β_m) preserves the dependence between the matrix entries; independent intervals for the entries would throw away useful structure.

Put $S = \{3, 5, 7, 11, 13, 17, 19, 23\}$ and $T = \{5, 7, 13, 19\}$. For $p \in T$ define the degree-fourteen polynomial

$$P_p(u) = \frac{u^7 \prod_{q \in S \setminus \{p\}} (u - 1/q)}{p^{-7} \prod_{q \in S \setminus \{p\}} (1/p - 1/q)}.$$

It has rational coefficients and satisfies $P_p(1/q) = \mathbf{1}_{\{p=q\}}$ for $q \in T$. Let $P_i(u) = \sum_r a_{i,r} u^r$, and form

$$(Q_m)_{ij} = \sum_{r,s} a_{i,r} a_{j,s} \alpha_{r+s+2-m}, \quad \alpha_n = 0 \quad (n < 0).$$

The actual infinite moment form on these four coefficient vectors is

$$\sum_{m=0}^{30} \beta_m Q_m.$$

All α_n through degree 30 are obtained from the rational recursion

$$\alpha_0 = 1, \quad \alpha_n = \frac{1}{n} \sum_{k=1}^n \left(\sum_{2 < p \leq 100} \{p^{-k} - (p+3)^{-k}\} \right) \alpha_{n-k}.$$

Thus the certificate has no unspecified numerical inputs.

Exact symmetric elimination verifies that Q_0 is negative definite. Rational Gram–Schmidt changes the four polynomials to a basis in which $Q_0 = -\text{diag}(a_1, a_2, a_3, a_4)$, all $a_i > 0$. In this basis, exact elimination also verifies that Q_1, Q_2 are negative definite. Define

$$E_{ij} = \sum_{m=3}^{30} \binom{m+2}{2} 100^{-m} |(Q_m)_{ij}|.$$

The four exact comparisons checked by the program are

$$7 \sum_j E_{ij} < a_i \quad (1 \leq i \leq 4). \quad (\text{J.2})$$

For orientation, the four ratios $\sum_j E_{ij}/a_i$, rounded only for display, are

$$0.001350780, \quad 0.003829068, \quad 0.027644920, \quad 0.140852054.$$

For any nonzero real vector v , use (J.1), discard the two negative tail contributions, and apply $2|v_i v_j| \leq v_i^2 + v_j^2$. The result is

$$v^T \left(\sum_{m=0}^{30} \beta_m Q_m \right) v \leq - \sum_i \left(a_i - \sum_j E_{ij} \right) v_i^2 < -\frac{6}{7} \sum_i a_i v_i^2 < 0.$$

The four polynomials are independent by their interpolation values. Hence they certify four negative directions in the infinite matrix.

The certificate's witnesses use the already known twin upper endpoints 5, 7, 13, 19. Its purpose is to show how a statement about infinitely many uncomputed primes can be certified by finite exact inequalities. It makes no claim to discover a new twin pair.

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